Dissecting the Aggregate Market Elasticity*

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Abstract

We study aggregate stock market elasticity in a general equilibrium model with heterogeneous investors, passive demand, and financial constraints. Without frictions, the elasticity of the endowment claim is infinite, as interest rate and risk premium responses offset each other. With frictions, price impact for the endowment claim remains modest (about 0.7 in our calibration). In contrast, the equity (dividend) claim exhibits large price impact (above 8), consistent with empirical evidence, as frictions dampen interest rate responses while leverage amplifies risk premia responses to portfolio flows. We introduce a state-global perturbation method that yields closed-form, state-dependent elasticities. Calibrated to match the magnitude of portfolio flows, the model simultaneously matches the equity premium, return volatility, and the level and countercyclical dynamics of price impact.

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JEL Classification: G12, G11, G21.

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1 Introduction

Why is the stock market so volatile? This question has long been central to asset pricing research. Classic tests of excess volatility attribute the puzzle to unobserved "dark matter" factors.[1] More recent evidence points instead to a demand-based channel: asset prices respond strongly to portfolio flows. Yet these flows are small relative to market capitalization, implying that markets must be highly *inelastic*, with even modest quantity shifts generating large price changes (Gabaix and Koijen, 2023). Moreover, the ratio of return volatility to flows—the "volatility multiplier"—is itself time-varying and countercyclical, rising sharply during periods of financial stress. This pattern points to a state-dependent relationship between prices and portfolio flows.

These facts raise a central question. How can small portfolio flows generate large price movements in equilibrium? More specifically, what determines the *aggregate* elasticity of the stock market, why does price impact vary over time, and which frictions are essential for matching its magnitude in the data?

To address these questions, we develop a general equilibrium model with heterogeneous investors, passive demand, and financial constraints. We study the *aggregate (macro) elasticity*, defined as the change in total stock market value in response to a \$1 flow between (short-term) bonds and stocks. Unlike micro elasticities, which capture substitution across individual stocks, aggregate elasticity reflects market-wide reallocations between risky and risk-free assets. As a result, understanding aggregate elasticity requires tracking how the risk-free rate and the risk premium respond jointly to portfolio flows.

This joint response is central for the elasticity of the *endowment claim*, the claim on aggregate output. We show that, in the absence of frictions, the aggregate market elasticity of this claim is *infinite*, regardless of how price-inelastic individual investors are.[2] Consider an outflow of funds from stocks to bonds. The resulting increase in the risk premium tends to lower stock prices. At the

[1] See LeRoy and Porter (1981), Shiller (1981, 1992), and Cochrane (1992); and Chen, Dou, and Kogan (2024) for a recent discussion.

[2] A portfolio-flow shock is an asymmetric shock to risky-asset demand, affecting only a subset of investors and triggering portfolio reallocation in equilibrium.

same time, the associated decline in wealth would reduce consumption. With a fixed endowment, this adjustment cannot occur in equilibrium. The interest rate therefore falls to clear the goods market. In frictionless economies, it adjusts until it exactly offsets the increase in the risk premium, leaving the price of the risky asset unchanged. This result holds regardless of how inelastic individual demand is. Aggregate elasticity is therefore infinite even when individual demands are highly inelastic.

This result highlights a central distinction between individual and aggregate elasticities. If cross-price effects are absent, so that demand for risky assets does not respond to the interest rate, aggregate elasticity reduces to a weighted average of individual elasticities, as in Gabaix and Koijen (2023). Once interest rates adjust endogenously, however, aggregate elasticity is governed by general equilibrium forces rather than by the average slope of individual demands. In particular, aggregate elasticity is governed by how portfolio flows affect saving incentives. When flows do not alter these incentives, movements in the risk-free rate fully offset movements in the risk premium, so asset prices remain unchanged.

Aggregate elasticity becomes finite when risk is misallocated. Portfolio reallocation then alters saving incentives, so interest rates can no longer fully offset changes in risk premia.[3] This generates a nonzero price impact, even in general equilibrium. Frictions thus reduce aggregate elasticity. Even with frictions—such as passive demand and leverage constraints—the near-cancellation between interest rates and risk premia remains quantitatively important, so the endowment claim exhibits only a modest price impact of 0.7 in our calibration. We establish these results first in a transparent two-period model, then extend them to a fully dynamic setting.

Studying aggregate elasticity in models with frictions is challenging because closed-form solutions are typically not available. We develop a tractable state-global perturbation method that yields closed-form, state-dependent elasticities. Elasticity is time-varying and shaped by both the wealth distribution and risk allocation across investors. Passive demand, preference heterogeneity, and

[3]Intuitively, risk misallocation means investors hold too much or too little risk relative to the efficient allocation. Investors who are already overexposed to risk are less willing to save in assets that further increase their risk exposure. As a result, interest rates do not need to adjust as much to restore equilibrium after a portfolio-flow shock.

leverage constraints each amplify price impact via risk allocation. Excess volatility results from the interaction of portfolio-flow shocks with finite (state-dependent) elasticity. Neither inelasticity nor flows alone generate excess volatility; both are required.

Our second main contribution is to provide a characterization of the aggregate elasticity of the equity claim. Dividends are modeled as a time-varying share of output, as in Menzly, Santos, and Veronesi (2004). In contrast to the endowment claim, the equity claim exhibits substantially larger price impact, with a multiplier above 8 in our calibration, consistent with Gabaix and Koijen (2023) and Koijen, Richmond, and Yogo (2024). This difference arises because the general equilibrium offset between the risk-free rate and the risk premium that renders the endowment claim highly elastic does not extend to the dividend claim. Dividends are a levered share of aggregate output, so the dividend risk premium responds more strongly to portfolio flows than the endowment risk premium. Consequently, changes in risk premia are not fully offset, so prices respond strongly to portfolio flows. This result highlights that modeling the dividend claim, rather than aggregate consumption as is standard in macro-finance, is essential for generating realistic price impact.

We provide a tractable way to incorporate the dividend claim in a dynamic heterogeneous-agent economy with frictions. We establish conditions under which a version of the mutual fund theorem holds in our economy with frictions, so investors optimally hold risky claims at market-capitalization weights. This implies that the stochastic discount factor from the aggregate (endowment-claim) economy can be used to price the dividend claim in a second step. As a result, the problem separates into an aggregate equilibrium and a simpler pricing problem, reducing its dimensionality.

For the quantitative analysis, we solve the full model using the deep neural network method of Duarte, Duarte, and Silva (2024) and calibrate it to Flow of Funds data. We target the magnitude of portfolio flows observed in the data. The model matches both the level and countercyclical dynamics of price impact, while reproducing key asset pricing moments such as the equity premium and return volatility. These results reflect the joint role of frictions, which dampen interest rate responses, and leverage, which amplifies risk premia responses to portfolio flows.

Related literature

Our work relates to the literature on macro demand elasticity. The papers closest to ours are Johnson (2006), who finds finite elasticity even in a frictionless Lucas economy by perturbing risky asset supply, and Gabaix and Koijen (2023), who introduce a behavioral element that fixes interest rates and generates large price impacts. We build on this literature by developing a GE model with heterogeneous investors and financial frictions, showing that both GE effects and constraints are central for obtaining inelastic markets.[4] This contrasts with the larger body of work on micro elasticity, which examines relative price changes across individual stocks (e.g., Shleifer, 1986; Harris and Gurel, 1986; Chang, Hong, and Liskovich, 2015; Pavlova and Sikorskaya, 2023; Schmickler, 2020) and finds much larger elasticities, consistent with stocks being closer substitutes for each other than for bonds.

We also connect to demand-system asset pricing (Koijen and Yogo, 2019) and its extensions incorporating cross-asset elasticities (Fuchs, Fukuda, and Neuhann, 2023; Haddad, He, Huebner, Kondor, and Loualiche, 2025a). Our framework shows that even small cross-elasticities between equities and bonds generate large effects on macro elasticities, and extends the analysis to dynamic settings where the persistence of shocks and the evolution of risk premia shape both short-run and long-run adjustments (Binsbergen, David, and Opp, 2025; He, Kondor, and Li, 2025; Davis, Kargar, Li, and Silva, 2026). Our model of passive investors also connects to the literature on intermittent rebalancing (Chien, Cole, and Lustig, 2012), where sluggish portfolio adjustments are themselves an important source of aggregate inelasticity. Finally, our paper relates to preferred-habitat models (Vayanos and Vila, 2021; Kekre, Lenel, and Mainardi, 2025; Ray, Droste, and Gorodnichenko, 2024), which emphasize intermediary frictions and market segmentation. By contrast, our framework highlights wealth effects as a distinct channel through which shifts in aggregate wealth alter portfolio demand and amplify price movements.

[4]For more on macro elasticity, see Johnson (2008, 2009), Deuskar and Johnson (2011), Li, Pearson, and Zhang (2020), Hartzmark and Solomon (2025), among others.

2 A Simple Model of the Aggregate Market Elasticity

In this section, we analyze the determination of the aggregate market elasticity in a simple general equilibrium, demand-based asset pricing model. To keep the discussion transparent, we specify investor demands directly, rather than deriving them from utility functions. A micro-founded version of these demands, embedded in a dynamic heterogeneous-agents model, will be presented in Section 3.

Environment. We consider a two-period economy with two assets: a risky asset and a riskless bond. There are two types of investors: passive (p) and active (a). A fraction ω_j of the population is of type $j \in \{a, p\}$. The risky asset pays a random dividend Y' in the second period. Its price is denoted by P, while the price of the riskless asset is R_f^{-1}.

The budget constraints of investor j are

$$PQ_j + R_f^{-1} B_j + C_j = W_j, \qquad C_j' = Y' Q_j + B_j,$$

where Q_j is the investor's holdings of the risky asset, B_j denotes holdings of the riskless bond, C_j is first-period consumption, and C_j' is second-period consumption. Initial wealth is given by $W_j = (P + Y) Q_{j,-1} > 0$.

We define investor j's portfolio share in the risky asset as

$$\alpha_j \equiv \frac{PQ_j}{W_j - C_j}.$$

Passive investors maintain a fixed risky asset share $\alpha_p = \overline{\alpha}_p \geq 0$. Active investors' portfolio share, in contrast, depends on the risk premium

$$\alpha_a = \overline{\alpha}_a(\pi),$$

for some increasing function $\overline{\alpha}_a(\cdot)$, where $\pi \equiv \log \frac{1}{R_f} \mathbb{E}\left[\frac{Y'}{P}\right]$ denotes the risk premium.

Consumption is specified as

$$C_j = \overline{c}_j(r, \pi) W_j,$$

where $r \equiv \log R_f$ is the log risk-free rate. The consumption-wealth ratio $\overline{c}_j(\cdot)$ depends on both r and π. If $\overline{c}_j$ decreases with r, investors save more when the interest rate is high (a substitution effect); if $\overline{c}_j$ increases with r, investors save less (an income effect). Similar logic applies to the risk premium. We assume that $\overline{c}_j(\cdot)$ does not depend on the passive portfolio share $\overline{\alpha}_p$.

The market clearing conditions are

$$\sum_{j \in \{p,a\}} \omega_j C_j = Y, \qquad \sum_{j \in \{p,a\}} \omega_j Q_j = 1, \qquad \sum_{j \in \{p,a\}} \omega_j B_j = 0,$$

with initial risky asset endowment $\sum_{j \in \{p,a\}} \omega_j Q_{j,-1} = 1$.

Market for risky assets. Let $p = \log \frac{P}{Y}$ denote the log price-dividend ratio, and let $\mu \equiv \log \frac{\mathbb{E}[Y']}{Y}$ denote the log dividend growth. The risk premium is

$$\pi = \mu - p - r. \tag{1}$$

Investor j's demand for the risky asset is

$$Q_j = \alpha_j \left[1 - \overline{c}_j(r, \pi)\right] \frac{W_j}{P}.$$

Using Equation (1) to eliminate π, we can express demand as a function of price of the risky asset, the risk-free rate, and, for passive investors, the exogenous portfolio share: $Q_j = \overline{Q}_j(p, r, \overline{\alpha}_p)$.

The market clearing condition for the risky asset is therefore

$$\underbrace{\omega_a \overline{Q}_a(p, r, \overline{\alpha}_p)}_{\text{active demand}} = \underbrace{1 - \omega_p \overline{Q}_p(p, r, \overline{\alpha}_p)}_{\text{net supply}}. \tag{2}$$

Equilibrium in the risky asset market requires that active investors absorb the net supply, defined

as total supply minus the amount held by passive investors. Since active demand does not depend on $\overline{\alpha}_p$, changes in the passive portfolio share affect only the net supply curve.

Market for goods. The response of the consumption-wealth ratio to the interest rate and the risk premium is central to our analysis. In the dynamic model of Section 3, the average consumption-wealth ratio depends only on the sum $(r+\pi)$, so its sensitivity to interest rates equals its sensitivity to risk premia. Assumption 1 imposes the same structure in this two-period setting, ensuring consistency with standard micro-founded models.

Assumption 1. *The average consumption-wealth ratio is a function of the expected return on the risky asset,* $(r+\pi)$,

$$\sum_{j\in\{p,a\}} x_j \overline{c}_j(r,\pi) = \overline{c}(r+\pi),$$

for some function $\overline{c}(\cdot)$, *where* $x_j \equiv \frac{\omega_j W_j}{\omega_a W_a + \omega_p W_p}$ *denotes investor* j*'s wealth share.*

Intuitively, since bonds are in zero net supply, the return on investors' overall portfolios equals the return on the risky asset. From goods market clearing and Equation (1), we obtain

$$\overline{c}(\mu - p) = \frac{1}{1+e^p}, \tag{3}$$

using the fact that $r+\pi = \mu - p$ and $\sum_j x_j \overline{c}_j = \frac{Y}{Y+P}$.

Assumption 1 implies that the system of equations determining p and r has an important *recursive property*: condition (3) depends only on p, while condition (2) depends on both p and r.

2.1 General equilibrium implications of portfolio flows

We now examine how the price-dividend ratio p and the riskless return r respond to a portfolio flow from the riskless bond into the risky asset. Let $\overline{\alpha}_p^\star$ denote the passive portfolio share in the initial equilibrium, with $(p^\star, r^\star)$ the corresponding equilibrium prices.

For a small perturbation, $\overline{\alpha}_p = \overline{\alpha}_p^\star e^{\hat{\alpha}_p}$, we linearize demand around the initial equilibrium. Defining $\hat{p} \equiv p - p^\star$, $\hat{r} \equiv r - r^\star$, and $q_j \equiv \log(Q_j/Q_j^\star)$, we obtain

$$q_j \;=\; -\zeta_{j,p}\,\hat{p} \;-\; \zeta_{j,r}\,\hat{r} \;+\; f_j, \tag{4}$$

where the own-price and cross-price elasticities are given by $\zeta_{j,p} \equiv -\frac{\partial \log \overline{Q}_j}{\partial p}$ and $\zeta_{j,r} \equiv -\frac{\partial \log \overline{Q}_j}{\partial r}$. The *flow shock* is given by

$$f_j \;\equiv\; \frac{\partial \log \overline{Q}_j}{\partial \log \overline{\alpha}_p} \times \hat{\alpha}_p,$$

capturing an exogenous reallocation from bonds to stocks. Since active demand does not depend on $\overline{\alpha}_p$, we have $f_a = 0$. Formally, a flow shock is an asymmetric demand shock affecting some investors but not others, inducing equilibrium portfolio rebalancing.

An important implication of (4) is that demand for the risky asset depends on both the price-dividend ratio p and the bond price through the risk-free rate r. Since active investors respond to the risk premium $\pi = \mu - p - r$, their demand depends on both prices. Therefore, the prices of the risky and risk-free assets must be jointly determined in equilibrium.

Equilibrium. Let $s_j \equiv \omega_j Q_j^\star$ denote the equilibrium quantity share of the risky asset held by type j. Linearizing the market-clearing condition $\sum_j \omega_j Q_j = 1$ gives

$$s_a\left(-\zeta_{a,p}\,\hat{p} - \zeta_{a,r}\,\hat{r}\right) \;=\; s_p\left(\zeta_{p,p}\,\hat{p} + \zeta_{p,r}\,\hat{r} - f_p\right), \tag{5}$$

as $s_a q_a + s_p q_p = 0$.

The left panel of Figure 1 shows the equilibrium in the risky-asset market. The downward-sloping curve represents active demand, while the upward-sloping curve is net supply, i.e., total supply minus passive holdings.[5]

[5]Passive demand is $Q_p \;=\; \overline{\alpha}_p\,[1 - c_p(r,\pi)]\,\frac{W_p}{P}$, given $W_p = (P+Y)Q_{p,-1}$ and $\pi = \mu - p - r$, so its linearized contribution $-s_p q_p$ makes net supply upward sloping in p provided $\frac{\partial c_p}{\partial p}$ is not too large.

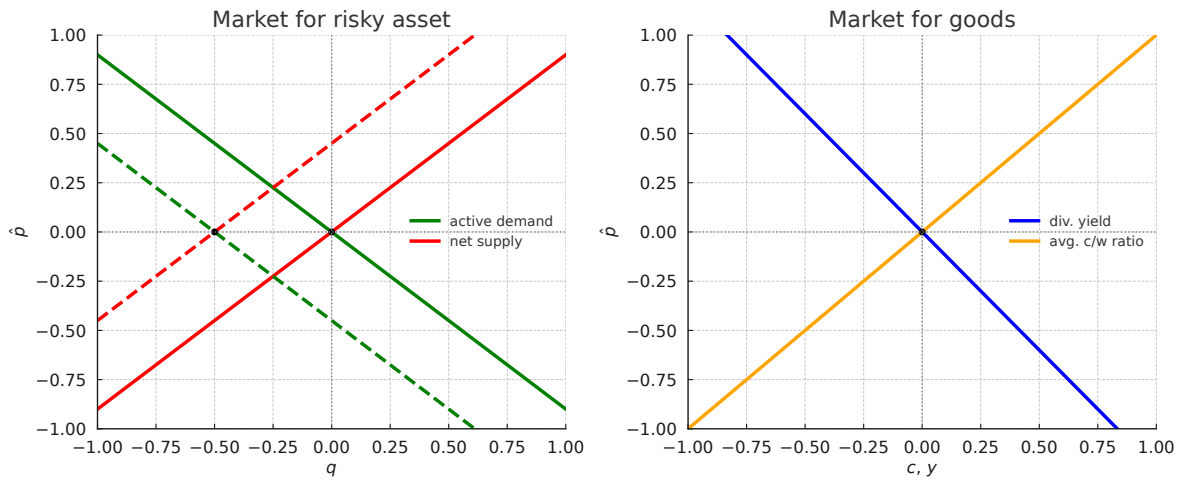


Figure 1. Equilibrium in the risky asset (left) and goods (right) markets.

Linearizing the goods-market condition $\overline{c}(\mu - p) = 1/(1 + e^p)$ around $p^\star$ yields

$$\chi_p \, \hat{p} \;=\; -\frac{e^{p^\star}}{(1 + e^{p^\star})^2} \, \hat{p}, \qquad \text{where} \qquad \chi_p \equiv -\overline{c}'(\mu - p^\star). \tag{6}$$

The right panel of Figure 1 shows the goods market. We focus on the case $\chi_p > 0$, so the average consumption-wealth ratio is increasing in p (investors save less when expected returns are low). The right-hand side in (6) is the (local) slope of the dividend-yield curve, which is downward sloping in p. By the recursive property, only p enters the goods market clearing condition.

The inelastic markets hypothesis. Consider a *partial equilibrium* version of the model in which we hold the interest rate fixed at $r = r^\star$ and ignore goods market clearing. Using the linearized risky asset market condition (5) with $\hat{r} = 0$, the price of the risky asset satisfies

$$\hat{p} \;=\; \underbrace{\frac{1}{\zeta_p}}_{\substack{\text{inverse market}\\ \text{elasticity}}} \times \underbrace{f}_{\text{flow shock}}, \tag{7}$$

where

$$\zeta_p \;\equiv\; s_a \, \zeta_{a,p} + s_p \, \zeta_{p,p}, \qquad f \;\equiv\; s_p \, f_p, \qquad s_j \equiv \omega_j Q_j^\star.$$

Equation (7) shows that the price-dividend ratio responds to flow shocks, with price impact given by the inverse of the (aggregate) market elasticity—an average of investors' own-price elasticities. This is the core implication of the *inelastic markets hypothesis* of Gabaix and Koijen (2023), who estimate large price responses to stock-market flows, and study a model where investors are relatively inelastic.

In the risky asset panel of Figure 1, an increase in the portfolio share of passive investors shifts the net supply curve to the left. The new equilibrium lies further up the active demand curve, implying lower expected returns and a higher price-dividend ratio. When demand is inelastic, even small inflows to stocks generate a sizable increase in $\hat{p}$.

The crucial role of cross-elasticity. The partial equilibrium analysis abstracts from cross-elasticity. In general equilibrium, however, both r and p adjust, and the aggregate market elasticity is fundamentally different, as the next proposition shows.

Proposition 1 (Infinite market elasticity)**.** *Define the aggregate cross-elasticity* $\zeta_r \equiv s_a\,\zeta_{a,r} + s_p\,\zeta_{p,r}$. *If* $\zeta_r \neq 0$*, then a portfolio-flow shock has* no *price impact:* $\hat{p} = 0$. *The shock is absorbed by offsetting movements in the interest rate and the risk premium:*

$$\hat{r} = \frac{1}{\zeta_r} f, \qquad \hat{\pi} = -\frac{1}{\zeta_r} f,$$

where $\hat{\pi} \equiv \pi - \pi^\star$.

Proof. From Equation (6), we obtain $\hat{p} = 0$. If $\zeta_r \neq 0$, then $\hat{r} = \frac{1}{\zeta_r} f$ from Equation (5). Since $\hat{\pi} = -\hat{p} - \hat{r}$ with $\hat{p} = 0$, it follows that $\hat{\pi} = -f/\zeta_r$. □

With any nonzero cross-elasticity $\zeta_r \neq 0$, the aggregate market becomes infinitely elastic, regardless of how inelastic individual demands may be. Even if $\zeta^q_{j,p} \approx 0$ for all j, a small but positive ζ_r is sufficient to make aggregate elasticity unbounded.

The disconnect between individual and aggregate elasticities arises from a general equilibrium effect. In the goods market (right panel of Figure 1), the price-dividend ratio is determined by

consumption behavior. This mechanism is well known in the macro-finance literature, most clearly in models with unit EIS, where p is tied directly to the discount rate. Our framework generalizes this logic. To restore equilibrium in the risky asset market (left panel), the interest rate adjusts so that active demand equals net supply at the *original price*. Graphically, this appears as the dashed active-demand curve shifting to intersect the new net-supply curve at the initial price.

Consider a positive flow shock to the risky asset (e.g., funds shift toward stocks, so $f > 0$). Lower bond demand increases r, while increased demand for the risky asset reduces the risk premium. In equilibrium these effects offset exactly, leaving the price-dividend ratio unchanged. If the rise in r were smaller than the fall in the risk premium, there would be excess demand for goods (equivalently, excess bond supply). Simultaneous clearing of risky and risk-free asset markets therefore requires infinite aggregate elasticity in this model.

The need for a micro-founded demand system. The analysis underscores the central role of cross-price elasticity. The aggregate market can be infinitely elastic—even if individual investors are inelastic—so long as the average consumption-wealth ratio is unaffected by the flow shock f. When the flow shock shifts both the net-supply curve and the average consumption-wealth ratio, prices adjust and aggregate elasticity becomes finite.[6]

Understanding aggregate elasticity therefore requires a framework that links portfolio reallocation shocks to both risky asset demand and bond demand (i.e., savings behavior). To this end, we develop a dynamic heterogeneous-agent asset pricing model and introduce a new methodology for deriving investor demand within this setting.

3 A Dynamic Asset Pricing Model with Passive Investors

This section develops a dynamic heterogeneous-agent asset pricing model that provides a micro-founded demand system. Building on the framework of Section 2, the economy features both passive

[6]An analogous argument applies to uncertainty. With unit EIS, the consumption–wealth ratio is fixed, so higher uncertainty leaves prices unchanged. For uncertainty to reduce prices, EIS must exceed one so that higher uncertainty lowers the consumption–wealth ratio (see Bansal and Yaron, 2004).

and active investors, but now embeds their behavior in a fully dynamic general equilibrium setting.

Passive investors hold exogenously specified, stochastic portfolio shares in risky assets. Shocks to their portfolio share generate portfolio-flow (i.e., noisy trading) shocks imperfectly correlated with cash-flow shocks, so the economy is driven by both fundamental and non-fundamental demand disturbances. Active investors, in contrast, optimally rebalance their portfolios subject to financial frictions. There are two types of active investors with different risk aversion: a high risk-aversion (unconstrained) investor and a low risk-aversion investor subject to a leverage constraint.

A key extension relative to the simple model is that investors trade two distinct claims—a *dividend claim* (equity) and a *risky corporate bond*—whose payoffs sum to the aggregate endowment. Explicitly modeling the dividend process allows us to study the response of equity prices to portfolio flows, instead of focusing on the aggregate claim. As we show below, this distinction is central, as the price impact for the dividend claim can differ substantially from the price impact for the aggregate claim.

3.1 Environment

Financial markets. We begin by specifying the aggregate endowment process. The aggregate endowment Y_t follows a geometric Brownian motion:

$$\frac{dY_t}{Y_t} = \mu\, dt + \sigma\, dZ_t, \tag{8}$$

where $\mu \in \mathbb{R}$ and $\sigma \in \mathbb{R}^{1\times d}$ are constants.[7] The process $Z = \{Z_t\}_{t\geq 0}$ is a standard d-dimensional Brownian motion on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. In the baseline model, $d = 3$, capturing three sources of risk: (i) aggregate endowment shocks, (ii) portfolio-flow shocks to passive investors, and (iii) shocks to the dividend share.

We consider a representative firm that issues two securities whose payoffs sum to the aggregate endowment. The first is a *dividend claim* (equity), with payoff flow $D_t = s_t Y_t$. Following Menzly

[7]Throughout the paper, we adopt the convention that loadings on the Brownian motion are row vectors.

et al. (2004), the dividend share $s_t \in (0,1)$ follows the mean-reverting process:

$$ds_t = \kappa_s(\bar{s} - s_t)\,dt + s_t(1 - s_t)\,\sigma_s\,dZ_t, \tag{9}$$

given $s_0 \in (0,1)$, with $\bar{s} \in (0,1)$, $\kappa_s > 0$, and $\sigma_s \in \mathbb{R}^{1\times d}$. This specification ensures $s_t \in [0,1]$ at all times and yields a stationary share process, so consumption and dividends are cointegrated. As a result, dividends inherit both aggregate risk and additional variation from movements in the dividend share. By Itô's lemma, dividends follow:

$$\frac{dD_t}{D_t} = \mu_{D,t}\,dt + \sigma_{D,t}\,dZ_t,$$

where

$$\sigma_{D,t} = \sigma + (1 - s_t)\,\sigma_s, \qquad \mu_{D,t} = \mu + \kappa_s\left(\frac{\bar{s}}{s_t} - 1\right) + (1 - s_t)\,\sigma\sigma_s^\top.$$

This specification captures several key features of the data. First, it generates the well-documented fact that dividends are more volatile than consumption (see, e.g., Abel 1999; Campbell 1999). Second, the dividend share predicts future dividend growth, consistent with Menzly et al. (2004). Third, dividend volatility increases after negative shocks to s_t, capturing a leverage effect.

The second claim is a *corporate bond* (risky debt), with payoff flow $B_t = (1 - s_t)Y_t$. Since $D_t + B_t = Y_t$, the bond payoff is smoother than aggregate consumption, reflecting the safer nature of debt and its lower exposure to aggregate risk. This decomposition allows us to distinguish between claims with different risk exposures, with important implications for price impact.

Let $P_{D,t}$ and $P_{B,t}$ denote the ex-dividend prices of the dividend and bond claims, with returns:

$$dR_{D,t} = \frac{dP_{D,t} + D_t\,dt}{P_{D,t}} = \mu_{R_D,t}\,dt + \sigma_{R_D,t}\,dZ_t,$$
$$dR_{B,t} = \frac{dP_{B,t} + B_t\,dt}{P_{B,t}} = \mu_{R_B,t}\,dt + \sigma_{R_B,t}\,dZ_t.$$

Since $D_t + B_t = Y_t$, the combined claim replicates the aggregate endowment, with price $P_{Y,t} =$

$P_{D,t} + P_{B,t}$. The return on the aggregate claim follows:

$$dR_{Y,t} = \frac{dP_{Y,t} + Y_t\, dt}{P_{Y,t}} = \mu_{R_Y,t}\, dt + \sigma_{R_Y,t}\, dZ_t.$$

Investors also have access to a risk-free asset paying rate r_t and, to ensure markets are dynamically complete, a derivative in zero net supply that spans risks not directly traded through the dividend and bond claims. In particular, we assume the derivative does not load on dividend-share shocks. As in a futures contract, there is no initial cash exchange to enter into the derivative contract. The derivative provides risk exposure $\sigma_{R_F,t} \in \mathbb{R}^{1\times d}$, orthogonal to the aggregate claim, $\sigma_{R_F,t}\, \sigma_{R_Y,t}^\top = 0$, so that it provides independent risk exposure, with normalization $\|\sigma_{R_F,t}\| = \|\sigma_{R_Y,t}\|$.

Stochastic discount factor. Absence of arbitrage implies the existence of a stochastic discount factor (SDF), Λ_t, which prices all assets in the economy:

$$\frac{d\Lambda_t}{\Lambda_t} = -r_t\, dt - \eta_t\, dZ_t,$$

where $\eta_t \in \mathbb{R}^{1\times d}$ is the market price of risk vector.

Absence of arbitrage implies that risk premia satisfy $\pi_{k,t} = \sigma_{R_k,t}\, \eta_t^\top$ for each claim $k \in \{D, B, F\}$. Let Σ_t denote the 3×3 matrix collecting the risk exposures of all traded assets:

$$\Sigma_t = \begin{pmatrix} \sigma_{R_D,t} \\ \sigma_{R_B,t} \\ \sigma_{R_F,t} \end{pmatrix},$$

and $\pi_t \equiv [\pi_{D,t}, \pi_{B,t}, \pi_{F,t}]^\top$, so that $\pi_t = \Sigma_t\, \eta_t^\top$.

In equilibrium, prices depend on an aggregate state vector $X_t \in \mathbb{R}^N$, so we write $r_t = r(X_t)$, $\eta_t = \eta(X_t)$, and return coefficients as functions of X_t. The state vector evolves as

$$dX_t = \mu_{X,t}\, dt + \sigma_{X,t}\, dZ_t,$$

with drift $\mu_{X,t} \in \mathbb{R}^N$ and exposure matrix $\sigma_{X,t} \in \mathbb{R}^{N\times d}$, both determined in equilibrium.

Demography and preferences. The economy consists of three investor types, indexed by $j = 0, 1, 2$, with population mass ω_j: one passive type ($j = 0$) and two active types ($j = 1, 2$). Investors die with Poisson intensity κ, and each instant a mass $\kappa\,\omega_j$ of type-j agents are born, so the total population remains constant and normalized to one. Newborns inherit parental wealth, so wealth is redistributed across cohorts without affecting aggregate resources. This mortality structure ensures a nondegenerate stationary wealth distribution.

Preferences are recursive as in Duffie and Epstein (1992), with type-j risk aversion γ_j and a common elasticity of intertemporal substitution (EIS) ψ. The aggregator is given by:

$$f_j(C,V) = \rho\frac{(1-\gamma_j)V}{1-\psi^{-1}}\left[\left(\frac{C}{\big((1-\gamma_j)V\big)^{\frac{1}{1-\gamma_j}}}\right)^{1-\psi^{-1}} - 1\right],$$

where ρ denotes the discount factor.

Each investor chooses consumption $C_{j,t}$ and portfolio allocations across the dividend claim, the bond claim, and the derivative, subject to the constraints specified below.

Passive investors. Passive investors allocate a fraction $1-\overline{\alpha}_{p,t}$ of their wealth to the risk-free asset and a fraction $\overline{\alpha}_{p,t}$ to a *market index* fund, which holds the dividend and bond claims in proportion to their market-capitalization weights. Passive investors do not trade the derivative.

The share of wealth invested in the market index fund, $\overline{\alpha}_{p,t}$, evolves according to an Ornstein-Uhlenbeck process:

$$d\overline{\alpha}_{p,t} \;=\; \theta_p\big(\overline{\alpha} - \overline{\alpha}_{p,t}\big)dt \;+\; \sigma_{\alpha_p}\,dZ_t, \tag{10}$$

with long-run mean $\overline{\alpha} > 0$, mean-reversion $\theta_p > 0$, and loading $\sigma_{\alpha_p} \in \mathbb{R}^{1\times d}$ on the d-dimensional Brownian motion Z_t. Innovations to $\overline{\alpha}_{p,t}$ represent *portfolio flow shocks*, i.e., fluctuations in the fraction of wealth passive investors allocate to risky claims. As in Section 2, a flow shock is an asymmetric demand shock affecting only passive investors and triggering portfolio reallocation.

Portfolio flows may comove with cash-flow shocks or arise independently. We focous on the case where $\sigma = (\sigma_1, 0, 0)$ for aggregate endowment risk, $\sigma_{\alpha_p} = (\sigma_{\alpha_p,1}, \sigma_{\alpha_p,2}, 0)$ for the passive-share process, and $\sigma_s = (\sigma_{s,1}, 0, \sigma_{s,3})$ for the dividend share process. The first shock represents aggregate endowment risk, the second represents portfolio-flow shocks, and the third represents dividend-share shocks. We allow portfolio-flow shocks and dividend-share shocks to load on the aggregate endowment risk factor. This implies that aggregate endowment risk can be correlated with both portfolio flows and dividend-share shocks.

The exposure of the passive share to aggregate shocks can reflect *imperfect rebalancing* or *performance-induced flows*. Without rebalancing, a positive return on the risky asset mechanically increases the passive share when $\overline{\alpha}_{p,t} < 1$. Similarly, investors may increase risky exposure after good performance. This specification aligns with empirical evidence on passive behavior: households exhibit substantial portfolio inertia (Ameriks and Zeldes, 2004; Brunnermeier and Nagel, 2008); flows chase performance (Chevalier and Ellison, 1997; Dannhauser and Pontiff, 2024); and regulatory changes can sharply increase equity allocations via TDF adoption (Parker, Schoar, Cole, and Simester, 2022), illustrating portfolio flow shocks not tied to cash flow fundamentals.

Active investors. Investors of type $j = 1, 2$ are *active investors*. Active investors optimally rebalance their portfolios, choosing allocations across the dividend claim, the bond claim, and the derivative, $\alpha_{j,t} \equiv [\alpha_{j,D,t}, \alpha_{j,B,t}, \alpha_{j,F,t}]^\top$. Type-2 investors have low risk aversion, $\gamma_2 < \gamma_1$, and therefore have an incentive to lever up their exposure to risky assets. They borrow short-term to increase their risk exposure, subject to a leverage constraint that limits total portfolio risk:

$$\|\sigma_{j,t}\| \leq \overline{\sigma},$$

where $\sigma_{j,t} \equiv \alpha_{j,t}^\top \Sigma_t = \alpha_{j,D,t}\, \sigma_{R_D,t} + \alpha_{j,B,t}\, \sigma_{R_B,t} + \alpha_{j,F,t}\, \sigma_{R_F,t}$ is the investor's total risk exposure and $\|\cdot\|$ denotes the Euclidean norm. This constraint resembles a Value-at-Risk (VaR) limit commonly applied to banks and leveraged institutions (see, e.g., Adrian and Shin 2014 and Brunnermeier and Pedersen 2009). Type-1 investors are more risk averse and do not face a leverage constraint.

Investors' problem. An investor of type $j = 0, 1, 2$ chooses consumption $C_{j,t}$ and portfolio allocations $\alpha_{j,t}$ to solve the problem:

$$V_{j,t} = \max_{C_{j,t}, \alpha_{j,t}} \mathbb{E}_t \left[\int_t^\infty f_j(C_{j,s}, V_{j,s})\, ds \right], \qquad s.t. \tag{11}$$

$$\frac{dW_{j,t}}{W_{j,t}} = \left[r_t + \pi_t^\top \alpha_{j,t} - \frac{C_{j,t}}{W_{j,t}} \right] dt + \sigma_{j,t}\, dZ_t,$$

with $W_{j,t} \geq 0$ and $\alpha_{j,t} \in \Omega_{j,t}$, where $\sigma_{j,t} = \alpha_{j,t}^\top \Sigma_t$ is the investor's risk exposure, $W_{j,t}$ denotes the investor's wealth, and $\Omega_{j,t}$ is the investment set for type-j investor.

Portfolio allocations are restricted to lie in an *investment set* $\Omega_{j,t}$, which depends on investor type and captures differences in financial constraints across investors.[8] For passive investors, $j = 0$, holdings of the market index fund induce indirect holdings of the dividend and bond claims, so their investment set is given by:

$$\Omega_{0,t} = \{\alpha_{0,t} : \ \alpha_{0,D,t} = \overline{\alpha}_{p,t} \frac{P_{D,t}}{P_{Y,t}}, \ \alpha_{0,B,t} = \overline{\alpha}_{p,t} \frac{P_{B,t}}{P_{Y,t}}, \ \alpha_{0,F,t} = 0\},$$

Type-1 investors are unconstrained, so their investment set is given by $\Omega_{1,t} = \mathbb{R}^3$. For type-2 investors, the investment set reflects their leverage constraint:

$$\Omega_{2,t} = \{\alpha_{2,t} : \|\Sigma_t^\top \alpha_{2,t}\| \leq \overline{\sigma}\}.$$

Market clearing and equilibrium. We define equilibrium as follows.

Definition 1. *A competitive equilibrium consists of stochastic processes, all adapted to the filtration generated by Z_t. These include the aggregate endowment Y, the dividend share s, the prices P_D and P_B, the risk-free rate r, the market price of risk η_t, the weight $\overline{\alpha}_p$, and, for each investor $j \in \{0, 1, 2\}$, wealth W_j, consumption C_j, and portfolio allocations α_j. These processes satisfy the following conditions:*

[8] The differences in investment sets here are reminiscent of the difference in the investment universe emphasized by Koijen and Yogo (2019).

(i) The aggregate endowment Y_t evolves according to (8), with $Y_0 > 0$. The dividend share s_t evolves according to (9).

(ii) Given $\{P_{D,t}, P_{B,t}, r_t, \eta_t\}$ and the passive share $\overline{\alpha}_{p,t}$, the choices $(C_{j,t}, \alpha_{j,t})$ solve investor j's problem in (11). The weight on the market index fund, $\overline{\alpha}_{p,t}$, evolves according to (10).

(iii) Markets clear for goods, the dividend and bond claims, riskless bonds, and the derivative:

$$\sum_{j=0}^{2} \omega_j C_{j,t} = Y_t, \qquad \sum_{j=0}^{2} \omega_j Q_{j,k,t} = 1, \qquad \sum_{j=0}^{2} \omega_j B_{j,t} = 0, \qquad \sum_{j=0}^{2} \omega_j F_{j,t} = 0,$$

for $k \in \{D, B\}$. Holdings are defined as follows. The number of units of the dividend and bond claims held by investor j are $Q_{j,D,t} = \alpha_{j,D,t} W_{j,t}/P_{D,t}$ and $Q_{j,B,t} = \alpha_{j,B,t} W_{j,t}/P_{B,t}$. Riskless bond holdings are given by $B_{j,t} = (1 - \alpha_{j,D,t} - \alpha_{j,B,t}) W_{j,t}$. The notional position in the derivative is $F_{j,t} = \alpha_{j,F,t} W_{j,t}$. [9]

3.2 Equilibrium characterization

We now characterize equilibrium in the dynamic model. We derive the optimality conditions that govern portfolio choice and consumption decisions, and show how they jointly determine prices and risk premia. We then establish a version of the *mutual fund theorem* in our economy with frictions. This result implies that investors hold the dividend and bond claims in proportion to their market-capitalization weights, so that aggregate equilibrium objects are independent of the dividend-share shock. As a result, the equilibrium admits a *block-recursive* structure, in which we first solve for the aggregate equilibrium and then recover the prices of individual claims. This representation will be central for our analytical characterization of aggregate market elasticity.

[9] As there is no initial cash exchange for the derivative, wealth invested in the risk-free asset equals total wealth minus the amount invested in the dividend and bond claims.

3.2.1 Equilibrium conditions

Value function and consumption-wealth ratio. With homothetic preferences, investor j's value function can be written as

$$V_{j,t} = \left(\frac{c_{j,t}}{\rho^{\psi}}\right)^{\frac{1-\gamma_j}{1-\psi}} \frac{W_{j,t}^{1-\gamma_j}}{1-\gamma_j}, \qquad \text{where} \qquad \frac{dc_{j,t}}{c_{j,t}} = \mu_{c_j,t}\, dt + \sigma_{c_j,t}\, dZ_t.$$

The function $c_{j,t} = c_j(X_t)$ denotes the consumption-wealth ratio, so that $C_{j,t}/W_{j,t} = c_{j,t}$.

From the Hamilton-Jacobi-Bellman equation, the consumption-wealth ratio satisfies

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \pi_t^{\top}\alpha_{j,t} - \frac{\gamma_j}{2}\|\sigma_{j,t}\|^2\right] + \xi_{j,t}, \tag{12}$$

where $\sigma_{j,t} = \alpha_{j,t}^{\top}\Sigma_t$ is the investor's risk exposure. The first term reflects time preference, while the second term captures the impact of current investment opportunities on consumption.

The term $\xi_{j,t}$ is given by

$$\xi_{j,t} = \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{j,t}^{\top} + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j,t}\|^2}{2}.$$

This term captures the effect of changes in future investment opportunities on consumption.

As in Section 2, the consumption-wealth ratio depends on both the interest rate r_t and the risk premium π_t. Importantly, it also depends on the investor's portfolio choice $\alpha_{j,t}$. Finally, $\xi_{j,t}$ acts as a state-dependent shifter, and it is equal to zero when investment opportunities are constant.

Optimal risk exposure share. The optimal risk exposure for active investors $j = 1, 2$ is given by

$$\sigma_{j,t} = \frac{1}{1+\nu_{j,t}}\left[\frac{\eta_t}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi-1}\sigma_{c_j,t}\right],$$

where $\nu_{j,t}$ is the Lagrange multiplier on the leverage constraint, so $\nu_{j,t} \geq 0$ and $\nu_{1,t} = 0$. When the leverage constraint is not binding, the optimal risk exposure is the sum of two components. The first

is the myopic demand, $\frac{\eta_t}{\gamma_j}$, which depends on the investor's risk tolerance and the vector of market prices of risk η_t. This term captures the compensation per unit of exposure to each risk factor: $\eta_t^\top = \Sigma_t^{-1}\pi_t$. The second is the hedging demand, $\frac{1-\gamma_j^{-1}}{\psi-1}\,\sigma_{c_j,t}$, which reflects the investor's desire to hedge fluctuations in future investment opportunities. When the constraint binds, the multiplier $\nu_{j,t} > 0$ scales down exposure uniformly across all risk factors to satisfy $\|\sigma_{j,t}\| \leq \overline{\sigma}$. This acts like an endogenous increase in effective risk aversion.

The Lagrange multiplier $\nu_{2,t}$ satisfies the complementary slackness condition:

$$\nu_{2,t}\left(\|\sigma_{2,t}\| - \overline{\sigma}\right) = 0.$$

For passive investors, risk exposure is given by

$$\sigma_{0,t} = \overline{\alpha}_{p,t}\sigma_{R_Y,t},$$

where $\sigma_{R_Y,t}$ is the risk exposure of the market index fund. Hence, $\overline{\alpha}_{p,t}$ contols the overall risk exposure of passive investors.

Given an investor's risk exposure, the corresponding portfolio shares are recovered from

$$\alpha_{j,t} = \left(\sigma_{j,t}\,\Sigma_t^{-1}\right)^\top,$$

which maps risk exposures into asset holdings.

Market clearing conditions. Market clearing implies that aggregate wealth equals the price of the aggregate claim:

$$\sum_{j=0}^{2}\omega_j W_{j,t} = P_{Y,t}.$$

Let $x_{j,t}$ denote the wealth share of type-j investors:

$$x_{j,t} \equiv \frac{\omega_j W_{j,t}}{P_{Y,t}}.$$

The market clearing conditions for goods and risky assets are given by:

$$\sum_{j=0}^{2} x_{j,t} c_{j,t} = \frac{1}{p_{Y,t}}, \qquad \sum_{j=0}^{2} x_{j,t} \sigma_{j,t} = \sigma_{R_Y,t}.$$

These conditions echo those in the simple model. The first condition states that the wealth-weighted average consumption-wealth ratio equals the dividend yield of the aggregate claim. The second condition states that the wealth-weighted average risk exposure equals the risk exposure of the economy's endowment claim.

Market price of risk. We can write the market clearing condition for risky assets as follows:

$$\underbrace{\sum_{j=1}^{2} \frac{x_{j,t}}{x_{a,t}} \sigma_{j,t}}_{\substack{\text{active demand} \\ \text{for risk}}} = \underbrace{\chi_{a,t} \sigma_{R_Y,t}}_{\substack{\text{net supply} \\ \text{of risk}}},$$

where $\chi_{a,t} \equiv \frac{1-x_{0,t}\overline{\alpha}_{p,t}}{1-x_{0,t}}$ captures the share of aggregate risk that must be absorbed by active investors, and $x_{a,t} \equiv 1 - x_{0,t}$ is the wealth share of active investors.

This is the dynamic counterpart of the risk market clearing condition in Section 2. The expression above states that active investors absorb the net supply of risk, defined as total risk minus the portion held by passive investors.

Substituting optimal risk exposures into the market clearing condition yields the market price of risk:

$$\eta_t = \gamma_{a,t} \left[\chi_{a,t} \sigma_{R_Y,t} - h_{a,t} \right], \tag{13}$$

where $h_{a,t} = \sum_{j=1}^{2} \frac{x_{j,t}}{x_{a,t}} \frac{1-\gamma_j^{-1}}{\psi-1} \frac{\sigma_{c_j,t}}{1+\nu_{j,t}}$ is the average hedging demand of active investors, adjusted for leverage constraints, and $\gamma_{a,t}$ is the average effective risk aversion:

$$\gamma_{a,t} \equiv \left[\sum_{j=1}^{2} \frac{x_{j,t}}{x_{a,t}} \frac{1}{\gamma_j (1+\nu_{j,t})} \right]^{-1}.$$

Equation (13) shows that the market price of risk is determined by three forces.

First, the average effective risk aversion of active investors, $\gamma_{a,t}$. This varies over time with the wealth distribution and the extent to which leverage constraints bind.

Second, the net supply of risk to active investors, captured by $\chi_{a,t}\sigma_{R_Y,t}$. As passive investors increase their holdings of the market index, the amount of risk that must be absorbed by active investors falls, reducing the compensation for risk.

Third, the average hedging demand of active investors, $h_{a,t}$. Higher hedging demand implies active investors are willing to bear more risk, lowering the equilibrium compensation for risk.

Equation (13) makes clear that portfolio flows affect risk premia through their impact on the net supply of risk absorbed by active investors.

Interest rate. Combining goods market clearing with the expression for the consumption–wealth ratio yields the equilibrium interest rate:

$$r_t = \rho + \psi^{-1}\left(\mu_{P_Y,t} + \xi_t\right) + (1-\psi^{-1})\sum_{j=0}^{2} x_{j,t}\frac{\gamma_j}{2}\|\sigma_{j,t}\|^2 - \pi_{Y,t}, \tag{14}$$

where $\xi_t \equiv \sum_{j=0}^{2} x_{j,t}\,\xi_{j,t}$.

The first term reflects time preference. The second term captures intertemporal substitution. The third term and fourth terms capture the effect of uncertainty on the interest rate.

This expression highlights that the interest rate is determined jointly by risk premia and the allocation of risk across investors, providing the key general equilibrium link between portfolio flows, risk allocation, and saving behavior.

Pricing condition. Let $p_{D,t} \equiv P_{D,t}/D_t$ and $p_{B,t} \equiv P_{B,t}/B_t$ denote the price-dividend ratios of the dividend and corporate bond claims, respectively. The price-dividend ratio of the aggregate claim is $p_{Y,t} \equiv P_{Y,t}/Y_t$, which satisfies $p_{Y,t} = s_t p_{D,t} + (1-s_t)p_{B,t}$.

The pricing condition for each claim k is given by

$$\frac{1}{p_{k,t}} = r_t + \pi_{k,t} - \left(\mu_{k,t} + \mu_{p_k,t} + \sigma_{k,t}\,\sigma_{p_k,t}^{\top}\right), \tag{15}$$

where $\pi_{k,t} = \sigma_{R_k,t}\,\eta_t^{\top}$ is the risk premium on claim k implied by no arbitrage. The terms $\mu_{p_k,t}$ and $\sigma_{p_k,t}$ denote the drift and diffusion of the price-dividend ratio, respectively.

State variables. The aggregate state vector consists of both endogenous and exogenous variables. The endogenous state variables are the wealth shares of active investors, $x_{1,t}$ and $x_{2,t}$, which summarize the distribution of wealth. The exogenous state variables are the passive portfolio share $\overline{\alpha}_{p,t}$ and the dividend share s_t.

Hence, the aggregate state vector is $X_t = (x_t, \overline{\alpha}_{p,t}, s_t)$, where $x_t = (x_{1,t}, x_{2,t})$. The processes $\overline{\alpha}_{p,t}$ and s_t evolve according to (10) and (9), respectively. To obtain $\mu_{X,t}$ and $\sigma_{X,t}$, we need to solve for the drift and diffusion of the endogenous state variables, $x_{1,t}$ and $x_{2,t}$.

By Itô's lemma, the law of motion for $x_{j,t}$ is given by

$$\frac{dx_{j,t}}{x_{j,t}} = \left[(\eta_t - \sigma_{R_Y,t})\left(\sigma_{j,t} - \sigma_{R_Y,t}\right)^{\top} - \left(c_{j,t} - \frac{1}{p_{Y,t}}\right) + \kappa\frac{\omega_j - x_{j,t}}{x_{j,t}}\right]dt + (\sigma_{j,t} - \sigma_{R_Y,t})\,dZ_t.$$

Endogenous volatility. Return volatility has both a cash-flow and a valuation component, $\sigma_{R_k,t} = \sigma_{k,t} + \sigma_{p_k,t}$, where $\sigma_{k,t}$ reflects cash-flow volatility and $\sigma_{p_k,t}$ arises from movements in the price-dividend ratio $p_{k,t}$ for asset $k \in \{D, B\}$. By Itô's lemma,

$$\sigma_{p_k,t} = \frac{p_{k,x}(X_t)}{p_{k,t}(X_t)}\sigma_{x,t} + \frac{p_{k,\overline{\alpha}_p}(X_t)}{p_{k,t}(X_t)}\sigma_{\overline{\alpha}_p,t} + \frac{p_{k,s}(X_t)}{p_{k,t}(X_t)}\sigma_{s,t}. \tag{16}$$

Equation (16) shows that endogenous volatility depends on the sensitivity of the price-dividend ratio to three groups of variables: (i) the wealth distribution, (ii) the passive portfolio share, and (iii) the dividend share. The first term is standard in heterogeneous-agent models (see Panageas, 2020). The third term is typical of models with multiple assets. The second term arises only in the presence

of exogenous portfolio flow shocks. This second term is quantitatively important when markets are sufficiently inelastic, that is, when price-dividend ratios are highly sensitive to changes in passive demand. Thus, (16) provides a direct link between market elasticity and return volatility, as lower elasticity amplifies the sensitivity of prices to portfolio flows and therefore increases endogenous volatility.

Given the volatility of individual claims, the volatility of the aggregate claim follows from aggregation:

$$\sigma_{R_Y,t} = \sum_{k=D,B} \frac{P_{k,t}}{P_{Y,t}} \sigma_{R_k,t}.$$

It remains to determine $\sigma_{R_F,t}$. We make three assumptions about $\sigma_{R_F,t}$: (i) it does not load on the dividend-share shock, so $\sigma_{R_F,t,3} = 0$; (ii) it is orthogonal to the aggregate claim, so $\sigma_{R_F,t}\, \sigma_{R_Y,t}^{\top} = 0$; (iii) it has the same norm as the aggregate claim, so $\|\sigma_{R_F,t}\| = \|\sigma_{R_Y,t}\|$. These conditions uniquely determine $\sigma_{R_F,t}$ given $\sigma_{R_Y,t}$.

Markov equilibrium. Equations (13) and (14) allow us to express the market price of risk η_t and the interest rate r_t as functions of the consumption-wealth ratios $c_j(X_t)$ and the valuation ratios $p_k(X_t)$. Using Itô's lemma, the drift and diffusion terms $(\mu_{c_j,t}, \mu_{p_k,t})$ and $(\sigma_{c_j,t}, \sigma_{p_k,t})$ can be written as functions of the state dynamics $(\mu_{X,t}, \sigma_{X,t})$.

Substituting these expressions into the HJB equation (12) and the pricing condition (15), together with the laws of motion for the state variables, yields a system of five partial differential equations.

These equations jointly determine the consumption-wealth ratios $\{c_j(X_t)\}_{j=0}^{2}$ and the price-dividend ratios $\{p_k(X_t)\}_{k\in\{D,B\}}$ as functions of four state variables: the two active wealth shares $x_{1,t}$ and $x_{2,t}$, the passive portfolio share $\overline{\alpha}_{p,t}$, and the dividend share s_t.

3.2.2 Block recursivity and the mutual fund theorem

We show that the system of PDEs admits a block-recursive structure. First, we solve for the consumption-wealth ratios and the price-dividend ratio of the aggregate claim in a reduced set of states $\hat{X}_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t})$. Second, we solve for the price-dividend ratios of the dividend and

corporate bond claims in the full set of states $X_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t}, s_t)$.

This provides a substantial simplification, as the equilibrium allocation and pricing kernel are identical to those of an economy with a single claim on the aggregate endowment and a derivative. At the same time, the block-recursive structure allows us to recover the pricing implications for individual claims. In particular, we can characterize how portfolio flows affect the market elasticity of the dividend claim.

An important implication of the block-recursive structure is that a version of the *mutual fund theorem* holds in our economy. Despite financial frictions, investors optimally hold risky claims in proportion to their market-capitalization weights, as in the classic mutual fund theorem.

Proposition 2 (Block recursivity and the mutual fund theorem)**.** *Define the reduced set of states as* $\hat{X}_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t})$. *Then,*

(i) ***Block recursivity:*** *The consumption-wealth ratios,* $\{c_{j,t}(\hat{X}_t)\}_{j=0}^{2}$, *and the price-dividend ratio for the aggregate claim,* $p_Y(\hat{X}_t)$, *are a function of the reduced set of states* $\hat{X}_t$, *and they satisfy the following system of PDEs:*

$$c_j = \psi\rho + (1-\psi)\left[r + \eta^\top \sigma_j - \frac{\gamma_j}{2}\|\sigma_j\|^2\right] + \mu_{c_j} + (1-\gamma_j)\sigma_{c_j}\sigma_j^\top + \frac{\psi - \gamma_j}{1-\psi}\frac{\|\sigma_{c_j}\|^2}{2}$$
$$\frac{1}{p_Y} = r + \eta^\top(\sigma + \sigma_{p_Y}) - \left(\mu + \mu_{p_Y} + \sigma\,\sigma_{p_Y}^\top\right),$$

where $(r, \eta, \sigma_j, \mu_{c_j}, \sigma_{c_j}, \mu_{p_Y}, \sigma_{p_Y})$ *are all functions of* $\hat{X}_t$, (c_j, p_Y) *and their derivatives.*

(ii) ***Mutual fund theorem:*** *The portfolio shares of active investors are given by:*

$$\alpha_{j,k} = \alpha_{j,Y}\frac{P_k}{P_Y}, \quad \forall k \in \{D, B\},$$

where $\alpha_{j,Y}$ *denotes the holdings of the market index fund for investor* j, *which satisfies the condition*

$$\sigma_j = \alpha_{j,Y}\sigma_{R_Y} + \alpha_{j,F}\sigma_{R_F}.$$

Proposition 2 shows that the equilibrium with separate dividend and bond claims is equivalent to that of an economy with a single claim on the aggregate endowment and a derivative.

Two assumptions are key to this result. First, the aggregate endowment process Y_t is independent of the dividend share s_t. This assumption follows Menzly et al. (2004), but differs from the two-trees model of Cochrane, Longstaff, and Santa-Clara (2008), where the dividend share affects aggregate volatility.

Second, markets are dynamically complete. In equilibrium, investors are not exposed to the dividend-share shock, which can be perfectly diversified by holding the dividend and bond claims in proportion to their market-capitalization weights. Investors can therefore obtain any desired exposure to the fundamental and portfolio-flow shocks by trading the market index fund and the derivative.

Without the derivative, investors would need to deviate from market-capitalization weights to adjust their exposure to these risks, at the cost of bearing dividend-share risk.

Pricing the individual claims. Having solved for the aggregate objects $\{c_{j,t}(\hat{X}_t)\}_{j=0}^{2}$ and $p_Y(\hat{X}_t)$, we next recover the valuation of individual claims. The price-dividend ratios of the dividend and bond claims are functions of the full state vector $X_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t}, s_t)$, as they load on the dividend-share shock.

Given the aggregate equilibrium objects $r(\hat{X}_t)$ and $\eta(\hat{X}_t)$, and using Itô's lemma to express the drift and diffusion of valuation ratios as functions of the state dynamics and their derivatives, the pricing condition in Equation (15) yields a linear PDE for $p_{D,t}(X_t)$ and $p_{B,t}(X_t)$.

Thus, individual claim prices are obtained in a second step, conditional on the aggregate equilibrium. This separation highlights that portfolio flows affect aggregate quantities through the reduced state vector, while their impact on individual asset prices operates through valuation in the full state space.

This completes the characterization of the equilibrium with separate dividend and bond claims. We now turn to an analytical characterization of the aggregate market elasticity in this economy.

4 The Determinants of the Aggregate Market Elasticity

In this section, we analyze how the *aggregate market elasticity* governs the price impact of portfolio flows. The elasticity is determined by the coupled PDE system (**??**)–(15), which generally has no closed-form solution. To study its determinants, we extend the perturbation method of Kogan and Uppal (2001) and derive asymptotic closed-form approximations.

State-global perturbations Our perturbation approach studies economies in a neighborhood of a benchmark that admits a closed-form solution. A convenient benchmark arises when preferences are homogeneous, $\gamma_j = \gamma$, and passive investors are fully invested in the risky asset, $\overline{\alpha}_{p,t} = 1$. In this case, the economy collapses to a Lucas economy, as shown in Lemma 1. Throughout, a superscript $\star$ denotes benchmark values.

Lemma 1 (Benchmark economy)**.** *Suppose* $\gamma_j = \gamma$, $\overline{\alpha}_{p,t} = 1$, *and* $\rho > \left(1-\psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right)$. *Then,*

$$\pi^\star = \gamma\|\sigma\|^2, \qquad r^\star = \rho + \psi^{-1}\mu - \left(1+\psi^{-1}\right)\frac{\gamma\|\sigma\|^2}{2}, \qquad p^\star = \left[\rho - \left(1-\psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right)\right]^{-1},$$

$c_j^\star = (p^\star)^{-1}$, *and* $\alpha_j^\star = 1$ *for* $j = 0, \ldots, J$.

We analyze small deviations from this benchmark. Formally, consider a family of economies indexed by $\epsilon > 0$, where ϵ scales departures from the benchmark along the following dimensions.

First, investors have heterogeneous risk aversion:

$$\gamma_j = \gamma\left(1 + \hat{\gamma}_j \epsilon\right), \qquad \sum_{j=0}^{J} \omega_j \hat{\gamma}_j = 0, \tag{17}$$

so that γ is the population-weighted average and $\hat{\gamma}_j$ measures deviations. When $\epsilon = 0$, preferences are homogeneous; when $\epsilon = 1$, we recover the heterogeneous-investor economy.

Second, passive investors need not be fully invested in the risky asset. Their portfolio share is

$$\overline{\alpha}_{p,t} = 1 + \hat{\alpha}_{p,t}\,\epsilon, \tag{18}$$

where $\hat{\alpha}_{p,t}$ measures deviations from full investment benchmark and evolves as an Ornstein-Uhlenbeck process, consistent with Equation (10).

Third, the leverage constraint coefficient is perturbed according to

$$\overline{\sigma} = \|\sigma\|\big(1 + \hat{\sigma}\,\epsilon\big),$$

so that the tightness of the constraint is of order $O(\epsilon)$, matching the order of leverage demand. Finally, we assume a small mortality rate, $\kappa = \hat{\kappa}\,\epsilon$.

Equilibrium objects now depend on both the state X_t and the parameter ϵ. We expand them to second order:

$$p(X, \epsilon) = p_0(X) + p_1(X)\epsilon + p_2(X)\epsilon^2 + O(\epsilon^3),$$
$$c_j(X, \epsilon) = c_{j,0}(X) + c_{j,1}(X)\epsilon + c_{j,2}(X)\epsilon^2 + O(\epsilon^3),$$

where $p_k(X)$ and $c_{j,k}(X)$, $k \in \{0, 1, 2\}$, are functions of the state X.

Unlike standard DSGE perturbations, which linearize in both X and ϵ, our method is *state-global*: we solve directly for functions of X rather than coefficients around a steady state.[10] We refer to this approach as a *state-global perturbation*. This approach builds on the perturbation method of Kogan and Uppal (2001) and, more closely, the extension by Silva (2020). This method allows us to characterize how aggregate market elasticity varies with the state of the economy.

We proceed in two steps. First, we compute the first-order corrections $p_1(X)$ and $c_{j,1}(X)$. Second, we solve for the second-order terms $p_2(X)$ and $c_{j,2}(X)$. Zeroth-order terms are given by Lemma 1, i.e., $p_0(X) = p^\star$ and $c_{j,0}(X) = c_j^\star$.

[10] See Kargar, Passadore, and Silva (2023) for a comparison with standard perturbation methods.

4.1 The first-order demand system

Demand for risky asset. Investor j's demand for the risky asset is

$$Q_j(X;\epsilon) = \alpha_j(X;\epsilon)\,\frac{x_j}{\omega_j},$$

where x_j is the wealth share. Define the proportional deviation from the benchmark as

$$q_{j,t} \;\equiv\; \frac{Q_j(X_t;\epsilon) - Q_j(X_t;0)}{Q_j(X_t;0)} \;=\; \alpha_{j,1}(X_t)\,\epsilon + O(\epsilon^2),$$

where $\alpha_{j,1}(X)$ is the first-order correction to the portfolio weight.

Three results determine $\alpha_{j,1}(X)$. First, the volatility of the price-dividend ratio is second order:

$$\sigma_{p,t} = \frac{p_X(X_t)}{p(X_t)}\,\sigma_X(X_t) = O(\epsilon^2),$$

since $p_{X,0} = \sigma_{X,0} = 0$ (Lemma 1). Second, hedging demand is also $O(\epsilon^2)$ because $\sigma_{c_j}(X_t)$ is second order. Third, expanding the pricing condition (15) gives

$$\hat{\pi}_t = -\,\hat{r}_t - \frac{1}{p^\star}\hat{p}_t + O(\epsilon^2),$$

where $\hat{\pi}_t \equiv \pi_t - \pi^\star$, $\hat{r}_t \equiv r_t - r^\star$, and $\hat{p}_t \equiv \frac{p_t - p^\star}{p^\star}$, using $\mu_{p,t} = O(\epsilon^2)$.

Substituting into (**??**), the first-order demand system is

$$q_{j,t} = -\,\zeta_{j,p}\,\hat{p}_t - \zeta_{j,r}\,\hat{r}_t + f_{j,t} + O(\epsilon^2), \tag{19}$$

where, for unconstrained investors,

$$\zeta_{j,p} = \frac{1}{p^\star}\frac{1}{\gamma\|\sigma\|^2}, \qquad \zeta_{j,r} = \frac{1}{\gamma\|\sigma\|^2}, \qquad f_{j,t} = -\,\hat{\gamma}_j\,\epsilon.$$

For passive and constrained investors, own- and cross-price elasticities vanish, $\zeta_{j,p} = \zeta_{j,r} = 0$, with

demand shifters $f_{j,t} = \hat{\sigma}\,\epsilon$ for constrained investors and $f_{0,t} = \hat{\alpha}_{p,t}\,\epsilon$ for passive investors.

Equation (19) shows that the first-order approximation to the dynamic heterogeneous agent model in Section 3 provides microfoundations for the demand system in Section 2. As in the simple model, unconstrained active investors exhibit nonzero own- and cross-price elasticities, here explicitly linked to structural parameters. Shocks to $\overline{\alpha}_{p,t}$ move the passive investors' demand shifter $f_{0,t}$—the dynamic analog of the comparative statics in Section 2. In addition, the model generates further demand shifts from occasionally binding leverage constraints faced by active investors.

Consumption-wealth ratio. A first-order expansion of (**??**) yields

$$c_{j,t} \;=\; \psi\rho \;+\; (1-\psi)\left[r_t + \pi_t - \frac{\gamma_j\|\sigma\|^2}{2}\right] \;+\; O(\epsilon^2).$$

Consistent with Assumption 1, the risk-free rate and the risk premium enter symmetrically at first order. Substituting the expansion $\hat{\pi}_t = -\,\hat{r}_t - p^{\star\,-1}\hat{p}_t + O(\epsilon^2)$ gives

$$\hat{c}_{j,t} \;\equiv\; c_{j,t} - c_j^{\star} \;=\; \chi_{j,0} \;+\; \chi_{j,p}\,\hat{p}_t \;+\; O(\epsilon^2),$$

with coefficients

$$\chi_{j,p} = (\psi-1)\,\frac{1}{p^{\star}}, \qquad \chi_{j,0} = (\psi-1)\,\frac{\gamma\|\sigma\|^2}{2}\,\hat{\gamma}_j\,\epsilon.$$

Hence, conditional on p_t, the consumption-wealth ratio is independent of r_t, as in Section 2. Its slope with respect to p_t depends on the EIS: it rises with p_t when $\psi > 1$, the standard assumption in macro-finance models. Finally, to first order, the consumption–wealth ratio is unaffected by $\overline{\alpha}_{p,t}$, a property that will be central for aggregate market elasticity.

Aggregate market elasticity. Given the demand system, equilibrium prices follow from market clearing in the risky-asset and goods markets:

$$\sum_{j=0}^{J} x_{j,t}\,q_{j,t} = 0, \qquad \sum_{j=0}^{J} x_{j,t}\,\hat{c}_{j,t} = -\frac{1}{p^{\star}}\,\hat{p}_t.$$

Aggregating across investors, and dropping the j subscript on coefficients that are common across types, yields the linear system

$$\begin{bmatrix} \zeta_p & \zeta_r \\ \chi_p + \frac{1}{p^\star} & 0 \end{bmatrix} \begin{bmatrix} \hat{p}_t \\ \hat{r}_t \end{bmatrix} = \begin{bmatrix} f_t \\ -\chi_{0,t} \end{bmatrix},$$

where $f_t \equiv \sum_{j=0}^{J} x_{j,t} f_{j,t}$ and $\chi_{0,t} \equiv \sum_{j=0}^{J} x_j \chi_{j,0}$.

Definition 2 (Aggregate market elasticity)**.** *The aggregate market elasticity $\epsilon_{M,t}$ is defined as the inverse of the proportional change in the price of the risky asset in response to a flow shock f_t:*

$$\epsilon_{M,t} \equiv \left[\frac{\partial \hat{p}_t}{\partial f_t} \right]^{-1}.$$

Proposition 3 (First-order impact of portfolio flows)**.** *Suppose $\rho > \left(1 - \psi^{-1}\right)\left(\mu - \frac{\gamma \|\sigma\|^2}{2}\right)$. Then:*

(i) The price impact is zero to first order: $\epsilon_{M,t}^{-1} = \frac{\partial \hat{p}_t}{\partial f_t} = 0$.

(ii) The interest rate and risk premium responses satisfy

$$\frac{\partial \hat{r}_t}{\partial f_t} = \frac{1}{\zeta_r}, \qquad \frac{\partial \hat{\pi}_t}{\partial f_t} = -\frac{1}{\zeta_r}.$$

Proposition 3 shows that the main lesson from the simple model in Section 2 extends to the dynamic economy: portfolio inflows into the risky asset reduce the risk premium, while the corresponding shift out of bonds raises the risk-free rate. To first order, these effects offset exactly, leaving the price-dividend ratio unchanged. Hence, the aggregate market is infinitely elastic, regardless of the individual investors' own-price elasticities $\zeta_{j,p}$.

Why is the market infinitely elastic? As shown in Figure 1, price impact is absent as long as the consumption-wealth ratio does not respond directly to the portfolio-flow shock. Otherwise, price changes would generate excess demand or supply in the goods market.

To see why flows leave consumption unchanged to first order, differentiate (**??**) with respect to $\alpha_{j,t}$:

$$\frac{\partial c_{j,t}}{\partial \alpha_{j,t}} = (1-\psi)\,\gamma_j\,\|\sigma_{R,t}\|^2 \left[\frac{\pi_t}{\gamma_j\|\sigma_{R,t}\|^2} + \frac{1-\gamma_j^{-1}}{\psi-1}\,\frac{\sigma_{c_j,t}\,\sigma_{R,t}^\top}{\|\sigma_{R,t}\|^2} - \alpha_{j,t}\right].$$

The term in brackets is equal to zero at the privately optimal portfolio, as implied by the investor's first-order condition. By the envelope theorem, small changes in $\alpha_{j,t}$ around the optimum have negligible welfare effects on welfare and hence do not affect $c_{j,t}$.

In the first-order approximation, this derivative is evaluated at the benchmark Lucas economy, where all investors hold optimal portfolios. The derivative is thus zero, and by Proposition 3, price impact is negligible for small (first-order) deviations from the frictionless allocation: the consumption-wealth ratio does not react to flow shocks. In the simple model of Section 2, this was imposed as an *assumption*; here it emerges as a *result* of expanding around the efficient allocation.

This reasoning also clarifies when aggregate elasticity becomes finite: when the initial allocation of risk is *inefficient*. To study such economies requires larger departures from the benchmark, captured by a second-order approximation of the demand system.

4.2 Inefficient passive demand

We next turn to a second-order approximation of the demand system to study how risk allocation affects aggregate market elasticity. To focus on the mechanism most clearly, consider the special case without preference heterogeneity or leverage constraints.

The role of risk misallocation. In this setting, demand for the risky asset is still given by (19) up to second order. Without heterogeneity, the relevant demand shifter is

$$f_t \equiv x_{0,t}\left(\overline{\alpha}_{p,t} - 1\right),$$

where $x_{0,t}$ denotes the wealth share of passive investors.

The key difference relative to the benchmark arises from the consumption-wealth ratio.

Aggregating (**??**) across investors, letting $c_t \equiv \sum_{j=0}^{J} x_{j,t} c_{j,t}$, and imposing market clearing $\sum_{j=0}^{J} x_{j,t}\alpha_{j,t} = 1$, yields

$$c_t \;=\; \psi\rho \;+\; (1-\psi)\left[r_t + \pi_t - \frac{\gamma\|\sigma\|^2}{2}\sum_{j=0}^{J} x_{j,t}\alpha_{j,t}^2\right] \;+\; O(\epsilon^3).$$

Under the efficient allocation $\alpha_{j,t} = 1$, so $\sum_j x_{j,t}\alpha_{j,t}^2 = 1$. Any deviation from this efficient allocation creates dispersion in portfolios, leading to $\sum_j x_{j,t}\alpha_{j,t}^2 > 1$ by Jensen's inequality, which lowers the risk-adjusted expected return $r_t + \pi_t - \frac{\gamma\|\sigma\|^2}{2}\sum_j x_{j,t}\alpha_{j,t}^2$. When $\psi > 1$, this reduction weakens incentives to save and raises the consumption-wealth ratio for a given $r_t + \pi_t$.

Movements in the passive portfolio therefore shift the aggregate consumption–wealth ratio. Its second-order expansion is

$$\hat{c}_t \;\equiv\; c_t - c^\star \;=\; \chi_{0,t} \;+\; \chi_p\,\hat{p}_t \;+\; O(\epsilon^3),$$

with coefficients

$$\chi_p = (\psi-1)\,\frac{1}{p^\star}, \qquad \chi_{0,t} = (\psi-1)\,\frac{\gamma\|\sigma\|^2}{2}\,\frac{f_t^2}{x_{0,t}(1-x_{0,t})}.$$

Because $\chi_{0,t}$ is quadratic in the flow f_t, it is $O(\epsilon^2)$. For example, when passive investors are initially underexposed to the risky asset ($\overline{\alpha}_{p,t} < 1$ and $\psi > 1$), an inflow into stocks reduces the consumption-wealth ratio:

$$\frac{\partial\chi_{0,t}}{\partial f_t} \;=\; -\,(\psi-1)\,\gamma\|\sigma\|^2\,\frac{1-\overline{\alpha}_{p,t}}{1-x_{0,t}}.$$

Note that $\chi_{0,t}$ is locally insensitive to portfolio flow changes when passive portfolio is efficient ($\overline{\alpha}_{p,t} = 1$), consistent with the first-order analysis.

Figure 2 illustrates the effects of portfolio flows when passive investors are initially underexposed to risk, i.e., $\overline{\alpha}_{p,t} < 1$. An inflow into the risky asset shifts both the net supply curve (left panel)

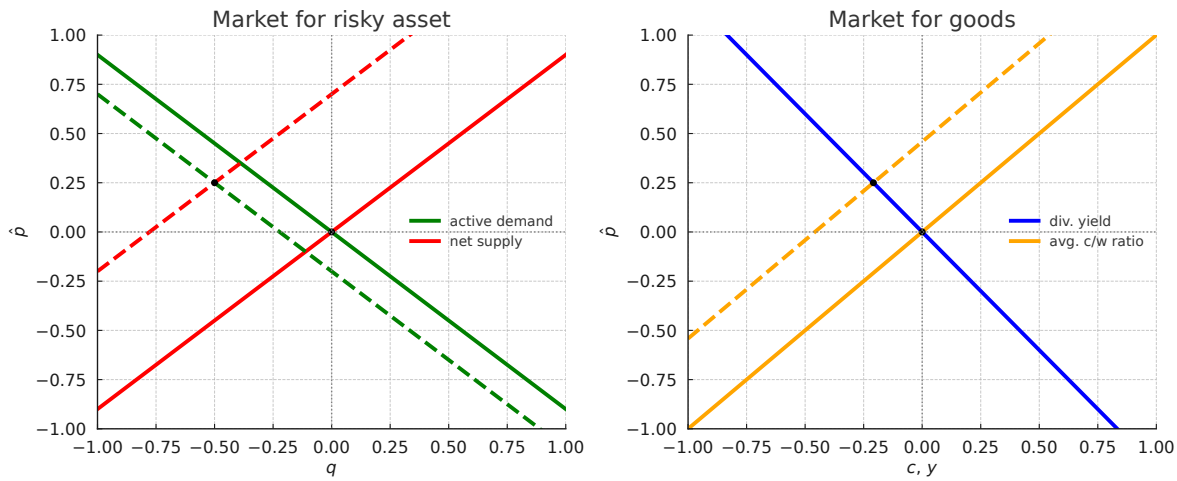


Figure 2. Equilibrium with inefficient risk allocation: risky asset (left) and goods (right) markets.

and the average consumption-wealth ratio schedule (right panel) leftward. The interest rate adjusts so that active demand meets net supply at the goods market clearing price. In this case, portfolio flows have a positive price impact and the aggregate elasticity is *finite*.

Proposition 4 (Aggregate elasticity: inefficient passive demand)**.** *Without preference heterogeneity or leverage constraints, the inverse aggregate elasticity is*

$$\varepsilon_{M,t}^{-1} = -(\chi_p + y^\star)^{-1} \frac{\partial \chi_{0,t}}{\partial f_t} = \left(1 - \psi^{-1}\right) \frac{\gamma \|\sigma\|^2}{y^\star} \frac{1 - \overline{\alpha}_{p,t}}{x_{a,t}} + \mathcal{O}\left(\epsilon^2\right),$$

where $x_{a,t} = 1 - x_{0,t}$ is the wealth share of active investors and $y^\star = 1/p^\star$ is the dividend-price ratio in the benchmark economy.

Proposition 4 shows that, unlike in Gabaix and Koijen (2023), the risky-asset elasticities ζ_p and ζ_r are not the direct drivers of aggregate elasticity. Because the demand system is recursive (interest rates only enters the risky asset demand) the overall price impact is governed instead by how portfolio flows impact the consumption-wealth ratio.

The macro elasticity is finite whenever $\overline{\alpha}_{p,t} \neq 1$ and $\psi \neq 1$. In the unit EIS case, the price-dividend ratio is pinned down by the subjective discount rate, so portfolio flows generate offsetting movements in the risk premium and the risk-free rate. With efficient risk allocation ($\overline{\alpha}_{p,t} = 1$), there is again no price impact.

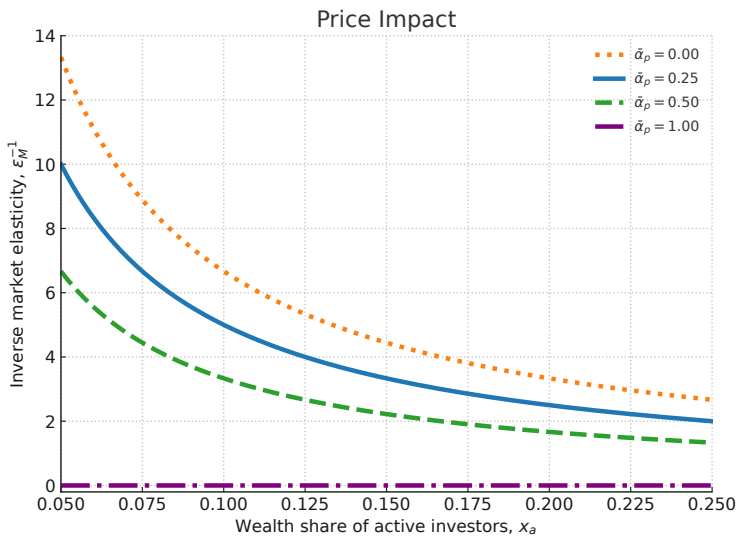


Figure 3. Price impact: inefficient passive demand.

The price impact is positive when $\psi > 1$ and $\overline{\alpha}_{p,t} < 1$: interest rate adjustments no longer fully offset the risk premium response, so the latter dominates. This mirrors representative agent models in which uncertainty shocks affect prices only when $\psi > 1$ (e.g., Bansal and Yaron, 2004).

The state-global perturbation analysis also makes clear that elasticity is state dependent. Figure 3 plots the price impact as a function of active investors' wealth for different passive-share levels. Price impact rises when active investors are under capitalized, reflecting their limited risk-bearing capacity, and when the passive share deviates further from its efficient level. In the extreme case where passive investors hold no equities ($\overline{\alpha}_{p,t} = 0$), we recover a version of Basak and Cuoco (1998) in which price impact is maximized.

4.3 Preference heterogeneity and leverage constraints

We now reintroduce preference heterogeneity and leverage constraints into the model. These features shift the conditions under which passive investors' portfolios are efficient and therefore alter the aggregate market elasticity. The next proposition provides the main result.

Proposition 5 (Aggregate elasticity: heterogeneity and leverage constraints)**.** *The inverse aggregate*

market elasticity is

$$\varepsilon_{M,t}^{-1} \;=\; \left(1-\psi^{-1}\right)\frac{\gamma\,\|\sigma\|^2}{y^\star}\,\frac{\overline{\alpha}_{p,t}^{\text{opt}}-\overline{\alpha}_{p,t}}{x_{a,t}-x_{c,t}}\left(1-x_{c,t}\right) \;+\; O\left(\epsilon^2\right), \tag{20}$$

where $x_{c,t}$ is the wealth share of constrained active investors and the passive investors' optimal portfolio share is

$$\overline{\alpha}_{p,t}^{\text{opt}} \;=\; 1 \;-\; \frac{x_{a,t}-x_{c,t}}{1-x_{c,t}}\,\frac{\gamma_0-\mathbb{E}_t^u[\gamma_j]}{\gamma} \;-\; \frac{x_{c,t}}{1-x_{c,t}}\left(\frac{\overline{\sigma}}{\|\sigma\|}-1\right), \tag{21}$$

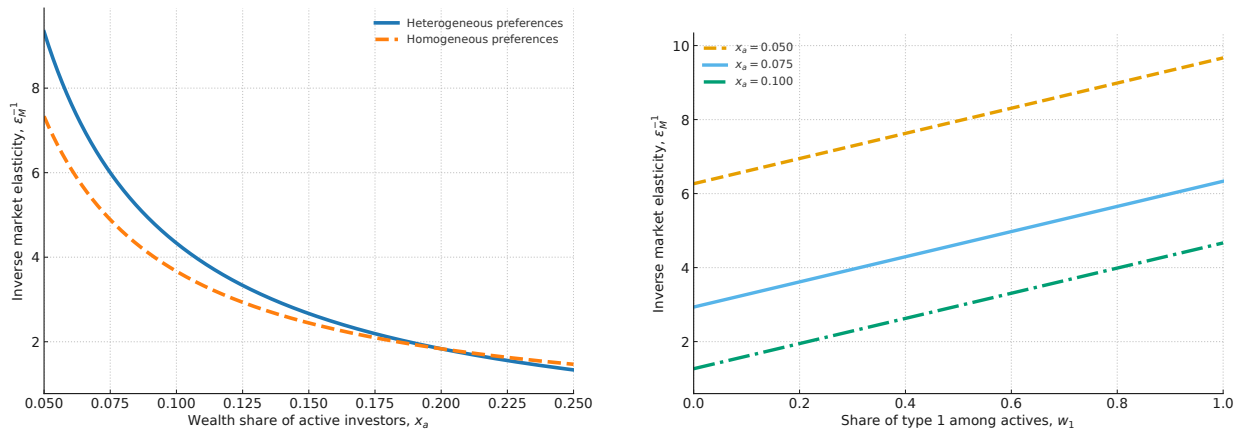
with $x_{u,t} \equiv x_{a,t} - x_{c,t}$ the wealth share of unconstrained active investors and

$$\mathbb{E}_t^u[\gamma_j] \;\equiv\; \sum_{j\in\mathcal{J}^u}\frac{x_{j,t}}{x_{u,t}}\,\gamma_j.$$

Proposition 5 characterizes aggregate market elasticity in the full model. As in the frictionless benchmark, the market remains infinitely elastic when passive investors hold their optimal portfolio share, $\overline{\alpha}_{p,t} = \overline{\alpha}_{p,t}^{\text{opt}}$. The optimal portfolio share corresponds to the value of $\overline{\alpha}_{p,t}$ such that the passive investor would have no incentive to change their portfolio. Under homogeneous preferences ($\gamma_j = \gamma$) and slack leverage constraints ($x_{c,t} = 0$), this condition reduces to $\overline{\alpha}_{p,t} = 1$. More generally, preference heterogeneity and binding leverage constraints shift the efficient passive share away from unity, with its level determined by the wealth distribution and the relative risk aversion of different investor types.

Preference heterogeneity. With slack leverage constraints ($x_{c,t} = 0$), (21) simplifies to

$$\overline{\alpha}_{p,t}^{\text{opt}} = 1 - x_{a,t}\,\frac{\gamma_0-\mathbb{E}_t^u[\gamma_j]}{\gamma},$$



(a) Homogeneous vs. heterogeneous preferences

(b) Role of wealth distribution of active investors

Figure 4. Price impact comparisons.
Panel (a) compares heterogeneous and homogeneous preferences. Panel (b) shows the sensitivity of the price impact to wealth distribution among active investors when $J = 2$ for different passive portfolio shares.

so, unlike the homogeneous preference case, the efficient passive share $\overline{\alpha}_{p,t}^{\text{opt}}$ differs from unity whenever $\gamma_0 \neq \mathbb{E}_t^u[\gamma_j]$. The inverse elasticity in (20) becomes

$$\varepsilon_{M,t}^{-1} = \left(1 - \psi^{-1}\right) \frac{\gamma \|\sigma\|^2}{y^\star} \frac{\overline{\alpha}_{p,t}^{\text{opt}} - \overline{\alpha}_{p,t}}{x_{a,t}} + O(\epsilon^2).$$

Optimal risk sharing implies that when passive investors are more risk averse than the average unconstrained active ($\gamma_0 > \mathbb{E}_t^u[\gamma_j]$),

$$\frac{\partial \overline{\alpha}_{p,t}^{\text{opt}}}{\partial x_{a,t}} = -\frac{\gamma_0 - \mathbb{E}_t^u[\gamma_j]}{\gamma} < 0,$$

so the efficient passive equity share rises as the active wealth share falls, typically after adverse aggregate shocks that erode active wealth that is more exposed to risk. If passive investors do not rebalance in such episodes, market inelasticity increases because $x_{a,t}$ declines (reducing risk-bearing capacity) while the gap $\overline{\alpha}_{p,t}^{\text{opt}} - \overline{\alpha}_{p,t}$ widens (moving the economy farther from the efficient allocation). Consequently, price impact rises as $x_{a,t}$ falls, as shown in the left panel (a) of Figure 4: the heterogeneous- and homogeneous-preference curves coincide at $x_a = 0.20$, but only the former steepens markedly as x_a declines.

Panel (b) of Figure 4 shows how the distribution of wealth across active investors shapes price impact. With two active types ($J = 2$) and $\gamma_1 > \gamma_2$, the average risk active aversion is

$$\mathbb{E}_t^u[\gamma_j] \;=\; w_{1,t}\gamma_1 + \big(1 - w_{1,t}\big)\gamma_2, \qquad w_{1,t} \equiv \frac{x_{1,t}}{x_{a,t}}.$$

When leverage constraints are slack ($x_{c,t} = 0$), the efficient passive share satisfies $\overline{\alpha}_{p,t}^{\text{opt}} = 1 - x_{a,t}\,(\gamma_0 - \mathbb{E}_t^u[\gamma_j])/\gamma$, so holding $x_{a,t}$ fixed,

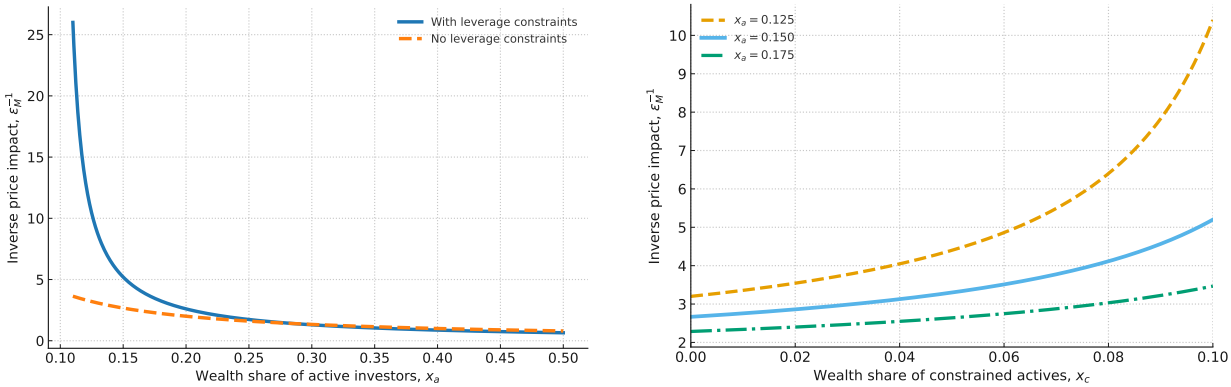
$$\frac{\partial\,\overline{\alpha}_{p,t}^{\text{opt}}}{\partial w_{1,t}} = \frac{x_{a,t}}{\gamma}\,(\gamma_1 - \gamma_2) \;>\; 0.$$

Thus, as the wealth share of the more risk averse active type rises, $\mathbb{E}_t^u[\gamma_j]$ increases and the passive sector should optimally hold more of the risky asset. If the passive portfolio is not adjusted, the deviation $\overline{\alpha}_{p,t}^{\text{opt}} - \overline{\alpha}_{p,t}$ widens, worsening risk allocation and raising price impact, consistent with the upward shift in the panel (b) of Figure 4.

Leverage constraints. When some active investors are constrained ($x_{c,t} > 0$), the marginal investors are the *unconstrained* active agents, and their risk-bearing capacity,

$$x_{u,t} \;\equiv\; x_{a,t} - x_{c,t},$$

becomes the key determinant of price impact. The panel (a) of Figure 5 plots price impact against the active wealth share $x_{a,t} = 1 - x_{0,t}$, with and without binding constraints. To isolate the role of constraints, we set $\gamma_0 = \mathbb{E}_t^u[\gamma_j]$, removing pure preference heterogeneity effects. As $x_{a,t}$ declines, e.g., after a negative aggregate shock when $\overline{\alpha}_{p,t} < 1$, price impact rises more sharply under binding constraints because $x_{u,t}$ shrinks and the risk-bearing capacity of marginal investors is depleted. Panel (b) holds $x_{a,t}$ fixed and varies the constrained share $x_{c,t}$. The price impact increases monotonically with $x_{c,t}$. This is intuitive: price impact is larger when a larger share of active investors is constrained.



(a) Role of leverage constraints *(assumes $x_{c,t} = 0.10$)*

(b) Varying the wealth share of constrained actives, $x_{c,t}$

Figure 5. Price impact under leverage constraints. Panel (a) compares the cases with and without leverage constraints (holding $x_{c,t} = 0.10$). Panel (b) plots the price impact as a function of $x_{c,t}$ for different values of $x_{a,t}$.

Taking stock. Proposition 5 clarifies the main determinants of aggregate market elasticity. The elasticity is *finite* only when risk is misallocated: small portfolio flows around the frictionless allocation have no first-order price impact. Preference heterogeneity amplifies price impact in downturns because, as the active wealth share falls, the efficient passive share rises, widening the gap between efficient and actual portfolios. Binding leverage constraints further increase price impact by shrinking the marginal (unconstrained) risk-bearing capacity, $x_{u,t}$. Taken together, these forces imply rich state dependence of aggregate elasticity in economies with heterogeneous investors and financial frictions.

5 Quantitative Implications

We now evaluate the quantitative implications of the dynamic model of Section 3. The exercise has two goals. The first is to ask whether a calibration disciplined by U.S. data on sectoral portfolio holdings and passive equity flows can simultaneously deliver an empirically plausible price impact and the standard asset-pricing moments. The second is to quantify how much of the variation in return volatility, return predictability, and the variance of the price–dividend ratio is attributable to how portfolio flows interact with preference heterogeneity, leverage constraints, and passive-sector risk misallocation. We do the latter by comparing the baseline economy to a set of counterfactuals

that turn off these frictions one at a time. We solve the full dynamic system numerically using the neural-network method of Duarte et al. (2024), which handles the four state variables (passive demand, dividend share, and the wealth shares of the constrained and unconstrained active sectors). Appendix J describes the numerical procedure in detail.

5.1 Calibration

We aggregate Flow of Funds (FoF) sectors into three ultimate sectors: a passive sector, a constrained active sector, and an unconstrained active sector.[11] The passive sector is modeled as an exogenous CIR process for the portfolio share $\overline{\alpha}_{p,t}$ in Equation (10). The two active sectors differ both in risk aversion and in whether they are subject to the leverage constraint. Table 1 reports the calibrated parameter values, and the next subsection describes the moments used to discipline them.

Figure 6 motivates two of the calibration targets. The left panel plots quarterly aggregate equity flows, scaled by lagged market capitalization as in Appendix D3 of Gabaix and Koijen (2023), together with realized return volatility. The right panel reports the "volatility multiplier," defined as the ratio of return volatility to flow volatility. The multiplier is a reduced-form analog of the price impact that our calibration aims to reproduce.

5.2 Moments

We discipline the calibration with two groups of moments. The first group targets the cross-sectional distribution of risky holdings. We choose the sector masses ω_j and the risk-aversion coefficients γ_j jointly so that the model matches the average sectoral wealth shares in the Flow of Funds and the sectoral "betas," defined as the slope coefficient from a time-series regression of sector j's dollar risky-asset holdings on aggregate risky supply. These sector betas are the empirical counterpart of the comovement between sector j's dollar position and the aggregate; we

[11] We follow the aggregation in Koijen et al. (2024). Appendix I describes the sector mapping and the treatment of pass-through vehicles in detail.

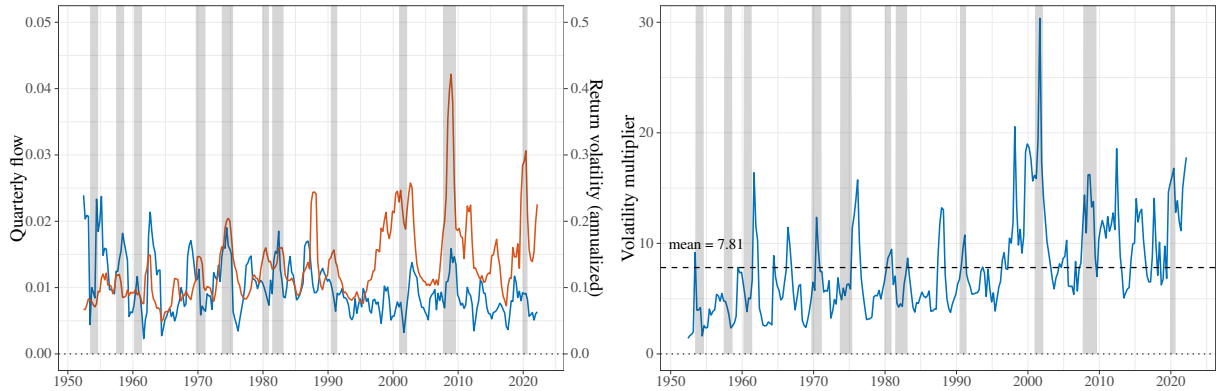


Figure 6. Quarterly flows and return volatility.
The left panel plots quarterly aggregate equity flows in blue and realized return volatility in red. Quarterly flows are scaled by lagged aggregate market capitalization, following Appendix D3 of Gabaix and Koijen (2023). The right panel plots the "volatility multiplier," defined as the ratio of return volatility to flow volatility; this multiplier is a reduced-form analog of the price impact targeted in the calibration. Source: Flow of Funds and CRSP.

compute them for the two active sectors, $j \in \{c, u\}$. The average risk aversion $\overline{\gamma}$ is fixed at 6.15, a standard value in the macro-finance literature. The leverage-constraint tightness $\overline{\sigma}$ is set so that the asset-to-equity ratio of the constrained sector matches its Flow of Funds counterpart.

The second group targets the passive-demand process. Its long-run mean $\overline{\alpha}$ is set to the household-weighted aggregate implied by our FoF aggregation, which yields $\overline{\alpha} = 0.52$. This value is intermediate between the one-third estimate in Chinco and Sammon (2024), inferred from index-reconstitution events, and the roughly two-thirds estimate in Koijen et al. (2024), which uses active-share measures in the spirit of Cremers and Petajisto (2009). The volatility σ_p is disciplined by the observed dispersion of quarterly aggregate equity flows shown in the left panel of Figure 6. Following Appendix D3 of Gabaix and Koijen (2023), we scale each sector's dollar equity flow by lagged aggregate market capitalization, purge mechanical revaluation, and aggregate gross flows across sectors, dividing by two to avoid double counting. The resulting series replicates the quarterly flows in their Figure D.7. The persistence θ_p is informed by evidence on household portfolio inertia. Using PSID data, Brunnermeier and Nagel (2008) regress the actual change in the household risky-asset share on a counterfactual change that assumes no rebalancing, and find an inertia coefficient near 0.75, implying that portfolio shares adjust slowly to capital gains and to

	Parameter	Choice
	Preferences & distribution	
ψ	EIS	1.5
ρ	Discount rate	0.01
γ_0	Risk aversion of passive investors	10.86
γ_1	Risk aversion of unconstrained investors	25.52
γ_2	Risk aversion of constrained investors	6.15
ω_0	Share of passive agents	0.50
ω_1	Share of unconstrained agents	0.25
ω_2	Share of constrained agents	0.25
	Technology	
μ	Endowment growth rate	0.018
σ	Endowment volatility	0.036
σ_s	Dividend share volatility	0.11
$\bar{s}$	Mean dividend share	0.50
κ_s	Mean reversion of dividend share	0.10
	Passive demand	
$\overline{\alpha}$	Mean	0.52
θ_p	Mean reversion parameter	0.75
σ_p	Passive demand shock exposures	0.02
ν	Corr. of passive demand and fundamental shock	0.50
	Leverage constraints	
$\overline{\sigma}$	Tightness of the leverage constraint	0.11

Table 1. Parameter values.
This table reports the parameter values used to calibrate the baseline economy. Preference parameters (ψ, γ_j, ρ) and sectoral masses (ω_j) are chosen to match average sectoral wealth shares and sectoral "betas" from time-series regressions of risky-asset holdings on aggregate risky supply. Technology parameters (μ, σ, σ_s, $\bar{s}$, κ_s) are standard macro–finance targets. Passive-demand parameters ($\overline{\alpha}$, θ_p, σ_p, ν) are disciplined by the average passive portfolio share, household portfolio inertia (Brunnermeier and Nagel, 2008), and the volatility and correlation of quarterly aggregate equity flows with market returns (Gabaix and Koijen, 2023). The leverage-constraint tightness $\overline{\sigma}$ targets the empirical asset-to-equity ratio of the constrained sector.

flow-induced changes in wealth.[12] We choose θ_p so that the half-life of the passive-demand process in the simulated model matches the half-life implied by this inertia coefficient. Finally, we estimate the loading ν of passive-flow shocks on the aggregate shock indirectly by regressing the quarterly passive-flow series on contemporaneous market returns. This yields a model-implied correlation between flows and returns of 0.005, close to the empirical value of 0.011.

[12] Parker, Schoar, and Sun (2023) show that the adoption of target-date funds has further stabilized household portfolio shares.

Moment	Model	Data
Panel A. Dividend-claim moments		
Risk premium (%)	6.60	6.00
Risk-free rate (%)	0.60	1.00
Return volatility (%)	22.40	18.00
Flow volatility (%)	1.00	1.70
Price impact	6.85	4.91
Standard error (price impact)	0.01	—
Panel B. Sector balance-sheet moments		
Assets/Equity (unconstrained)	0.52	0.53
Assets/Equity (constrained)	1.98	1.62
Asset share (passive, %)	45.40	38.80
Asset share (unconstrained, %)	22.30	20.10
Asset share (constrained, %)	32.20	29.20
Panel C. Endowment-claim moments		
Risk premium (%)	2.00	—
Return volatility (%)	4.50	—
Price impact	0.57	—
Standard error (price impact)	0.01	—

Table 2. Unconditional moments.
This table reports model-implied and empirical unconditional moments for the dividend claim and the endowment claim in the baseline economy. Model moments are computed from a 1,000-year monthly simulation of the calibrated economy. The price impact is estimated as the slope coefficient $\hat{b}_1$ of the regression $d\log(P/D)_t = a_0 + b_1\,\overline{\alpha}^s_{p,t} + e_t$, where $\overline{\alpha}^s_{p,t}$ is the innovation to passive demand. Asset/Equity denotes the ratio of risky-asset holdings to equity for the constrained and unconstrained sectors; asset shares report the fraction of the aggregate risky market held by each sector. Rates, volatilities, and asset shares are reported in percent; Asset/Equity ratios, price impacts, and their standard errors are unit-free. Standard errors for price-impact estimates are reported in separate rows. All values are annualized.

Following Lettau and Wachter (2011), we set the correlation of dividend-share shocks with the fundamental at 0.3 and the dividend-share volatility at 11%.

5.3 Unconditional moments

We simulate the calibrated economy at monthly frequency for 1,000 years and compare unconditional moments of the simulated data with their empirical counterparts in Table 2. For the dividend claim, the model delivers a risk premium of 6.6%, a risk-free rate of 0.6%, and a return volatility of 22.4%, all close to their empirical targets. We estimate the price impact of the dividend

claim by regressing the change in the log price–dividend ratio on the innovation to passive demand,

$$d\log(P/D)_t = a_0 + b_1\,\overline{\alpha}^s_{p,t} + e_t, \tag{22}$$

where $\overline{\alpha}^s_{p,t} = \overline{\alpha}_{p,t} - \overline{\alpha}_{p,t-1} - \mathbb{E}[d\overline{\alpha}_{p,t}]$ is the flow innovation. Equation (22) is the model counterpart of the empirical price-impact regression in Gabaix and Koijen (2023). In the data, an OLS projection of returns on observed flows is contaminated by a simultaneity bias: prices and aggregate flows are jointly determined, so flows respond to common macro shocks (e.g. to fundamentals or to the equity premium) that also move returns. To address this, Gabaix and Koijen (2023) build a granular instrumental variable (GIV) following Gabaix and Koijen (2024). They decompose sector-level flows into a common factor structure, $\xi_{i,t} = \lambda_i'\eta_t + u_{i,t}$, extract the idiosyncratic component $\hat{u}_{i,t}$ after partialing out observed and principal-component-estimated latent factors η_t, and form the GIV as the difference between the size-weighted and equal-weighted averages of these idiosyncratic shocks, $z_t = \sum_i (S_{i,t-1} - \frac{1}{N})\hat{u}_{i,t}$. They then run an instrumental-variables regression of returns on aggregate flows using z_t as the instrument, recovering a multiplier of about 5 for the U.S. aggregate equity market. In our setting, the passive-demand innovation $\overline{\alpha}^s_{p,t}$ is observed and exogenous by construction (the passive share follows the exogenous CIR process in equation (10)), so in simulated data the OLS slope of equation (22) directly identifies the model-implied counterpart of the GIV estimand without requiring an instrument. The slope $\hat{b}_1 = 6.85$ matches the empirical benchmark of about 6.90.[13] The model also matches sectoral asset-to-equity ratios and risky-asset shares. The constrained sector is levered (an asset/equity ratio of 1.98 in the model versus 1.62 in the data), the unconstrained sector holds a conservative share (0.52 versus 0.53), and the passive sector accounts for about 45% of risky holdings.

[13]Price impact in the model is $d\log(P/D)/d\overline{\alpha}_p$. Gabaix and Koijen (2023) estimate a multiplier of roughly 5, while Hartzmark and Solomon (2025) estimate 1.9. Averaging the two gives 3.45, and scaling by the passive share of roughly one half yields $3.45 \times 2 \approx 6.90$.

5.4 Counterfactuals: the role of frictions

To quantify the role of each friction, we compare the baseline economy to five counterfactuals that peel the frictions away progressively. The first two work through risk aversion and leverage. In *heterogeneous, unconstrained*, we keep the baseline heterogeneous risk aversions but set $\overline{\sigma}$ = 10,000, so leverage constraints are slack. *Homogeneous, unconstrained* additionally sets $\gamma_p = \gamma_c = \gamma_u = 6.15$, eliminating both preference heterogeneity and the leverage constraint. The remaining three experiments vary the passive-demand process. *Homog.-optimal passive share* sets $\overline{\alpha} = 1$, which is the passive share that is optimal under homogeneous preferences and no frictions; with heterogeneous risk aversion the true optimal passive portfolio need not equal unity, and we defer a general optimal-passive-portfolio benchmark to future work.[14] *Low passive share* sets $\overline{\alpha} = 0.23$, which halves the mass of the passive sector relative to the baseline. *Uncorrelated flow* sets $\nu = 0$, so passive-demand shocks are orthogonal to fundamentals. All other parameters are held at their baseline values from Table 1, and Table 3 reports annualized moments across the five economies.

The baseline economy produces a large dividend-return volatility ($\sigma_{R,D} = 22.55\%$) and a sizable price impact ($\hat{b}_1 = 6.85$), reflecting limited aggregate risk-bearing capacity. Turning off leverage constraints while keeping preference heterogeneity (the "heterogeneous, unconstrained" column) still generates a substantial price impact of 4.91 and a return volatility of 17.0%. Heterogeneity alone therefore delivers roughly three-quarters of the baseline price impact through state-dependent risk sharing. Once preference heterogeneity is also removed (the "homogeneous, unconstrained" column), return volatility collapses to 3.9% and the price impact falls to 0.10, values near zero and consistent with the frictionless limit.

The next two experiments vary the size of the passive sector and trace out the two sides of the misallocation channel. When $\overline{\alpha} = 1$, passive demand is at the level that would be optimal in the homogeneous-preferences economy, constraints and heterogeneity are effectively muted, and the price impact nearly vanishes (0.25). Reducing $\overline{\alpha}$ to 0.23 forces the active sector to absorb a much

[14] Computing the optimal passive portfolio in the heterogeneous-preferences economy would require solving a version of the model in which the passive investor chooses $\overline{\alpha}$ to maximize its own objective. We plan to report that benchmark in a later version.

	Baseline	Heterog., unconstr.	Homog., unconstr.	Passive-eff. $\overline{\alpha} = 1$	Low-passive $\overline{\alpha} = 0.23$	Uncorr. flow $\nu = 0$
$\sigma_{R,Y}$ (%)	4.54	4.7	3.7	3.76	—	—
π_Y (%)	2.03	2.2	1.2	1.26	—	—
r (%)	0.57	0.4	1.3	1.27	—	—
Γ	9.7	9.6	6.1	10.1	—	—
π_D (%)	6.61	5.9	1.2	1.52	—	—
$\sigma_{R,D}$ (%)	22.55	17.0	3.9	5.17	—	—
PI (Y)	0.571	0.681	0.093	0.102	—	—
PI (D)	6.854	4.910	0.101	0.248	—	—
Cond. PI (D)	8.503	5.096	0.103	0.100	—	—
Disc.-rate beta β_r	1.09	1.12	-0.0005	-0.11	—	—

Table 3. Counterfactual experiments.
This table reports annualized model-implied moments for the baseline economy and five counterfactuals that shut down or modify individual frictions. Counterfactuals are ordered so that frictions are peeled away progressively. In *Heterog., unconstr.* we keep the baseline heterogeneous risk aversions but set $\overline{\sigma} = 10{,}000$, so leverage constraints are slack. In *Homog., unconstr.* we additionally set $\gamma_c = \gamma_p = \gamma_u = 6.15$, eliminating both preference heterogeneity and leverage constraints. *Homog.-opt. passive* ($\overline{\alpha} = 1$) is the passive share that is optimal only under homogeneous preferences and no frictions; with heterogeneous preferences the true optimal passive share need not equal one, and we defer a general optimal-passive benchmark to future work. *Low passive share* sets $\overline{\alpha} = 0.23$, roughly halving the passive sector's size. *Uncorr. flow* sets $\nu = 0$, so passive-demand shocks are orthogonal to fundamentals. All other parameters are as in Table 1. Price impact (Y) and (D) denote the slope of the regression in equation (22) for the endowment and dividend claim, respectively. Conditional price impact restricts the regression to "bad states" in which $(P/D)_t$ is below its 25th percentile. The discount-rate coefficient β_r reports the share of the variance of the log price–dividend ratio attributable to discount-rate news at a 40-quarter horizon.

larger share of fundamental risk. The unconditional price impact of the dividend claim remains only modestly below the baseline (4.52), but the price impact of the endowment claim roughly doubles (1.20 versus 0.57). Decorrelating passive flows from fundamentals ($\nu = 0$) flips the sign of the dividend-claim price impact. Because flows are no longer informative about aggregate risk, pure flow-driven trading pushes prices in the opposite direction of fundamentals, as in the flow-driven risk literature (Deuskar and Johnson, 2011).

These results complement two closely related strands of the recent literature. Haddad, Huebner, and Loualiche (2025b) derive strong restrictions on how market elasticity should respond to changes in the mass of passive investors under competitive price-taking, and show that the empirical response is inconsistent with a fully competitive market. This points to limits on intermediation and on effective risk-bearing capacity. Our "low passive share" and "homog.-optimal passive" experiments

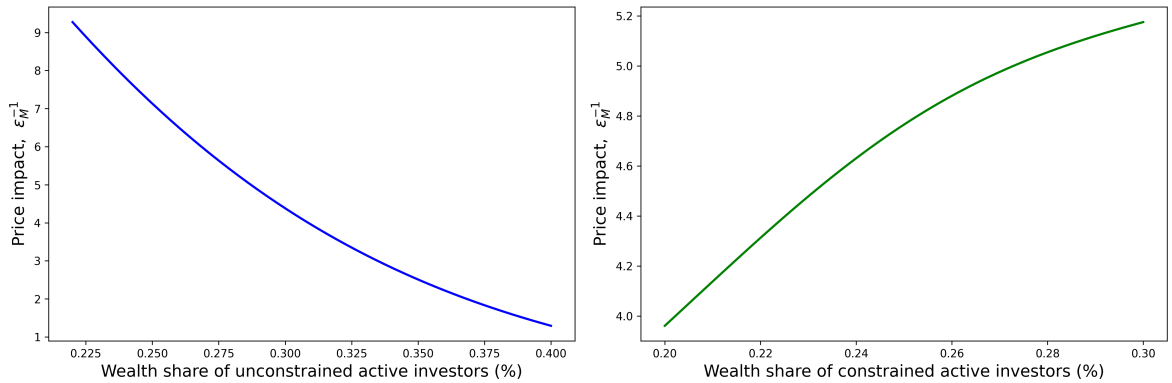


Figure 7. Conditional price impact.
Model-implied price impact of the dividend claim evaluated off the ergodic mean. In each panel, $\overline{\alpha}_{p,t}$ and s_t are fixed at their unconditional means and one wealth-share coordinate is varied. Conditional price impact is computed as the change in $\log(P/D)$ per unit of passive-flow shock, where the shock is propagated one quarter forward through the state-variable dynamics.

provide a structural counterpart: moving along the $\overline{\alpha}$ dimension traces out how aggregate price impact depends on the size of the passive sector, and the non-monotonic response of return volatility across columns indicates that a larger passive sector need not raise price impact once the active sector is near its efficient portfolio. Lochstoer and Muir (2026) provide causal evidence that the secular rise of market-index trading, through index funds, S&P 500 futures, and extended trading hours, has roughly doubled U.S. stock market volatility since the 1950s, even as macroeconomic fundamentals have become calmer. In their account, index-level trading forces arbitrageurs to absorb larger systematic demand shocks, which raises market variance. Our counterfactuals offer a complementary structural mechanism for the same force. A larger and more inelastic passive sector concentrates aggregate risk on the active sector, and the interaction of heterogeneous risk aversion with leverage constraints amplifies flow shocks into large, state-dependent price movements, even when expectations are rational. Heterogeneity and frictions thus appear to be the central ingredients for matching the price-impact and volatility facts.

5.5 Return predictability and the variance of the price–dividend ratio

We next examine the predictive relationship between the price–dividend ratio and future excess returns in the simulated economy. Figure 8 plots cumulative future excess returns against the

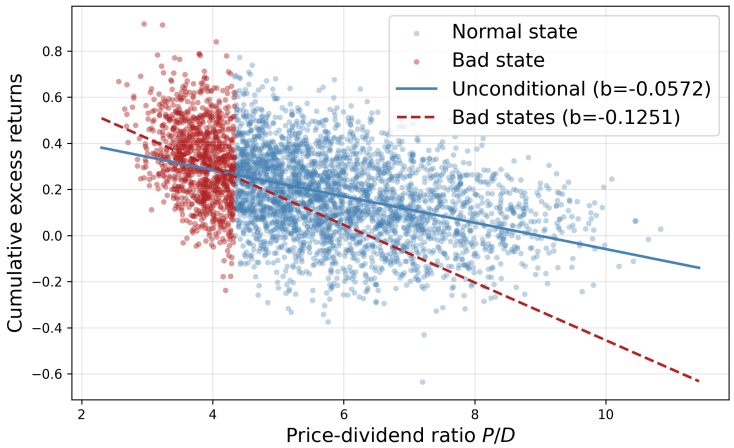


Figure 8. Return predictability.
Scatter of cumulative future excess returns ($T = 4$ quarters) against the current price–dividend ratio in the simulated baseline economy. Blue dots correspond to normal states and red dots to bad states defined as $(P/D)_t$ below the 25th percentile of its long-run mean. Lines are the corresponding OLS fits.

current P/D ratio, and Table 4 reports the corresponding predictive-regression coefficient

$$\sum_{k=1}^{T} r_{t+k} - r_t = a + b\,(P/D)_t + \varepsilon_{t+T}, \qquad T = 4 \text{ quarters},$$

estimated both on the full sample and on the subsample of "bad states" in which $(P/D)_t$ lies below its 25th percentile. The slope is negative in both cases, and it is roughly 2.5 times larger in magnitude in bad states. This pattern is consistent with countercyclical risk premia: when the P/D ratio is depressed, expected returns are high, and the P/D ratio is correspondingly more informative about future returns.

Following Campbell and Shiller (1988) and Cochrane (2011), we decompose the unconditional variance of $pd_t \equiv \log(P/D)_t$ into a discount-rate component and a cash-flow component,

$$\mathrm{Var}(pd_t) \;=\; \mathrm{Cov}\left(pd_t,\; \sum_{k=1}^{\infty} \rho^{k-1} \Delta d_{t+k}\right) \;-\; \mathrm{Cov}\left(pd_t,\; \sum_{k=1}^{\infty} \rho^{k-1} r_{t+k}\right). \tag{23}$$

Dividing through by $\mathrm{Var}(pd_t)$ yields the discount-rate share

$$\beta_r \;=\; -\,\frac{\mathrm{Cov}\left(pd_t, \sum_{k=1}^{\infty} \rho^{k-1} r_{t+k}\right)}{\mathrm{Var}(pd_t)},$$

which we evaluate at a 40-quarter horizon. In the baseline economy $\beta_r = 1.16$, so essentially all of the variation in the P/D ratio is driven by expected returns. The cash-flow component actually *offsets* rather than reinforces pd_t: because the dividend share is positively correlated with aggregate fundamentals, expected future dividend growth is higher precisely when pd_t is low, producing a negative cash-flow covariance that mechanically pushes β_r above one.

The counterfactuals show that this pattern comes from heterogeneity and frictions. In the homogeneous, unconstrained economy $\beta_r = -0.02$, so variation in pd_t reflects cash flows and not discount rates. Removing only leverage constraints leaves $\beta_r = 1.24$, close to the baseline, so heterogeneous risk aversion alone is enough to deliver discount-rate-dominated variation in pd_t. Setting $\overline{\alpha} = 1$ yields $\beta_r = -0.21$: at this level passive demand would be optimal under homogeneous preferences with no frictions, so there are no state-dependent movements in the price of risk, and pd_t again reflects cash flows. The large β_r in the baseline is thus a consequence of the frictions, not of the passive-demand process in isolation. These magnitudes are in line with the empirical findings in Cochrane (2011), whose Table II shows that β_r exceeds unity for aggregate U.S. equities and that cash-flow news contributes little or nothing to the variance of pd_t.

The accounting identity in (23) also connects the predictability of returns to the volatility of prices. The "excess volatility" puzzle documented by Shiller (1981) and LeRoy and Porter (1981), under which prices are too volatile to be justified by subsequent dividends, is mechanically the same fact as return predictability: if β_r is close to one, then almost all of the observed variance of pd_t must show up in predictable movements in future returns. Our counterfactuals make this duality sharp. Economies in which heterogeneity or leverage constraints bind simultaneously deliver large return volatility, a sizable price impact, and a predictive slope on $(P/D)_t$ close to the data, whereas the frictionless counterfactuals deliver low return volatility, a negligible price impact, and $\beta_r \approx 0$. Matching the empirical excess-volatility and predictability facts therefore requires heterogeneity and frictions in the active sector, not just time variation in fundamentals or beliefs.

Dependent variable:	Cumulative excess return, $\sum_{k=1}^{4} r_{t+k} - r_t$	
Sample:	Full	Bad state
$(P/D)_t$	−0.05	−0.12
	(0.0005)	(0.0048)
R^2	0.19	0.06
Observations	79,120	19,780

Table 4. Return predictability regression.
This table reports OLS estimates of the predictive regression $\sum_{k=1}^{T} r_{t+k} - r_t = a + b\,(P/D)_t + \varepsilon_{t+T}$, where $T = 4$ quarters, estimated on a 1,000-year monthly simulation of the calibrated baseline economy. Column (1) uses the full sample. Column (2) restricts to "bad states" in which $(P/D)_t$ lies below the 25th percentile of its long-run distribution. Standard errors are in parentheses.

5.6 Conditional price impact and dynamics

Averaging across states masks the state dependence of aggregate elasticity emphasized in Section 4. To trace this dependence, we compute the model-implied *conditional* price impact off the ergodic mean, plotted in Figure 7. We fix passive demand $\overline{\alpha}_{p,t}$ and the dividend share s_t at their unconditional means, vary one wealth-share coordinate at a time, and apply a fixed one-quarter passive-flow shock. The shock is propagated through the dynamic transition of the state variables. We then recompute $\log(P/D)$ at the shocked and unshocked states, and measure the conditional price impact as the resulting change in $\log(P/D)$ per unit of passive-flow shock.

The left panel of Figure 7 shows that the price impact rises sharply as the wealth share of the unconstrained active sector $x_{u,t}$ falls. When the sector most capable of absorbing risk is small, the market is less elastic and flow shocks translate into larger price movements. The right panel shows that the price impact also rises monotonically in the wealth share of the constrained active sector $x_{c,t}$. As more of the active sector hits its leverage limit, the residual risk-bearing capacity in $x_{u,t}$ shrinks and price impact grows. These patterns mirror the analytical characterization in Proposition 5 and explain why the conditional price impact reported in Table 3 (the row labeled "Cond. price impact (D)" with a value of 8.50) is substantially larger than its unconditional counterpart.

Combined with the counterfactual results, Figure 7 makes clear that the very states in which risk-bearing capacity is depleted, with low x_u or high x_c, are the states in which discount rates are most volatile and the P/D ratio is most informative about future returns. The large unconditional

β_r in Table 3 is therefore driven by these tail states.

5.7 Takeaways

The quantitative exercise delivers three main messages. A calibration disciplined by sectoral FoF data and passive equity flows matches the level of the aggregate price impact, the volatility of dividend-claim returns, and the level of the risk premium. Preference heterogeneity is quantitatively central: it alone accounts for roughly three-quarters of the baseline price impact and produces a discount-rate-dominated variance decomposition of the P/D ratio, even when leverage constraints are slack. Leverage constraints and a below-efficient passive share are complementary amplifiers that operate through state-dependent risk sharing. Finally, the excess-volatility puzzle and return predictability are two sides of the same coin in our economy. The same frictions that raise aggregate price impact are the ones that push β_r above one, consistent with Cochrane (2011), and counterfactuals that eliminate heterogeneity or shift $\overline{\alpha}$ to its optimum under homogeneous preferences collapse all three facts at once.

6 Conclusion

This paper develops a general equilibrium model with heterogeneous investors, passive demand, and financial constraints to study the determinants of aggregate market elasticity. We show how general equilibrium adjustments, particularly the joint movement of the risk-free rate and the risk premium, fundamentally shape the price impact of portfolio flows.

A central insight is the role of cross-price elasticity. When demand for risky assets responds to interest rates, shifts in the risk premium are offset by changes in the risk-free rate. In this case, the market can be infinitely elastic even if individual investors are highly inelastic. Aggregate elasticity becomes finite only when risk is initially misallocated: portfolio flows then alter aggregate savings behavior, preventing interest rates from fully offsetting risk-premium movements.

Macro elasticity is both state-dependent and time-varying. Passive investors amplify price

impacts, leverage constraints further tighten risk-bearing capacity, while preference heterogeneity increases elasticity by improving risk allocation. Inelasticity alone does not generate excess volatility, nor do flows in infinitely elastic markets; both elements are needed to explain observed fluctuations and the countercyclical volatility multiplier.

Our analytical results rely on a state-global perturbation method that provides closed-form characterizations of elasticity, while our quantitative analysis solves the full dynamic model numerically. The calibrated model highlights how the wealth distribution across households, intermediaries, and long-term investors shapes aggregate elasticity and volatility.

Overall, our findings suggest that market inelasticity is best understood as a symptom of inefficient risk allocation, rather than a structural feature of financial markets. Improving the allocation of risk across sectors may therefore be more effective for reducing excess volatility than targeting individual investor behavior. Future work could extend this framework to study asset pricing anomalies, monetary policy transmission, and the role of market structure in shaping aggregate elasticity.

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Appendix

A Derivations for Section 2

A.1 The bond demand view

Let $b_j \equiv \frac{B_j}{W_j}$ denote the bond-to-wealth ratio. The market clearing condition for bonds can be expressed as follows

$$\underbrace{x_a b_a}_{\textbf{active bond demand}} = \underbrace{-x_p b_p}_{\textbf{net bond supply}}$$

The linearized bond demand for a passive investor is given by

$$b_j - b_j^\star = \left[r - r^\star + \frac{c_j'(\mu - p^\star)}{1 - c_j(\mu - p^\star)}(p - p^\star) \right] b_j^\star - \alpha_j^\star e^{r^\star}(1 - c_j(\mu - p^\star))\hat{\alpha}_j$$

For simplicity, focus on the case $\overline{\alpha}_p = 1$, so $b_p^\star = b_a^\star = 0$. The passive bond demand is then given by

$$x_p b_p = f^b,$$

where $f^b \equiv -x_p[1 - c_p(\mu - p^\star)]\hat{\alpha}_p$. The active bond demand is given by

$$x_a b_a = -\zeta_p^b(p - p^\star) - \zeta_r^b(r - r^\star),$$

where $\zeta_p^b = \zeta_r^b = -x_a g_a'(\mu - p^\star - r^\star)[1 - c_a(\mu - p^\star)]$.

The demand system can be written as follows:

$$\begin{bmatrix} \zeta_p^q & \zeta_r^q \\ \zeta_p^b & \zeta_r^b \end{bmatrix} \begin{bmatrix} p - p^\star \\ r - r^\star \end{bmatrix} = \begin{bmatrix} f^q \\ f^b \end{bmatrix},$$

where we denote here $f^q \equiv f$ for symmetry.

Inverting the system above, we obtain

$$\begin{bmatrix} p - p^\star \\ r - r^\star \end{bmatrix} = \frac{1}{\zeta_p^q \zeta_r^b - \zeta_r^q \zeta_p^b} \begin{bmatrix} \zeta_r^b & -\zeta_r^q \\ -\zeta_p^b & \zeta_p^q \end{bmatrix} \begin{bmatrix} f^q \\ f^b \end{bmatrix}.$$

The price is given by

$$p - p^\star = \frac{\zeta_r^b f^q - \zeta_r^q f^b}{\zeta_p^q \zeta_r^b - \zeta_r^q \zeta_p^b} = x_p \hat{\alpha}_p \frac{\zeta_r^b + \zeta_r^q(1 - c_p(\mu - p^\star))}{\zeta_p^q \zeta_r^b - \zeta_r^q \zeta_p^b} = 0,$$

using the fact that $\zeta_r^q = x_a g_a'(\mu - p^\star - r^\star)$ and $c_a(\mu - p^\star) = c_p(\mu - p^\star)$.

B Derivations for Section 3

B.1 Investors' problem

The Hamilton-Jacobi-Bellman (HJB) equation for investor j can be written as

$$0 = \max_{C_j, \alpha_j} f_j(C_j, V_j) + V_{j,W}\left[rW_j + \pi^\top \alpha_j W_j - C_j\right] + V_{j,X}\mu_X$$
$$+ \frac{1}{2}V_{j,WW}W_j^2\|\sigma_j\|^2 + V_{j,WX}W_j\sigma_X\sigma_j^\top + \frac{1}{2}\sum_{k=1}^{3}\sigma_{X,k}^\top V_{j,XX}\sigma_{X,k},$$

subject to

$$\alpha_j = \overline{\alpha}_p[\alpha_{D,t}^m, \alpha_{B,t}^m, 0]^\top \qquad \text{for } j = 0, \qquad\qquad \sigma_j\sigma_j^\top \leq \overline{\sigma}^2 \qquad \text{for } j = 2.$$

given $\sigma_j = \alpha_j^\top \Sigma$, where $\alpha_{k,t}^m \equiv \frac{P_{k,t}}{P_{Y,t}}$ for $k \in \{D, B\}$. For ease of notation, we dropped time subscripts. Note that $V_{j,X}$ and $V_{j,WX}$ are $1 \times N$ vectors, $V_{j,XX}$ is a $N \times N$ matrix, and both $V_{j,W}$ and $V_{j,WW}$ are scalars. The drift μ_X is a $N \times 1$ vector, the diffusion σ_X is a $N \times 3$ matrix, while σ_R is a 1×3 vector. The notation $\sigma_{X,k}$ denotes the $k-$th column of σ_X, that is, the exposure to the $k-$th Brownian motion.

The optimality condition for consumption and portfolio share are given by

$$f_{j,C}(C_j, V_j) = V_{j,W}$$
$$V_{j,W}\pi = -V_{j,WW}W_j\Sigma\Sigma^\top\alpha_j - \Sigma(V_{j,WX}\sigma_X)^\top + \tilde{\nu}_{j,t}\Sigma\Sigma^\top\alpha_j.$$

where the second optimality condition holds for $j = 1, 2$, and $\tilde{\nu}_{j,t}$ is the Lagrange multiplier on the leverage constraint.

The optimal consumption is given by

$$C_j = \rho^\psi((1-\gamma_j)V_j)^{\frac{1-\gamma_j\psi}{1-\gamma_j}}V_{j,W}^{-\psi}.$$

The optimal portfolio share for an active investor is given by

$$\sigma_j = -\frac{V_{j,W}}{V_{j,WW}W_j + \tilde{\nu}_{j,t}}\eta_j - \frac{V_{j,WX}}{V_{j,WW}W_j + \tilde{\nu}_{j,t}}\sigma_X.$$

using the fact that $\pi = \Sigma\eta^\top$.

Given the homotheticity of preferences, the value function for investor j can be written as

$$V_{j,t} = \left(\frac{c_{j,t}}{\rho^{\psi}}\right)^{\frac{1-\gamma_j}{1-\psi}} \frac{W_{j,t}^{1-\gamma_j}}{1-\gamma_j}. \tag{B.1}$$

This particular parametrization of the value function implies that the consumption-wealth ratio is given by

$$\frac{C_{j,t}}{W_{j,t}} = c_{j,t}.$$

The optimal risk exposure for active investors is given by

$$\sigma_{j,t} = \frac{1}{1+\nu_{j,t}} \left[\frac{\eta_t}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi-1}\sigma_{c_j,t}\right],$$

where $\nu_{j,t} \equiv \tilde{\nu}_{j,t}/(V_{j,WW,t}W_{j,t})$ is the normalized Lagrange multiplier on the leverage constraint.

Plugging the consumption-wealth ratio into the HJB equation and using Equation (B.1), we obtain

$$\begin{aligned} 0 = {} & \frac{\rho}{1-\psi^{-1}}\left[\rho^{-1}c_j - 1\right] + r + \eta\sigma_j^{\top} - c_j + \frac{1}{1-\psi}\left[\frac{\xi_{j,X}}{\xi_j}\mu_X + \frac{1}{2}\sum_{k=1}^{d}\sigma_{X,k}^{\top}\frac{\xi_{j,XX}}{\xi_j}\sigma_{X,k}\right] \\ & - \frac{\gamma_j}{2}\|\sigma_j\|^2 + \frac{1-\gamma_j}{1-\psi}\frac{c_{j,X}}{c_j}\sigma_X\sigma_j^{\top} + \frac{1}{2}\frac{\psi-\gamma_j}{(1-\psi)^2}\sum_{k=1}^{d}\sigma_{X,k}^{\top}\frac{c_{j,X}^{\top}}{c_j}\frac{c_{j,X}}{c_j}\sigma_{X,k}. \end{aligned}$$

Rearranging the expression above, we obtain

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \eta_t\sigma_{j,t}^{\top} - \frac{\gamma_j}{2}\|\sigma_{j,t}\|^2\right] + \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{j,t}^{\top} + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j,t}\|^2}{2},$$

where the law of motion of $c_{j,t}$ is given by

$$\frac{dc_{j,t}}{c_{j,t}} = \mu_{c_j,t}dt + \sigma_{c_j,t}dZ_t,$$

and the drift and diffusion of $c_{j,t}$ are given by Ito's lemma:

$$\mu_{c_j,t} = \frac{c_{j,X}}{c_j}\mu_{X,t} + \frac{1}{2}\sum_{k=1}^{d}\sigma_{X,k,t}^{\top}\frac{c_{j,XX,t}}{c_{j,t}}\sigma_{X,k,t}, \qquad \sigma_{c_j,t} = \frac{c_{j,X}}{c_j}\sigma_{X,t}.$$

B.2 Pricing condition

Let $p_{D,t} = \frac{P_{D,t}}{D_t}$, $p_{B,t} = \frac{P_{B,t}}{B_t}$, and $p_{Y,t} = \frac{P_{Y,t}}{Y_t}$ denote the price-dividend ratios of the dividend claim, the bond claim, and the aggregate claim, respectively. The pricing condition for asset $k \in \{D, B\}$ is given by

$$r_t + \sigma_{R_k,t}\eta_t^\top = \frac{1}{p_{k,t}} + \mu_{k,t} + \mu_{p_k,t} + \sigma_{k,t}\sigma_{p_k,t}^\top,$$

where $\sigma_{R,t} = \sigma + \sigma_{p_k,t}$ and $(\mu_{p_k,t}, \sigma_{p_k,t})$ are given by Ito's lemma:

$$\mu_{p_k,t} = \frac{p_{k,X,t}}{p_{k,t}}\mu_{X,t} + \frac{1}{2}\sum_{k=1}^{d}\sigma_{X,k,t}^\top \frac{p_{k,XX,t}}{p_{k,t}}\sigma_{X,k,t}, \qquad \sigma_{p_k,t} = \frac{p_{k,X,t}}{p_{k,t}}\sigma_{X,t}.$$

The expected return and risk exposure on the aggregate claim are given by $\mu_{R_Y,t} = \alpha_{D,t}^m \mu_{R_D,t} + \alpha_{B,t}^m \mu_{R_B,t}$ and $\sigma_{R_Y,t} = \alpha_{D,t}^m \sigma_{R_D,t} + \alpha_{B,t}^m \sigma_{R_B,t}$, respectively.

B.3 Aggregate state variable

Define the share of wealth of type-j investors as follows

$$x_{j,t} \equiv \frac{\omega_j W_{j,t}}{\sum_{i=0}^{2}\omega_i W_{i,t}}.$$

We define the aggregate state variable as $X_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t}, s_t)$.

Market clearing conditions. We can write the market clearing condition for the equity claim, corporate bond claim, and short-term bonds as follows:

$$\sum_{j=0}^{2}\omega_j W_{j,t}\alpha_{j,k,t} = P_{k,t}, \qquad \sum_{j=0}^{2}\omega_j W_{j,t}(1 - \alpha_{j,D,t} - \alpha_{j,B,t}) = 0,$$

for $k \in \{D, B\}$.

Using the fact that $P_{Y,t} = P_{D,t} + P_{B,t}$, we obtain:

$$\sum_{j=0}^{2}\omega_j W_{j,t} = P_{Y,t}.$$

The market clearing condition for goods is given by

$$\sum_{j=0}^{2} x_{j,t} c_{j,t} = \frac{1}{p_{Y,t}}.$$

We can then write the market clearing condition for the equity claim, corporate bond claim, and the derivative as follows:

$$\sum_{j=0}^{2} x_{j,t} \alpha_{j,D,t} = \alpha_{D,t}^{m}, \qquad \sum_{j=0}^{2} x_{j,t} \alpha_{j,B,t} = \alpha_{B,t}^{m}, \qquad \sum_{j=0}^{2} x_{j,t} \alpha_{j,F,t} = 0.$$

Let $\alpha_t^m = [\alpha_{D,t}^m, \alpha_{B,t}^m, 0]^\top$ denote the market weights of the dividend, bond, and derivative claims. We can then write the market clearing conditions above in a compact way:

$$\sum_{j=0}^{2} x_{j,t} \alpha_{j,t} = \alpha_t^m.$$

Multiplying both sides by Σ and using the fact that $\sigma_{R_Y,t} = \omega_{D,t}\sigma_{R_D,t} + \omega_{B,t}\sigma_{R_B,t}$, we obtain

$$\sum_{j=0}^{2} x_{j,t} \sigma_{j,t} = \sigma_{R_Y,t}.$$

Law of motion of the aggregate state variable. The law of motion of $\overline{\alpha}_{p,t}$ is given by (10). The law of motion of s_t is given by (9). To compute the law of motion of $x_{j,t}$, first, note that the law of motion of wealth for a type-j investor can be written as

$$\frac{dW_{j,t}}{W_{j,t}} = \left[r_t + \eta_t \sigma_{j,t}^\top - c_{j,t} \right] dt + \sigma_{j,t} dZ_t.$$

From Ito's lemma, the law of motion of $x_{j,t}$ is given by

$$\frac{dx_{j,t}}{x_{j,t}} = \left[r_t + \pi_t^\top \alpha_{j,t} - c_{j,t} - \mu_{P_Y,t} - (\sigma_{j,t} - \sigma_{R_Y,t})\, \sigma_{R_Y,t}^\top + \kappa \frac{\omega_j - x_{j,t}}{x_{j,t}} \right] dt + (\sigma_{j,t} - \sigma_{R_Y,t})\, dZ_t.$$

using $\mu_{P_Y,t} = \mu_{R_Y,t} - \mu_{R_D,t}\alpha_{D,t}^m - \mu_{R_B,t}\alpha_{B,t}^m$.

B.4 The reduced system of PDEs and the mutual fund theorem

To compute the equilibrium, one needs to solve a system of 5 partial differential equations (PDEs), involving the three consumption-wealth ratios $\{c_{j,t}(X_t)\}_{j=0}^{2}$ and the two price-dividend

ratios $\{p_{k,t}(X_t)\}_{k=D,B}$. These functions depend on 4 state variables, the 2−dimensional vector x_t, the portfolio share of passive investors $\overline{\alpha}_{p,t}$, and the dividend share s_t.

The system of PDEs satisfies an important *block-recursive* structure. We can first solve for a reduced system of PDEs involving (c_j, p_Y) in the set of states $\hat{X}_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t})$. Given the solution of the reduced system of PDEs, we can then solve for $p_{D,t}$ and $p_{B,t}$ in the full set of states $X_t = (x_{1,t}, x_{2,t}, \overline{\alpha}_{p,t}, s_t)$.

B.4.1 Reduced system of PDEs

We next derive the system of PDEs. We will show that one can solve for a reduced system of PDEs involving (c_j, p_Y). This is the crucial in the obtaining the mutual fund theorem in our setting.

Lemma B.1. *The consumption-wealth ratios, $\{c_{j,t}(X_t)\}_{j=0}^2$, and the price-dividend ratio for the aggregate claim, $p_Y(X_t)$, satisfy the following system of PDEs:*

$$c_{j,t} = \psi\rho + (1-\psi)\left[r_t + \eta_t^\top \sigma_{j,t} - \frac{\gamma_j}{2}\|\sigma_{j,t}\|^2\right] + \mu_{c_j,t} + (1-\gamma_j)\sigma_{c_j,t}\sigma_{j,t}^\top + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j,t}\|^2}{2}$$

$$\frac{1}{p_{Y,t}} = r_t + \eta_t^\top (\sigma + \sigma_{p_Y,t}) - \left(\mu + \mu_{p_Y,t} + \sigma\,\sigma_{p_Y,t}^\top\right),$$

where $(r_t, \eta_t, \sigma_{j,t}, \mu_{c_j,t}, \sigma_{c_j,t}, \mu_{p_Y,t}, \sigma_{p_Y,t})$ are all functions of X_t, (c_j, p_Y) and their derivatives.

Proof. We proceed in four steps. First, we show how to solve for the diffusion of the endogenous state variables, $\sigma_{x_1,t}$ and $\sigma_{x_2,t}$, in terms of X_t, (c_j, p_Y) and their derivatives. Second, we show how to solve for the drift of the endogenous state variables, $\mu_{x_1,t}$ and $\mu_{x_2,t}$, again in terms of X_t, (c_j, p_Y) and their derivatives. Third, we show how to solve for the interest rate, r_t. Finally, we show how to obtain the PDEs for (c_j, p_Y).

In the derivations below, we drop the explicit dependence on X for brevity.

Step 1: Risk exposure of endogenous state variables. Consider the expression for the risk exposure of the aggregate claim and the consumption-wealth ratio:

$$\sigma_{R_Y} = \sigma + \sum_{k=1}^{2} \frac{p_{Y,x_k}}{p_Y}\sigma_{x_k} + \frac{p_{Y,\overline{\alpha}_p}}{p_Y}\sigma_{\overline{\alpha}_p} + \frac{p_{Y,s}}{p_Y}\sigma_s,$$

$$\sigma_{c_j} = \sum_{k=1}^{2} \frac{c_{j,x_k}}{c_j}\sigma_{x_k} + \frac{c_{j,\overline{\alpha}_p}}{c_j}\sigma_{\overline{\alpha}_p} + \frac{c_{j,s}}{c_j}\sigma_s,$$

Notice that we can write σ_{R_Y} as follows:

$$\sigma_{R_Y} = \chi_{\sigma_{R_Y},0} + \chi_{\sigma_{R_Y},1}\sigma_{x_1} + \chi_{\sigma_{R_Y},2}\sigma_{x_2},$$

where the coefficients $\chi_{\sigma_{R_Y},k}$, $k = 0, 1, 2$, are functions of X given by

$$\chi_{\sigma_{R_Y},0} = \sigma + \frac{p_{Y,\overline{\alpha}_p}}{p_Y}\sigma_{\overline{\alpha}_p} + \frac{p_{Y,s}}{p_Y}\sigma_s,$$
$$\chi_{\sigma_{R_Y},k} = \frac{p_{Y,x_k}}{p_Y}, k = 1, 2.$$

Similarly, we can write σ_{c_j} as follows:

$$\sigma_{c_j} = \chi_{\sigma_{c_j},0} + \chi_{\sigma_{c_j},1}\sigma_{x_1} + \chi_{\sigma_{c_j},2}\sigma_{x_2},$$

where the coefficients $\chi_{\sigma_{c_j},k}$, $k = 0, 1, 2$, are functions of X given by

$$\chi_{\sigma_{c_j},0} = \frac{c_{j,\overline{\alpha}_p}}{c_j}\sigma_{\overline{\alpha}_p} + \frac{c_{j,s}}{c_j}\sigma_s$$
$$\chi_{\sigma_{c_j},k} = \frac{c_{j,x_k}}{c_j}, k = 1, 2.$$

The market price of risk

$$\eta = \gamma_a \left[\chi_a \sigma_{R_Y} - \sum_{j=0}^{2} x_j \frac{1-\gamma_j^{-1}}{\psi - 1} \frac{\sigma_{c_j}}{1+\nu_j} \right],$$

where $\chi_a = \frac{1-(1-x_1-x_2)\overline{\alpha}_p}{x_1+x_2}$ is a known function of the aggregate state variables X.

We can then write η as follows:

$$\eta = \chi_{\eta,0} + \chi_{\eta,1}\sigma_{x_1} + \chi_{\eta,2}\sigma_{x_2},$$

where the coefficients $\chi_{\eta,k}$, $k = 0, 1, 2$, are functions of X given by

$$\chi_{\eta,k} = \gamma_a \left[\chi_a \chi_{\sigma_{R_Y},k} - \sum_{j=0}^{2} x_j \frac{1-\gamma_j^{-1}}{\psi - 1} \frac{\chi_{\sigma_{c_j},k}}{1+\nu_j} \right].$$

The risk exposure of investor $j = 0, 1, 2$ is given by

$$\sigma_0 = \overline{\alpha}_p \sigma_{R_Y}$$
$$\sigma_j = \frac{1}{1+\nu_j}\left[\frac{\eta}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi - 1}\sigma_{c_j} \right], \qquad j = 1, 2.$$

We can then solve for σ_j in terms of the risk exposure of the endogenous state variables:

$$\sigma_j = \chi_{\sigma_j,0} + \chi_{\sigma_j,1}\sigma_{x_1} + \chi_{\sigma_j,2}\sigma_{x_2},$$

where the coefficients $\chi_{\sigma_j,k}$, $k = 0, 1, 2$, are functions of X given by

$$\begin{aligned}
\chi_{\sigma_0,k} &= \overline{\alpha}_p \chi_{\sigma_{R_Y},k} \\
\chi_{\sigma_j,k} &= \frac{1}{1+\nu_j}\left[\frac{\chi_{\eta,k}}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi - 1}\chi_{\sigma_{c_j},k}\right], \qquad j = 1, 2.
\end{aligned}$$

The risk exposure of the endogenous state variables is given by

$$\sigma_{x_j} = x_j\left[\sigma_j - \sum_{k=0}^{2} x_k \sigma_k\right],$$

using the fact that $\sigma_{R_Y} = \sum_{k=0}^{2} x_k \sigma_k$.

This gives us a system of equations determining $\sigma_{x_j,t}$:

$$\begin{bmatrix} 1 - x_1 \sum_{k=0}^{2}\left(\chi_{\sigma_1,1} - \chi_{\sigma_k,1}\right) x_k & -x_1 \sum_{k=0}^{2}\left(\chi_{\sigma_1,2} - \chi_{\sigma_k,2}\right) x_k \\ -x_2 \sum_{k=0}^{2}\left(\chi_{\sigma_2,1} - \chi_{\sigma_k,1}\right) x_k & 1 - x_2 \sum_{k=0}^{2}\left(\chi_{\sigma_2,2} - \chi_{\sigma_k,2}\right) x_k \end{bmatrix} \begin{bmatrix} \sigma_{x_1} \\ \sigma_{x_2} \end{bmatrix} = \begin{bmatrix} x_1 \sum_{k=0}^{2}\left(\chi_{\sigma_1,0} - \chi_{\sigma_k,0}\right) x_k \\ x_2 \sum_{k=0}^{2}\left(\chi_{\sigma_2,0} - \chi_{\sigma_k,0}\right) x_k \end{bmatrix}.$$

Solving the system above, we obtain σ_{x_j} as a function of X and the functions c_j and p_Y and their derivatives. Given σ_{x_j}, we can solve for σ_j, η, σ_{c_j}, and σ_{R_Y} using the expressions above. The risk premium on the aggregate claim is given by $\pi_Y = \sigma_{R_Y}\eta^\top$.

Step 2: Drift of endogenous state variables. Given $(\sigma_j, \eta, \sigma_{R_Y})$, we can solve for the drift of the endogenous state variables:

$$\mu_{x_j} = x_j\left[(\eta - \sigma_{R_Y})\left(\sigma_j - \sigma_{R_Y}\right)^\top - \left(c_j - \frac{1}{p_Y}\right)\right] + \kappa(\omega_j - x_j),$$

This gives us μ_X and σ_X in terms of X, (c_j, p_Y) and their derivatives. We can then solve for the drift of c_j and p_Y using Ito's lemma:

$$\begin{aligned}
\mu_{c_j} &= \frac{c_{j,X}}{c_j}\mu_X + \frac{1}{2}\sum_{k=1}^{3} \sigma_{X,k}^\top \frac{c_{j,XX}}{c_j}\sigma_{X,k}, \\
\mu_{p_Y} &= \frac{p_{Y,X}}{p_Y}\mu_X + \frac{1}{2}\sum_{k=1}^{3} \sigma_{X,k}^\top \frac{p_{Y,XX}}{p_Y}\sigma_{X,k}.
\end{aligned}$$

Step 3: Interest rate. Given the expressions derived in the previous steps, we can solve for the shifter of the consumption-wealth ratio $\xi_{j,t}$:

$$\xi_j = \mu_{c_j} + (1-\gamma_j)\sigma_{c_j}\sigma_j^\top + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j}\|^2}{2}.$$

We can then solve for $\xi = \sum_{j=0}^{2} x_j \xi_j$ and the interest rate:

$$r = \rho + \psi^{-1}\left(\mu_{p_Y} + \mu + \sigma\sigma_{p_Y}^\top + \xi\right) + (1-\psi^{-1})\sum_{j=0}^{2} x_j \frac{\gamma_j}{2}\|\sigma_j\|^2 - \pi_Y.$$

Step 4: PDEs for the consumption-wealth ratio and price-dividend ratio. From the expression for the consumption-wealth ratio and the pricing condition for the aggregate claim, we can write the PDE for c_j and p_Y as follows:

$$\begin{aligned} c_j &= \psi\rho + (1-\psi)\left[r + \eta^\top \sigma_j - \frac{\gamma_j}{2}\|\sigma_j\|^2\right] + \mu_{c_j} + (1-\gamma_j)\sigma_{c_j}\sigma_j^\top + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j}\|^2}{2} \\ \frac{1}{p_Y} &= r + \eta^\top \sigma_{R_Y} - \left(\mu + \mu_{p_Y} + \sigma\,\sigma_{p_Y}^\top\right), \end{aligned}$$

Notice that $(r, \eta, \sigma_j, \mu_{c_j}, \sigma_{c_j}, \mu_{p_Y}, \sigma_{p_Y})$ are all functions of X, (c_j, p_Y) and their derivatives. Hence, these equations form a system of PDEs for (c_j, p_Y). □

A corollary of the above proposition is that the consumption-wealth ratio and price-dividend ratio are independent of the dividend share s.

Corollary 1. *The consumption-wealth ratio c_j and price-dividend ratio p_Y are a function of the reduced state vector $\hat{X} = (x_1, x_2, \overline{\alpha}_p)$.*

Proof. The dividend share s does not appear in the expressions for PDE system for (c_j, p_Y). They only appear indirectly through μ_s and σ_s. If (c_j, p_Y) are a function of $\hat{X}$, then their derivatives with respect to s are zero, and s completely drops out of the PDE system for (c_j, p_Y). □

B.4.2 The mutual fund theorem

Lemma **??** and its corollary show that the consumption-wealth ratio and price-dividend ratio of the aggregate claim are independent of the dividend share s. This implies that investor's consumption or wealth choices are not exposed to idiosyncratic dividend-share shocks.

C Derivations for Section 4

C.1 Proof of Lemma 1

Proof. The assumption $\epsilon = 0$ implies that there is no preference heterogeneity and passive investors are fully invested in the risky asset. We guess and verify that in this benchmark economy, there are no variation in expected returns. In particular, the wealth distribution plays no role in the economy. This implies that $\mu_{c_j,0}(X) = \sigma_{c_j,0}(X) = \mu_{p,0}(X) = \sigma_{p,0}(X) = 0$. In this case, the risk premium is given by

$$\pi_0(X) = \frac{\gamma}{x_{u,t}} \left[1 - x_{0,t} - x_{c,t}\overline{\alpha}_{c,t}\right] \|\sigma_{R_t}\|^2,$$

using the fact that $\varsigma_t = 0$, as $\sigma_{c_j,t} = 0$, and $\overline{\alpha}_{p,t} = 1$.

Given that $\sigma_{y,t} = 0$, we have that $\sigma_{R,0}(X) = \sigma$. Using the fact that $\overline{\sigma} = \|\sigma\|$ and $x_{u,t} = 1 - x_{0,t} - x_{c,t}$, we obtain the risk premium

$$\pi_0(X) = \gamma\|\sigma\|^2,$$

using $\alpha_{c,t} = 1$. Using $\sigma_{c_j,t} = 0$ and the expression for $\pi_0(X)$, we obtain that $\alpha_{j,0}(X) = 1$, for $j = 1, \ldots, J$, from Equation (**??**).

The interest rate is given by

$$r_0(X) = \rho + \psi^{-1}\mu - \gamma(1 + \psi^{-1})\frac{\|\sigma\|^2}{2}.$$

The consumption-wealth ratio $c_{j,0}(X)$ is given by

$$c_{j,0}(X) = \psi\rho + (1 - \psi)\left[r_0(X) + \pi_0(X)\alpha_{j,0}(X) - \frac{\gamma}{2}\alpha_{j,0}(X)^2||\sigma||^2\right].$$

Plugging in the expression for $r_0(X)$, $\theta_0(X)$ and $\sigma_{j,0}$, we obtain

$$c_{j,0}(X) = \rho - \left(1 - \psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right),$$

where we assume $\rho > \left(1 - \psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right)$.

From the market clearing condition for goods, we obtain:

$$\frac{1}{p_0(X)} = \rho - \left(1 - \psi^{-1}\right)\left(\mu - \frac{\gamma\|\sigma\|^2}{2}\right).$$

The drift and diffusion of the wealth shares are given

$$\mu_{X,j,0}(X) = x_j \left[r_0(X) + \pi_0(X)\alpha_{j,0}(X) - c_{j,0}(X) - \mu\right]$$
$$\sigma_{X,j,0}(X) = x_j(\alpha_{j,0}(X) - 1)\sigma_{R,0}(X),$$

where $\mu_{X,j,0} = \sigma_{X,j,0} = 0$, using the expression for returns, portfolio share, and consumption-wealth ratio. The result $\mu_{X,j,0} = 0$ uses the fact that $\kappa = 0$. □

C.2 Proof of Proposition 3

Proof. We consider next the first-order correction terms. Note that the diffusion terms for c_j and p are both equal to zero up to the first order, since $\sigma_{c_j,t} = O(\epsilon^2)$ and $\sigma_{p,t} = O(\epsilon^2)$. From Ito's lemma:

$$\sigma_{c_j,t} = \underbrace{\frac{c_{j,X}}{c_j}}_{O(\epsilon)} \underbrace{\sigma_{X,t}}_{O(\epsilon)} = O(\epsilon^2).$$

We have $c_{j,X} = O(\epsilon)$, because $c_{j,X,0} = 0$, as $c_{j,0}(X)$ does not depend on X. Also, $\sigma_{X,t} = O(\epsilon)$ because $\sigma_{X,0}(X) = 0$. This implies that $\sigma_{c_j,1}(X) = 0$. An analogous argument applies to p_t, so that we have $\sigma_{p,1}(X) = 0$ and $\|\sigma_{R,1}(X)\| = 0$.

Risk exposure of active investors. The risk exposure for active investors can be written as

$$\sigma_j(X,\epsilon) = \min\left\{\frac{\theta(X,\epsilon)}{\gamma_j} + \frac{1-\gamma_j^{-1}}{\psi - 1}\frac{\sigma_{c_j}(X,\epsilon)\sigma_R^\top(X,\epsilon)}{\|\sigma_R(X,\epsilon)\|}, \|\sigma\| + \hat{\sigma}\epsilon\right\},$$

where $\sigma_j(X;\epsilon) \equiv \alpha_j(X;\epsilon)\|\sigma_R(X;\epsilon)\|$ $\theta(X;\epsilon) \equiv \frac{\pi(X;\epsilon)}{\|\sigma_R(X;\epsilon)\|}$

Expanding the first term inside brackets in ϵ, we obtain

$$\sigma_j(X,\epsilon) = \min\left\{\frac{\theta_0(X)}{\gamma} + \left(\frac{\theta_1(X)}{\gamma} - \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j\right)\epsilon + \frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}(X)\sigma^\top}{\|\sigma\|}\epsilon^2 + O(\epsilon^3), \|\sigma\| + \hat{\sigma}\epsilon\right\},$$

Adding and subtracting $\|\sigma\| + \hat{\sigma}\epsilon$, and using $\frac{\theta_0(X)}{\gamma} = \|\sigma\|$, we obtain

$$\sigma_j(X,\epsilon) = \min\left\{\left(\frac{\theta_1(X)}{\gamma} - \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j - \hat{\sigma}\right)\epsilon + \frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}(X)\sigma^\top}{\|\sigma\|}\epsilon^2 + O(\epsilon^3), 0\right\} + \|\sigma\| + \hat{\sigma}\epsilon. \quad \text{(C.1)}$$

Consider first the case where the following condition is satisfied:

$$\frac{\theta_1(X)}{\gamma} - \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j - \hat{\sigma} = O(1), \quad \text{(C.2)}$$

If this is the case, then

$$\left(\frac{\theta_1(X)}{\gamma} - \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j - \hat{\sigma}\right)\epsilon \gg \left|\frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}(X)\sigma^\top}{\|\sigma\|}\right|\epsilon^2,$$

for small ϵ. So, the sign of the term inside the min operator in (C.1) is determined by the first term.

We can then write $\sigma_j(X,\epsilon)$ as follows:

$$\sigma_j(X,\epsilon) = \|\sigma\| + \min\left\{\frac{\theta_1(X)}{\gamma} - \frac{\theta_0(X)}{\gamma}\hat{\gamma}, \hat{\sigma}\right\}\epsilon + O(\epsilon^2).$$

In the region of the state space where condition (C.2) holds, one can determine whether an investor is constrained or unconstrained only based on the first-order terms. Suppose now that the following condition holds

$$\frac{\theta_1(X)}{\gamma} - \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j - \hat{\sigma} = O(\epsilon). \tag{C.3}$$

This condition states that, up to the first order, the leverage constraint is either always binding or slack by just a tiny amount parameterized by ϵ and ϵ^2 terms inside the min operator in (C.1). In this case, we can write $\sigma_j(X,\epsilon)$ as follows:

$$\sigma_j(X,\epsilon) = \|\sigma\| + \hat{\sigma}\epsilon + \min\left\{\left(\frac{\theta_1(X)}{\gamma} - \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j - \hat{\sigma}\right)\epsilon + \frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}(X)\sigma^\top}{\|\sigma\|}\epsilon^2, 0\right\} + O(\epsilon^3).$$

In this region of the state space where condition (C.3) is satisfied, we need the second-order term to determine whether an investor is constrained. This distinction will be relevant when computing the second-order correction.

For the first-order correction terms here, we focus on the case where condition (C.2) holds.

Aggregate risk aversion. The aggregate risk aversion of unconstrained investors, defined above, is given by

$$\gamma_u(X,\epsilon) = \frac{x_u}{\sum_{j\in\mathcal{J}^u}\frac{x_j}{\gamma(1+\hat{\gamma}\epsilon)}} = \gamma + \gamma\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j\,\epsilon + O(\epsilon^2)$$

Market price of risk. The market price of risk can be written as

$$\begin{aligned}\theta(X,\epsilon) &= \frac{\gamma_u(X,\epsilon)}{1-x_0-x_c}\left[(1-(1+\hat{\alpha}_p\epsilon)x_0)\|\sigma\| - (\|\sigma\|+\hat{\sigma}\epsilon)x_c\right] + O(\epsilon^2),\\ &= \theta_0(X) + \gamma\|\sigma\|\left[\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j - \left(\frac{\hat{\alpha}_p x_0 + \frac{\hat{\sigma}}{\|\sigma\|}x_c}{1-x_0-x_c}\right)\right]\epsilon + O(\epsilon^2).\end{aligned}$$

In the region of the state space where all active investors are unconstrained, we have

$$\theta(X,\epsilon) = \theta_0(X) + \gamma\|\sigma\|\left(\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j - \frac{\hat{\alpha}_p x_0}{1-x_0}\right)\epsilon + O(\epsilon^2).$$

The expression above shows the impact of fluctuations in the aggregate risk aversion and the effect of portfolio inflows in the market price of risk. If the average risk aversion in state X is lower than its level at $\epsilon = 0$, then the market price of risk will be lower than its level at $\epsilon = 0$, everything else constant.

Interest rate. The interest rate is given by

$$\begin{aligned}
r(X,\epsilon) &= r_0(X) + \left[-\theta_1(X)\|\sigma\| + \left(1-\psi^{-1}\right)\left(\sum_{j=0}^{J} x_{j,t}\frac{\gamma}{2}2\|\sigma\|\sigma_{j,1}(X) + \sum_{j=0}^{J} x_{j,t}\frac{\gamma}{2}\|\sigma\|^2\hat{\gamma}_j\right)\right]\epsilon + O(\epsilon^2) \\
&= r_0(X) + \gamma\|\sigma\|^2\left[-\frac{\theta_1(X)}{\gamma\|\sigma\|} + \left(1-\psi^{-1}\right)\sum_{j=0}^{J} x_{j,t}\frac{\hat{\gamma}_j}{2}\right]\epsilon + O(\epsilon^2) \\
&= r_0(X) - \gamma\|\sigma\|^2\left[\left(\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j - \sum_{j=0}^{J} x_{j,t}\frac{\hat{\gamma}_j}{2}\right) - \frac{\hat{\alpha}_p x_0 + \frac{\hat{\sigma}}{\|\sigma\|}x_c}{1-x_0-x_c} + \psi^{-1}\sum_{j=0}^{J} x_{j,t}\frac{\hat{\gamma}_j}{2}\right]\epsilon + O(\epsilon^2) \quad \text{(C.4)}
\end{aligned}$$

where we use the fact that $\sum_{j=0}^{J} x_{j,t}\sigma_{j,1}(X) = 0$ from the first-order correction of the risky asset market clearing condition.

The term in the square brackets in Equation (C.4) captures the first-order effect of frictions on the interest rate, $r_1(X)$. First, we see that the interest rate is decreasing in the difference between the average risk aversion of unconstrained investors and the average risk aversion of all investors in the economy. These two averages can differ for two reasons. First, passive investors may have a risk aversion different from the average unconstrained investor, the first term in parentheses in $r_1(X)$. Second, constrained investors are exactly the ones with low risk aversion, so unconstrained investors are on average more risk averse than all investors in the economy, which include the low risk aversion ones.

Consumption-wealth ratio. The consumption-wealth ratio for investor j is given by

$$\begin{aligned} c_j(X,\epsilon) &= c_{j,0}(X) + (1-\psi)\left[r_1(X) + \theta_0(X)\sigma_{j,1}(X) + \theta_1(X)\sigma_{j,0}(X)\right. \\ &\qquad \left. -\gamma\|\sigma\|\sigma_{j,1}(X) - \frac{\gamma}{2}\hat{\gamma}_j\|\sigma\|^2\right]\epsilon + O(\epsilon^2) \\ &= c_{j,0}(X) + (1-\psi)\gamma\|\sigma\|^2\left[\frac{r_1(X)}{\gamma\|\sigma\|^2} + \frac{\theta_1(X)}{\gamma\|\sigma\|} - \frac{\hat{\gamma}_j}{2}\right]\epsilon + O(\epsilon^2) \\ &= c_{j,0}(X) + (1-\psi)\gamma\|\sigma\|^2\left[\left(1-\psi^{-1}\right)\sum_{k=0}^{J} x_{k,t}\frac{\hat{\gamma}_k}{2} - \frac{\hat{\gamma}_j}{2}\right]\epsilon + O(\epsilon^2). \end{aligned}$$

Dividend yield. The dividend yield is given by

$$y(X,\epsilon) = y_0(X) + \left(1-\psi^{-1}\right)\gamma\|\sigma\|^2\sum_{j=0}^{J} x_{j,t}\frac{\hat{\gamma}_j}{2}\epsilon + O(\epsilon^2).$$

Notice that portfolio flows do not affect the dividend yield up to first order. The reason is that the interest rate and risk premium effects exactly cancel each other out. To derive the effect of portfolio flows on asset prices, we need to consider the second-order correction.

The price-dividend ratio is then given by

$$p(X,\epsilon) = p_0(X) - p_0(X)^2\left(1-\psi^{-1}\right)\gamma\|\sigma\|^2\sum_{j=0}^{J} x_{j,t}\frac{\hat{\gamma}_j}{2}\epsilon + O(\epsilon^2).$$

State dynamics. The diffusion of the wealth share of investor j, $j = 1,\ldots,J$, is given by

$$\begin{aligned} \sigma_{X,j}(X) &= x_j\sigma_{j,1}(X)\frac{\sigma}{\|\sigma\|}\epsilon + O(\epsilon^2) \\ &= x_j \min\left\{\sum_{k\in\mathcal{J}^u}\frac{x_k}{x_u}\hat{\gamma}_j - \left(\frac{\hat{\alpha}_p x_0 + \frac{\hat{\sigma}}{\|\sigma\|}x_c}{1-x_0-x_c}\right) - \hat{\gamma}_j, \frac{\hat{\sigma}}{\|\sigma\|}\right\}\sigma\epsilon + O(\epsilon^2). \end{aligned}$$

The drift of the wealth share of investor j, $j = 1,\ldots,J$, is given by

$$\begin{aligned} \mu_{X,j}(X,\epsilon) &= x_j\left[r_1(X) + \theta_1(X)\sigma_{j,0}(X) + \theta_0(X)\sigma_{j,1}(X) - \xi_{j,1}(X) - \sigma_{j,1}(X)\|\sigma\|\right]\epsilon + \kappa(\omega_j - x_j) + O(\epsilon^2) \\ &= x_j\left[(\psi-1)\gamma\|\sigma\|^2\left(\sum_{k=0}^{J} x_{k,t}\frac{\hat{\gamma}_k}{2} - \frac{\hat{\gamma}_j}{2}\right) + (\gamma-1)\|\sigma\|\sigma_{j,1}(X)\right]\epsilon + \kappa(\omega_j - x_j) + O(\epsilon^2) \end{aligned}$$

The law of motion of $\hat{\alpha}_p$ can be written as follows:

$$d\hat{\alpha}_p = \theta_p(\overline{\hat{\alpha}} - \hat{\alpha}_p)dt + \sigma_p dZ_t,$$

so the drift of $\overline{\alpha}_p$ is $\mu_{\overline{\alpha}_p} = \theta_p(\overline{\hat{\alpha}} - \hat{\alpha}_p)\epsilon$ and the diffusion is $\sigma_{\overline{\alpha}_p} = \sigma_p \epsilon$.

□

C.3 Second-order correction

In Proposition 6, we compute the second-order correction for our economy.

Proposition 6 (Second-order correction)**.** *Suppose* $\rho > \left(1 - \psi^{-1}\right)\left(\mu - \frac{\gamma\sigma^2}{2}\right)$*. Then,*

(i) The second-order correction for the consumption-wealth ratio and risk exposure are given by:

$$c_{j,2}(X) = (1-\psi)\frac{\gamma\sigma^2}{2}\left[\left(1-\psi^{-1}\right)\sum_{k=0}^{J} x_{k,t}\hat{\gamma}_k - \hat{\gamma}_j\right]$$
$$\sigma_{j,2}(X) = \frac{\theta_2(X)}{\gamma} - \frac{\theta_1(X)}{\gamma}\hat{\gamma}_j + \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j^2 - (1-\gamma^{-1})\sigma_{y,2}(X),$$

where

$$\sigma_{y,2}(X) = \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}\left(\hat{\gamma}_k - \hat{\gamma}_0\right)x_k\sigma_{k,1}(X)\frac{\sigma}{\|\sigma\|}.$$

(ii) The second-order correction for the Sharpe ratio, interest rate, and dividend yield are given by:

$$\begin{aligned}
\theta_2(X) &= -\frac{(\gamma-1)x_c + 1 - x_0}{1 - x_0 - x_c}\sigma_{y,2}(X) - \gamma\sigma\mathbb{E}^u[\hat{\gamma}_j]\frac{\hat{\alpha}_p x_0 + \frac{\hat{\sigma}}{\sigma}x_c}{1 - x_0 - x_c} - \gamma\sigma\mathrm{Var}^u[\hat{\gamma}_j] \\
r_2(X) &= -\theta_2(X)\sigma + \theta_0(X)\sigma_{y,2}(X) - \psi^{-1}(\mu_{y,2}(X) + \sigma\sigma_{y,2}(X)) \\
&\quad + \left(1-\psi^{-1}\right)\sum_{j=0}^{J} x_j\gamma\sigma\left[\sigma_{j,2}(X) + \frac{\sigma_{j,1}^2(X)}{2\sigma} + \hat{\gamma}_j\sigma_{j,1}\right] \\
&\quad + \psi^{-1}\sum_{j=0}^{J} x_j\left[\mu_{\xi_j,2}(X) + (1-\gamma)\sigma_{\xi_j,2}(X)\sigma\right] \\
y_2(X) &= \left(1-\psi^{-1}\right)\sum_{j=0}^{J} x_j\gamma\sigma\left(\frac{\sigma_{j,1}^2(X)}{2\sigma} + \hat{\gamma}_j\sigma_{j,1}(X)\right),
\end{aligned} \tag{C.5}$$

where

$$\mathbb{E}^u[\hat{\gamma}_j] \equiv \sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j, \quad \textit{and} \quad \mathrm{Var}^u[\hat{\gamma}_j] \equiv \sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j^2 - \left(\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j\right)^2.$$

Proof. **Step 1: Laws of motion for c_j and y.**
We start by considering the diffusion terms for c_j and y. The expansion of $\sigma_{c_j,t}$ in ϵ is given by

$$\begin{aligned}
\sigma_{c_j}(X,\epsilon) &= \frac{c_{j,X}(X,\epsilon)}{c_j(X,\epsilon)}\sigma_X(X,\epsilon) \\
&= \frac{c_{j,X,1}(X)}{c_{j,0}(X)}\sigma_{X,1}(X)\epsilon^2 + O(\epsilon^3) \\
&= -(\psi-1)\left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2c_{j,0}(X)}\sum_{k=1}^{J}(\hat{\gamma}_k-\hat{\gamma}_0)\,x_k\sigma_{k,1}(X)\epsilon^2 + O(\epsilon^3),
\end{aligned}$$

where we used the fact that $c_{j,\overline{\alpha}_p,1}(X) = 0$. Notice that $\sigma_{c_j,2}(X)$ does not depend on j, that is, it is the same for all investors. Moreover, $\sigma_{c_j,2}(X) > 0$, as $\sigma_{k,1}(X)$ is inversely related to $\hat{\gamma}_k$.

Similarly, the diffusion for dividend yield y can be written as

$$\begin{aligned}
\sigma_y(X,\epsilon) &= \frac{y_{X,1}(X)}{y_0(X)}\sigma_{X,1}(X)\epsilon^2 + O(\epsilon^2) \\
&= \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}(\hat{\gamma}_k-\hat{\gamma}_0)\,x_k\sigma_{k,1}(X)\epsilon^2 + O(\epsilon^3).
\end{aligned}$$

where we used the fact that $y_{\overline{\alpha}_p,1}(X) = 0$. The expression above is negative if $\psi > 1$, which implies that $\sigma_{R,2}(X) = -\sigma_{y,2}(X)$ is positive. In this case, a negative aggregate shock redistribute wealth to more risk averse investors, leading to a rise in the risk premium and a decline in the risk-free rate. If $\psi > 1$, the risk premium effect dominates, so the price-dividend ratio, $1/y$, falls in response to the shock. The movement in the price-dividend ratio amplifies the initial effect of the drop in dividends.

The drift of $c_{j,t}$ is given by

$$\begin{aligned}
\mu_{c_j}(X,\epsilon) &= \frac{c_{j,X}(X,\epsilon)}{c_j(X,\epsilon)}\mu_X(X,\epsilon) + \frac{1}{2}\sigma_X^\top(Z,\epsilon)\frac{c_{j,XX}(X,\epsilon)}{c_j(X,\epsilon)}\sigma_X(X,\epsilon) \\
&= \frac{c_{j,X,1}(X)}{c_{j,0}(X)}\mu_{X,1}(X)\epsilon^2 + O(\epsilon^3) \\
&= -(\psi-1)\left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2c_{j,0}(X)}\sum_{k=1}^{J}(\hat{\gamma}_k-\hat{\gamma}_0)\,\mu_{X_k,1}(X)\epsilon^2 + O(\epsilon^3).
\end{aligned}$$

The drift for y_t is given by

$$\begin{aligned}\mu_y(X,\epsilon) &= \frac{y_X(X,\epsilon)}{y(X,\epsilon)}\mu_X(X,\epsilon) + \frac{1}{2}\sigma_X^\top(Z,\epsilon)\frac{y_{XX}(X,\epsilon)}{y(X,\epsilon)}\sigma_X(X,\epsilon)\\ &= \frac{y_{X,1}(X)}{y_0(X)}\mu_{X,1}(X)\epsilon^2 + O(\epsilon^3)\\ &= \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}\left(\hat{\gamma}_k - \hat{\gamma}_0\right)\mu_{X_k,1}(X)\epsilon^2 + O(\epsilon^3),\end{aligned}$$

where we used the fact that $c_{j,\overline{\alpha}_p,1} = y_{\overline{\alpha}_p,1} = 0$.

Step 2: Risk exposures of investors.

We focus on the *inner region*, that is, the case where all investors are sufficiently far from the constraint boundary (on either side). For a constrained investor, the second-order term is zero. For an unconstrained investor, the second-order term is given by

$$\sigma_{j,2}(X) = \frac{\theta_2(X)}{\gamma} - \frac{\theta_1(X)}{\gamma}\hat{\gamma}_j + \frac{\theta_0(X)}{\gamma}\hat{\gamma}_j^2 + \frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}(X)\sigma^\top}{\|\sigma\|},$$

where investor j is unconstrained if the following condition holds:

$$\sum_{k\in\mathcal{J}^u}\frac{x_k}{x_u}\hat{\gamma}_k - \left(\frac{\hat{\alpha}_p x_0 + \hat{\sigma}x_c}{1-x_0-x_c}\right) - \hat{\gamma}_j < \hat{\sigma}.$$

Step 3: Aggregate risk aversion.

The aggregate risk aversion can be written as

$$\gamma_u(X,\epsilon) = \gamma\left[1 + \mathbb{E}^u\left[\hat{\gamma}_j\right]\epsilon - \mathrm{Var}^u\left[\hat{\gamma}_j\right]\epsilon^2\right] + O(\epsilon^3),$$

where

$$\mathbb{E}^u\left[\hat{\gamma}_j\right] \equiv \sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j, \quad \text{and} \quad \mathrm{Var}^u[\hat{\gamma}_j] \equiv \sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j^2 - \left(\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j\right)^2.$$

Step 4: Market price of risk.

The market price of risk is given by

$$\theta(X,\epsilon) = \frac{\gamma_u(X,\epsilon)}{x_u}\left[(1-(1+\hat{\alpha}_p\epsilon)x_{0,t})\|\sigma - \sigma_y(X,\epsilon)\| - \|\sigma\|(1+\hat{\sigma}\epsilon)x_{c,t} + \sum_{j\in\mathcal{J}_t^u}x_{j,t}\frac{1-\gamma_j^{-1}}{1-\psi}\sigma_{c_j}(X,\epsilon)\frac{\sigma^\top}{\|\sigma\|}\right],$$

The second-order term is then given by

$$\theta_2(X) = \frac{\gamma_{u,0}(X)}{x_u}\left[-(1-x_{0,t})\sigma_{y,2}(X)\frac{\sigma^\top}{\|\sigma\|} + \sum_{j\in\mathcal{J}_t^u} x_{j,t}\frac{1-\gamma^{-1}}{1-\psi}\sigma_{c_j,2}(X)\frac{\sigma^\top}{\|\sigma\|}\right] + \\ + \frac{\gamma_{u,1}(X)}{x_u}\|\sigma\|\left[-\hat{\alpha}_p x_0 - \hat{\sigma}x_c\right] + \frac{\gamma_{u,2}(X)}{x_u}\|\sigma\|\left[1-x_0-x_c\right].$$

The expression above can be written as

$$\theta_2(X) = -\frac{(\gamma-1)x_c + 1 - x_0}{1-x_0-x_c}\sigma_{y,2}(X)\frac{\sigma^\top}{\|\sigma\|} - \gamma\|\sigma\|\mathbb{E}^u[\hat{\gamma}_j]\frac{\hat{\alpha}_p x_0 + \hat{\sigma}x_c}{1-x_0-x_c} - \gamma\|\sigma\|\mathrm{Var}^u[\hat{\gamma}_j].$$

using the fact that $\sigma_{c_j,2}(X) = (1-\psi)\sigma_{y,2}$.

Step 5: Interest rate.

The interest rate is given by

$$r(X,\epsilon) = \rho - \theta(X,\epsilon)\|\sigma_R(X,\epsilon)\| + \psi^{-1}(\mu - \mu_y(X,\epsilon) - \sigma_y\sigma_R(X,\epsilon)^\top) \\ + \left(1-\psi^{-1}\right)\sum_{j=0}^{J} x_j\frac{\gamma_j}{2}\sigma_j^2(X,\epsilon) \\ + \psi^{-1}\sum_{j=0}^{J} x_{j,t}\left[\mu_{c_j}(X,\epsilon) + (1-\gamma_j)\sigma_{c_j}(X,\epsilon)\frac{\sigma_{R,t}^\top}{\|\sigma_R\|}\sigma_j(X,\epsilon) + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j,t}(X,\epsilon)\|^2}{2}\right].$$

The second-order term is given by

$$r_2(X) = -\theta_2(X)\|\sigma\| + \theta_0(X)\sigma_{y,2}(X)\frac{\sigma^\top}{\|\sigma\|} - \psi^{-1}(\mu_{y,2}(X) + \sigma_{y,2}(X)\sigma^\top) \\ + \left(1-\psi^{-1}\right)\sum_{j=0}^{J} x_j\gamma\|\sigma\|\left[\sigma_{j,2}(X) + \frac{\sigma_{j,1}^2(X)}{2\|\sigma\|} + \hat{\gamma}_j\sigma_{j,1}(X)\right] \\ + \psi^{-1}\sum_{j=0}^{J} x_j\left[\mu_{c_j,2}(X) + (1-\gamma)\sigma_{c_j,2}(X)\sigma^\top\right].$$

Step 6: Dividend yield.

The dividend yield, y, is given by

$$y(X,\epsilon) = \psi\rho + (1-\psi)\left[r(X,\epsilon) + \theta(X,\epsilon)\|\sigma_R(X,\epsilon)\| - \sum_{j=0}^{J} x_j \frac{\gamma_j}{2}\sigma_j^2(X,\epsilon)\right]$$
$$+ \sum_{j=0}^{J} x_j \left[\mu_{c_j}(X,\epsilon) + (1-\gamma_j)\sigma_{c_j}(X,\epsilon)\frac{\sigma_R(X;\epsilon)^\top}{\|\sigma_R(X;\epsilon)\|}\sigma_j(X,\epsilon) + \frac{\psi-\gamma_j}{1-\psi}\frac{\|\sigma_{c_j}(X,\epsilon)\|^2}{2}\right].$$

The second-order term in the expansion of $y(X,\epsilon)$ is given by

$$y_2(X) = (1-\psi)\left[r_2(X) + \theta_2(X)\|\sigma\| + \theta_0(X)\sigma_{R,2}(X)\frac{\sigma^\top}{\|\sigma\|} - \sum_{j=0}^{J} x_j \frac{\gamma}{2}\left(\sigma_{j,1}^2(X) + 2\sigma_{j,0}\sigma_{j,2}(X) + 2\hat{\gamma}_j\sigma_{j,0}\sigma_{j,1}(X)\right)\right]$$
$$+ \sum_{j=0}^{J} x_j \left[\mu_{c_j,2}(X) + (1-\gamma)\sigma_{c_j,2}(X)\sigma^\top\right].$$

Using the expression for the interest rate, we obtain

$$y_2(X) = \left(1-\psi^{-1}\right)\left[\mu_{y,2}(X) + \sigma_{y,2}(X)\sigma^\top + \sum_{j=0}^{J} x_j\gamma\|\sigma\|\left(\frac{\sigma_{j,1}^2(X)}{2\|\sigma\|} + \sigma_{j,2}(X) + \hat{\gamma}_j\sigma_{j,1}(X)\right)\right]$$
$$+ \psi^{-1}\sum_{j=0}^{J} x_j \left[\mu_{c_j,2}(X) + (1-\gamma)\sigma_{c_j,2}(X)\sigma^\top\right].$$

Given that $\mu_{c_j,2} = (1-\psi)\mu_{y,2}$ and $\sigma_{c_j,2} = (1-\psi)\sigma_{y,2}$, we obtain

$$y_2(X) = \left(1-\psi^{-1}\right)\left[\gamma\sigma_{y,2}(X)\sigma^\top + \sum_{j=0}^{J} x_j\gamma\|\sigma\|\left(\frac{\sigma_{j,1}^2(X)}{2\|\sigma\|} + \sigma_{j,2}(X) + \hat{\gamma}_j\sigma_{j,1}(X)\right)\right]$$
$$= \left(1-\psi^{-1}\right)\sum_{j=0}^{J} x_j\gamma\|\sigma\|\left(\frac{\sigma_{j,1}^2(X)}{2\|\sigma\|} + \hat{\gamma}_j\sigma_{j,1}(X)\right),$$

where in the second equality, we use the fact that, from the market clearing condition for the risky asset, we have $\sum_{j=0}^{J} x_j\sigma_{j,2}(X) = -\sigma_{y,2}(X)\frac{\sigma^\top}{\|\sigma\|}$.

Step 7: Risk premium.

Since the risk premium is given by $\pi_t = \theta_t\|\sigma_{R,t}\|$, we can write

$$\pi(X,\epsilon) = \theta_0(X)\|\sigma\| + \theta_1(X)\|\sigma\|\epsilon + \left(-\theta_0(X)\sigma_{y,2}(X)\frac{\sigma^\top}{\|\sigma\|} + \theta_2(X)\|\sigma\|\right)\epsilon^2 + O\left(\epsilon^3\right).$$

We can then write

$$\pi_2(X) = -\theta_0(X)\sigma_{y,2}(X)\frac{\sigma^\top}{\|\sigma\|} + \theta_2(X)\|\sigma\|. \tag{C.6}$$

□

D Derivation of the Market Elasticity

Let $p(X,\epsilon) \equiv 1/y(X,\epsilon)$ denote the price-dividend ratio. The second-order expansion of $p(X,\epsilon)$ is given by

$$p(X,\epsilon) = \frac{1}{y_0(X)} - \frac{y_1(X)}{y_0^2(X)}\epsilon + \left[\frac{y_1^2(X)}{y_0^3(X)} - \frac{y_2(X)}{y_0^2(X)}\right]\epsilon^2 + O(\epsilon^3). \tag{D.1}$$

Let $F(X) \equiv \frac{W_0(1+\hat{\alpha}_p\epsilon)-W_0}{P} = \hat{\alpha}_p\epsilon x_0$ denote the flow into the risky asset relative to the benchmark economy.

Proof. From Equation (D.1), the first-order impact of flows on the price-dividend ratio can be written as

$$\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1}{y_0^2(X)}\frac{\partial y_1(X)}{\partial \hat{\alpha}_p}\frac{1}{x_0} + O(\epsilon). \tag{D.2}$$

Since from Proposition 3, $y_1(X)$ does not depend on $\hat{\alpha}_p$, the right hand side of Equation (D.2) is zero, leading to an infinite aggregate elasticity to the first-order:

$$\varepsilon_M^{-1} = \frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = 0 + O(\epsilon).$$

Given that from Equation (15), we have $y_t = r_t + \pi_t - \mu_{P,t}$, we can write the first-order term for the dividend yield as

$$y_1(X) = r_1(X) + \pi_1(X) - \mu_{P,1}(X),$$

where $\mu_{P,t} = \mu - \mu_{y,t} - \sigma_{y,t}\sigma_{R,t}$ is the drift of the risky asset price P_t, and π_t is the risk premium. As shown in Proposition 3, the first-order term for the dividend yield is constant. This means $\mu_{y,1}(X) = \sigma_{y,1}(X) = 0$, leading to $\sigma_{R,1}(X) = 0$. Therefore, we have $\mu_{P,1}(X) = 0$, and

$$\frac{\partial y_1(X)}{\partial \hat{\alpha}_p} = \frac{\partial r_1(X)}{\partial \hat{\alpha}_p} + \frac{\partial \pi_1(X)}{\partial \hat{\alpha}_p}.$$

From Proposition 3, we have

$$\frac{\partial r_1(X)}{\partial \hat{\alpha}_p} = -\sigma \frac{\partial \theta_1(X)}{\partial \hat{\alpha}_p} = \frac{\gamma \sigma^2}{x_u} x_0,$$
$$\frac{\partial \pi_1(X)}{\partial \hat{\alpha}_p} = \sigma \frac{\partial \theta_1(X)}{\partial \hat{\alpha}_p} = -\frac{\gamma \sigma^2}{x_u} x_0.$$

Thus, up to the first order, the effect of portfolio flows on the risk-free rate is the exact opposite of its impact on the risk premium and portfolio flows do not affect the price-dividend ratio up to first order. Therefore, up to the first-order, the aggregate market elasticity is infinite. □

Using Equation (D.1), the derivative of the price-dividend ratio with respect to flows F is given by

$$\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1}{y_0^2(X)} \frac{\partial y_2(X)}{\partial \hat{\alpha}_p} \frac{\epsilon}{x_0} + O(\epsilon^2),$$

where $y_1(X)$ does not depend on $\hat{\alpha}_p$, leading to no price impact (infinite elasticity) up to the first-order.

D.1 Proof of Proposition 4

Proof. Consider the case where there is no preference heterogeneity and active investors do not face leverage constraints. In this case, $y_2(X)$ simplifies to

$$y_2(X) = \left(1 - \psi^{-1}\right) \gamma \sigma^2 \sum_{j=0}^{J} x_j \frac{\sigma_{j,1}^2(X)}{2\sigma^2},$$

using the fact that $\sigma_{y,2} = \sigma_{j,2} = 0$ when $\hat{\gamma}_j = 0$ for $j = 0, 1, \ldots, J$ when investors have the same preferences.

Using the expression for $\sigma_{j,1}$ in Proposition 3, the (inverse) aggregate market elasticity is given by

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = -\frac{1-\psi^{-1}}{2y_0(X)} \gamma \sigma^2 \left(2x_0 \hat{\alpha}_p + x_a \frac{2\hat{\alpha}_p x_0^2}{x_a^2}\right) \frac{\epsilon}{x_0} + O(\epsilon^2),$$

where $x_a \equiv 1 - x_0$ denotes the wealth share of active investors. This can be written as

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = \left(1 - \psi^{-1}\right) \frac{\gamma \sigma^2}{y_0(X)} \frac{1 - \overline{\alpha}_p}{x_a} + O(\epsilon^2),$$

where we use $\overline{\alpha}_p = 1 + \hat{\alpha}_p \epsilon$ from Equation (18). □

From Equation (C.6), the impact of flows of the risk premium can be written as

$$\frac{\partial \pi(X,\epsilon)}{\partial F(X,\epsilon)} = \sigma \frac{\partial \theta_1(X)}{\partial \hat{\alpha}_p} \frac{\epsilon}{x_0} + \frac{\partial \pi_2(X)}{\partial \hat{\alpha}_p} \frac{\epsilon}{x_0}$$

D.2 Proof of Proposition 5

Proof. Consider the case in which active investors have heterogeneous risk aversions, but face no leverage constraints. In this case, the expression for $y_2(X)$ in Equation (C.5) can be written as

$$y_2(X) = \left(1-\psi^{-1}\right)\gamma\sigma^2 \left[\sum_{j=1}^{J} x_j \left(\frac{\sigma_{j,1}^2(X)}{2\sigma^2} + \hat{\gamma}_j \frac{\sigma_{j,1}(X)}{\sigma}\right)\right] + \left(1-\psi^{-1}\right)\gamma\sigma^2 x_0 \left(\frac{\hat{\alpha}_p^2}{2} + \hat{\gamma}_0 \hat{\alpha}_p\right)$$

Note that with unconstrained active investors, the effect of endogenous volatility and hedging demand exactly cancel out. We first compute the derivative of the term involving $\sigma_{j,1}^2$:

$$\begin{aligned}
\sum_{j=0}^{J} x_j \frac{\partial \sigma_{j,1}^2}{\partial \hat{\alpha}_p} &= \sum_{j=0}^{J} 2x_j \sigma_{j,1} \frac{\partial \sigma_{j,1}}{\partial \hat{\alpha}_p} \\
&= 2x_0\sigma^2\hat{\alpha}_p + 2\sigma^2 \sum_{j=1}^{J} x_j \left(\sum_{k\in\mathcal{J}^u} \frac{x_k}{x_u}\hat{\gamma}_k - \left(\frac{\hat{\alpha}_p x_0}{1-x_0}\right) - \hat{\gamma}_j\right)\left(-\frac{x_0}{1-x_0}\right) \\
&= 2\sigma^2 \left[x_0\hat{\alpha}_p + \frac{\hat{\alpha}_p x_0^2}{1-x_0} + \left((1-x_0)\sum_{k=1}^{J}\frac{x_k}{x_u}\hat{\gamma}_k - \sum_{j=1}^{J} x_j\hat{\gamma}_j\right)\left(-\frac{x_0}{1-x_0}\right)\right] \\
&= 2\sigma^2 \frac{\hat{\alpha}_p x_0}{1-x_0}.
\end{aligned}$$

The derivatives of the term involving $\sigma_{j,1}(X)$ with respect to $\hat{\alpha}_p$ are given by:

$$\frac{1}{\sigma}\sum_{j=0}^{J} x_j\hat{\gamma}_j \frac{\partial \sigma_{j,1}(X)}{\partial \hat{\alpha}_p} = \sum_{j=1}^{J} x_j\hat{\gamma}_j \left(-\frac{x_0}{1-x_0}\right) + x_0\hat{\gamma}_0 = x_0\left(\hat{\gamma}_0 - \mathbb{E}^u[\hat{\gamma}_k]\right)$$

Thus, the (inverse) aggregate elasticity is then given by:

$$\frac{1}{p(X,\epsilon)}\frac{\partial p(X,\epsilon)}{\partial F(X,\epsilon)} = \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{y_0(X)}\left[\frac{1-\overline{\alpha}_p}{x_a} - \frac{\gamma_0 - \mathbb{E}^u\left[\gamma_j\right]}{\gamma}\right] + O(\epsilon^2),$$

where $x_a \equiv 1 - x_0$ is the wealth share of active investors, and we use $\gamma_j = \gamma\left(1+\hat{\gamma}_j\epsilon\right)$ from Equation (17). □

E Useful Formula

The following are useful for computing the derivatives above:

$$\gamma\sigma\frac{\partial\sigma_{y,2}(X)}{\partial\hat{\alpha}_p} = \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}\left(\hat{\gamma}_k-\hat{\gamma}_0\right)x_k\left(-\frac{\gamma\sigma^2 x_0}{1-x_0}\right).$$

Note that we can write $\sigma_{j,2}(X)$ as follows

$$\begin{aligned}\frac{\sigma_{j,2}(X)}{\sigma} &= \frac{\theta_2(X)}{\gamma\sigma}-\frac{\theta_1(X)}{\gamma\sigma}\hat{\gamma}_j+\frac{\theta_0(X)}{\gamma\sigma}\hat{\gamma}_j^2-(1-\gamma^{-1})\frac{\sigma_{y,2}(X)}{\sigma}\\ &= -\left(1+\frac{x_c}{x_u}\right)\frac{\sigma_{y,2}(X)}{\sigma}-\mathbb{E}^u[\hat{\gamma}_k]\frac{\hat{\alpha}_p x_0+\frac{\hat{\sigma}}{\sigma}x_c}{1-x_0-x_c}-\mathrm{Var}^u[\hat{\gamma}_k]+\\ &\quad-\hat{\gamma}_j\left[\sum_{k\in\mathcal{J}^u}\frac{x_k}{x_u}\hat{\gamma}_j-\frac{\hat{\alpha}_p x_0}{1-x_0}\right]+\hat{\gamma}_j^2.\end{aligned}$$

$$\begin{aligned}\sigma_{j,1}(X) &= \sigma\min\left\{\sum_{k\in\mathcal{J}^u}\frac{x_k}{x_u}\hat{\gamma}_k-\left(\frac{\hat{\alpha}_p x_0+\frac{\hat{\sigma}}{\|\sigma\|}x_c}{1-x_0-x_c}\right)-\hat{\gamma}_j,\frac{\hat{\sigma}}{\sigma}\right\}\\ \sigma_{j,2}(X) &= \frac{\theta_2(X)}{\gamma}-\frac{\theta_1(X)}{\gamma}\hat{\gamma}_j+\frac{\theta_0(X)}{\gamma}\hat{\gamma}_j^2-(1-\gamma^{-1})\sigma_{y,2}(X)\\ \sigma_{y,2}(X) &= \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}\left(\hat{\gamma}_k-\hat{\gamma}_0\right)x_k\sigma_{k,1}(X)\\ \theta_1(X) &= \gamma\sigma\left[\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j-\frac{\hat{\alpha}_p x_0}{1-x_0}\right]\\ \theta_2(X) &= -\frac{(\gamma-1)x_c+1-x_0}{1-x_0-x_c}\sigma_{y,2}(X)-\gamma\sigma\mathbb{E}^u[\hat{\gamma}_j]\frac{\hat{\alpha}_p x_0+\frac{\hat{\sigma}}{\sigma}x_c}{1-x_0-x_c}-\gamma\sigma\mathrm{Var}^u[\hat{\gamma}_j],\end{aligned}$$

where

$$\begin{aligned}\frac{\partial\sigma_{y,2}}{\partial\hat{\alpha}_p} &= \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sum_{k=1}^{J}\left(\hat{\gamma}_k-\hat{\gamma}_0\right)x_k\left(-\frac{\sigma x_0}{1-x_0}\right)\\ &= \left(1-\psi^{-1}\right)\frac{\gamma\sigma^2}{2y_0(X)}\sigma\left(\hat{\gamma}_0-\mathbb{E}^u[\hat{\gamma}_j]\right)x_0\\ \sum_{j=0}^{J}x_j\gamma\sigma\hat{\gamma}_j\frac{\partial\sigma_{j,1}(X)}{\partial\hat{\alpha}_p} &= \gamma\sigma^2\sum_{j=0}^{J}x_j\hat{\gamma}_j\left(-\frac{x_0}{1-x_0}\right)\end{aligned}$$

F Derivation of the perturbed solution

In this section, we compute the first-order and second-order correction of the equilibrium objects. It turns out that the system of equations determining the perturbed solution is block-recursive, so we are able to solve for the equilibrium objects one by one, provided we proceed in the appropriate order.

In contrast to the case considered in the text, we allow for portfolio-flow shocks. In particular, we assume that the portfolio share of the passive investor is given by $\alpha_{0,t} = 1 + \epsilon(\overline{\alpha}_{p,t} - 1)$, where $\overline{\alpha}_{p,t}$ follows the process

$$d\overline{\alpha}_{p,t} = \theta_p(\overline{\alpha} - \overline{\alpha}_{p,t})\epsilon dt + \sigma_p\sqrt{\overline{\alpha}_{p,t}}\epsilon dZ_t.$$

Notice that $\alpha_{0,t} = 1$ and $\overline{\alpha}_{p,t}$ is constant when $\epsilon = 0$. Finally, we assume that the mortality parameter is given by $\kappa = \hat{\kappa}\epsilon$.

F.1 First-order correction

Diffusion and drift terms. The diffusion term for the price-dividend ratio is given by

$$\sigma_{p,t} = \frac{p_x}{p}\sigma_x + \frac{p_{\overline{\alpha}_p}}{p}\sigma_p\sqrt{\overline{\alpha}_{p,t}}\epsilon = O(\epsilon^2).$$

Notice that $p_{x_j} = O(\epsilon)$ and $\sigma_{x_j} = O(\epsilon)$, as p and x_j are constant when $\epsilon = 0$, so the zeroth-order terms for p_{x_j}, $p_{\overline{\alpha}_p}$, and σ_{x_j} are equal to zero. This implies that the first-order correction for $\sigma_{p,t}$ is equal to zero. A similar argument shows that $\sigma_{c_j,t} = O(\epsilon^2)$.

The drift of p is given by

$$\mu_{p,t} = \frac{p_x}{p}\mu_x + \frac{p_{\overline{\alpha}_p}}{p}\theta_p(\overline{\alpha} - \overline{\alpha}_{p,t})\epsilon + \frac{1}{2}\sum_{k=1}^{d}\left[\sigma_{x,k}^{\top}\frac{p_{xx}}{p}\sigma_{x,k} + 2\sigma_{p,k}\sqrt{\overline{\alpha}_{p,t}}\epsilon\frac{p_{x\alpha_p}}{p}\sigma_{x,k} + \frac{p_{\overline{\alpha}_p\overline{\alpha}_p}}{p}\sigma_{p,k}^2\overline{\alpha}_{p,t}\epsilon^2\right],$$

where $\sigma_{x,k}$ is the k-th column of the $J \times d$ matrix σ_x. As p_x and $p_{\overline{\alpha}_p}$ are first-order in ϵ, and the same goes for μ_x and σ_x, then $\mu_{p,t} = O(\epsilon^2)$. A similar argument shows that $\mu_{c_j,t} = O(\epsilon^2)$. Notice these facts imply that $\varsigma_j = O(\epsilon^2)$ and $\xi_{j,t} = O(\epsilon^2)$.

Risk premium. The risk premium is given by

$$\pi_1(X) = \gamma||\sigma||^2\left[\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j - \frac{x_0\hat{\alpha}_p + x_c\frac{\hat{\sigma}}{||\sigma||}}{1 - x_0 - x_c}\right],$$

using the fact that $||\sigma_{R,t}|| = ||\sigma|| + O(\epsilon^2)$, and $\hat{\alpha}_{p,t} \equiv \overline{\alpha}_{p,t} - 1$.

Portfolio share. The portfolio share of an unconstrained investor is given by

$$\alpha_j(X, \epsilon) = 1 + \left[\frac{\pi_1(X)}{\gamma||\sigma||^2} - \hat{\gamma}_j\right]\epsilon + O(\epsilon^2),$$

the portfolio share of a constrained investor is given by

$$\alpha_j(X, \epsilon) = 1 + \frac{\hat{\sigma}}{||\sigma||}\epsilon + O(\epsilon^2),$$

and the portfolio share of the passive investor is given by $\alpha_0(X, \epsilon) = 1 + \hat{\alpha}_{p,t}\epsilon$. Notice that $\sum_{j=0}^{J} x_j \alpha_{j,1}(X) = 0$, consistent with market clearing.

Interest rate. The first-order correction for the interest rate is given by

$$r_1(X) = \left(1 - \psi^{-1}\right)\gamma||\sigma||^2 \sum_{j=0}^{J} x_j \left[\frac{\hat{\gamma}_j}{2} + \alpha_{j,1}(X)\right] - \pi_1(X),$$

using the fact that $\xi_t = O(\epsilon^2)$, $\mu_{p,t} = O(\epsilon^2)$, and $\sigma_{p,t} = O(\epsilon^2)$. Given the market clearing for the risky asset, we can write:

$$r_1(X) = \left(1 - \psi^{-1}\right)\gamma||\sigma||^2 \sum_{j=0}^{J} x_j \frac{\hat{\gamma}_j}{2} - \pi_1(X),$$

so $r_1(X) + \pi_1(X)$ is independent of $\overline{\alpha}_p$.

Price-dividend ratio. From the pricing condition, we obtain

$$-\frac{1}{p_0(X)^2} p_1(X) = r_1(X) + \pi_1(X).$$

Rearranging the expression above, and using the expression for the interest rate, we obtain

$$p_1(X) = -p_0(X)^2 \left(1 - \psi^{-1}\right)\gamma||\sigma||^2 \sum_{j=0}^{J} x_j \frac{\hat{\gamma}_j}{2},$$

which is independent of $\overline{\alpha}_{p,t}$.

Consumption-wealth ratio. The consumption-wealth ratio is given by

$$c_{j,1}(X) = (1-\psi)\left[r_1(X) + \pi_1(X) + \pi_0(X)\alpha_{j,1}(X) - \frac{1}{2}\gamma|||\sigma|||^2(\hat{\gamma}_j + 2\alpha_{j,1}(X))\right].$$

Using the expression for $r_1(X)$, we can write the expression above as follows:

$$c_{j,1}(X) = (1-\psi)\left[(1-\psi^{-1})\sum_{i=0}^{J} x_i \frac{\hat{\gamma}_i}{2} - \frac{\hat{\gamma}_j}{2}\right]\gamma|||\sigma|||^2.$$

Wealth dynamics. The diffusion term of x_j is given by

$$\sigma_{x_j}(X) = x_j\alpha_{j,1}(X)\epsilon\sigma + O(\epsilon^2).$$

The drift of x_j is given by

$$\mu_{x_j}(X) = x_j\left[r_1(X) + \pi_1(X) + \pi_0(X)\alpha_{j,1}(X) - c_{j,1}(X) - \alpha_{j,1}(X)||\sigma||^2 + \hat{\kappa}\frac{\omega_j - x_j}{x_j}\right]\epsilon + O(\epsilon^2).$$

We can write the first-order correction of μ_{x_j} as follows:

$$\mu_{x_j,1}(X) = x_j\left[(\psi-1)\frac{\gamma|||\sigma|||^2}{2}\left(\sum_{i=0}^{J} x_i\hat{\gamma}_i - \hat{\gamma}_j\right) + (\gamma-1)|||\sigma|||^2\alpha_{j,1}(X)\right] + \hat{\kappa}(\omega_j - x_j).$$

F.2 Second-order correction

Diffusion and drift terms. The diffusion term for the price-dividend ratio is given by

$$\sigma_{p,2}(X) = \frac{p_{x,1}(X)}{p_0(X)}\sigma_{x,1}(X) + \frac{p_{\overline{\alpha}_p,1}(X)}{p_0(X)}\sigma_p\sqrt{\overline{\alpha}_p}.$$

We can write the expression above as follows:

$$\sigma_{p,2}(X) = -p_0(X)(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\sum_{j=1}^{J}\left(\hat{\gamma}_j - \hat{\gamma}_0\right)x_j\alpha_{j,1}(X)\sigma,$$

where we used the fact that $p_{\overline{\alpha}_p,1}(X) = 0$.

Similarly, the diffusion for c_j is given by

$$\sigma_{c_j,2}(X) = -(\psi-1)\frac{\gamma|||\sigma|||^2}{2c_{j,0}(X)}(1-\psi^{-1})\sum_{i=1}^{J}(\hat{\gamma}_i - \hat{\gamma}_0)\,x_i\alpha_{i,1}(X)\sigma.$$

The second-order correction for the hedging demand is then given by $\varsigma_{j,2}(X) = \frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}\sigma^\top}{\|\|\sigma\|\|^2}$.

The second-order correction of the drift of p and c_j are given by

$$\mu_{p,2}(X) = \frac{p_{x,1}(X)}{p_0(X)}\mu_{x,1}(X), \qquad \mu_{c_j,2}(X) = \frac{c_{j,x,1}(X)}{c_{j,0}(X)}\mu_{x,1}(X),$$

which can be written as

$$\mu_{p,2}(X) = -p_0(X)(1-\psi^{-1})\frac{\gamma\|\sigma\|\|^2}{2}\sum_{j=1}^{J}(\hat{\gamma}_j - \hat{\gamma}_0)\mu_{x_j,1}(X)$$

$$\mu_{c_i,2}(X) = (1-\psi)(1-\psi^{-1})\frac{\gamma\|\sigma\|\|^2}{2c_{i,0}(X)}\sum_{j=1}^{J}(\hat{\gamma}_j - \hat{\gamma}_0)\mu_{x_j,1}(X).$$

The second-order correction for $\xi_{j,t}$ is then given by $\xi_{j,2}(X) = \mu_{c_j,2}(X) + (1-\gamma)\sigma_{c_j,2}\sigma^\top$.

Risk premium. The risk premium is given by

$$\pi_2(X) = \gamma\|\sigma\|\|^2\left[\frac{\gamma_{u,2}(X)}{\gamma} - \frac{\gamma_{u,1}(X)}{\gamma}\frac{x_0\hat{\alpha}_p + x_c\frac{\hat{\sigma}}{\|\sigma\|\|}}{1-x_0-x_c} + 2\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{\|\sigma\|^2} - \frac{x_c}{1-x_0-x_c}\overline{\alpha}_{c,2}(X) - \varsigma_2(X)\right].$$

Notice that we can write the aggregate risk aversion as follows:

$$\gamma_u(X) = \gamma\left[1 + \mathbb{E}^u[\hat{\gamma}_j]\epsilon - \delta^u[\hat{\gamma}_j]\epsilon^2\right] + O(\epsilon^3),$$

where $\mathbb{E}^u[\hat{\gamma}_j] \equiv \sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j$ and $\delta^u[\hat{\gamma}_j] \equiv \sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j^2 - \left(\sum_{j\in\mathcal{J}^u}\frac{x_j}{x_u}\hat{\gamma}_j\right)^2$, so $\gamma_{u,2}(X)/\gamma = -\delta^u[\hat{\gamma}_j]$.

Combining the previous two expressions, we obtain

$$\begin{aligned}\frac{\pi_2(X)}{\gamma\|\sigma\|\|^2} &= -\delta^u[\hat{\gamma}_j] - \mathbb{E}^u[\hat{\gamma}_j]\frac{x_0\hat{\alpha}_p + x_c\frac{\hat{\sigma}}{\|\sigma\|\|}}{1-x_0-x_c} + 2\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{\|\sigma\|^2} + \frac{x_c}{1-x_0-x_c}\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{\|\sigma\|\|^2} - \varsigma_2(X)\\ &= -\delta^u[\hat{\gamma}_j] - \mathbb{E}^u[\hat{\gamma}_j]\frac{x_0\hat{\alpha}_p + x_c\frac{\hat{\sigma}}{\|\sigma\|\|}}{1-x_0-x_c} + \left(1+\gamma^{-1}+\frac{x_c}{1-x_0-x_c}\right)\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{\|\sigma\|\|^2},\end{aligned}$$

where we used the fact that $\varsigma_2(X) = \frac{1-\gamma^{-1}}{\psi-1}\frac{\sigma_{c_j,2}\sigma^\top}{\|\sigma\|\|^2}$, $\sigma_{p,2} = \frac{\sigma_{c_j,2}}{\psi-1}$, and $\overline{\alpha}_{c,2}(X) = -\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}}{\|\sigma\|\|^2}$.

Portfolio share. The portfolio share of an unconstrained investor is given by

$$\alpha_{j,2}(X) = \frac{\pi_2(X)}{\gamma\|\sigma\|\|^2} - \frac{\pi_1(X)}{\gamma\|\sigma\|\|^2}\hat{\gamma}_j + \hat{\gamma}_j^2 - 2\sum_{k=1}^{d}\frac{\sigma_k\sigma_{p,2,k}(X)}{\|\sigma\|^2} + \varsigma_{j,2}(X),$$

the portfolio share of a constrained investor is $\alpha_{j,2}(X) = -\sum_{k=1}^{d} \frac{\sigma_k \sigma_{p,2,k}}{\|\sigma\|^2}$, and the portfolio share of the passive investor satisfies $\alpha_{0,2} = 0$.

We can write the expression above as follows:

$$\alpha_{j,2}(X) = \frac{\pi_2(X)}{\gamma\|\|\sigma\|\|^2} - \frac{\pi_1(X)}{\gamma\|\|\sigma\|\|^2}\hat{\gamma}_j + \hat{\gamma}_j^2 - (1+\gamma^{-1}) \sum_{k=1}^{d} \frac{\sigma_k \sigma_{p,2,k}(X)}{\|\sigma\|^2}.$$

Notice that $\sum_{j=0}^{J} x_j \alpha_{j,2}(X) = 0$, consistent with market clearing.

Interest rate. The interest rate is given by

$$r_2(X) = \psi^{-1}(\mu_{p,2}(X) + \sigma\sigma_{p,2}(X)^\top) + (1-\psi^{-1}).\frac{\gamma\|\sigma\|\|^2}{2} \sum_{j=0}^{J} x_j \left[2\hat{\gamma}_j \alpha_{j,1}(X) + \alpha_{j,1}^2(X) + 2\alpha_{j,2}(X)\right]$$
$$+ (1-\psi^{-1})\gamma\|\sigma\|^2 \sum_{k=1}^{d} \frac{\sigma_k \sigma_{p,2,k}(X)}{\|\sigma\|^2} - \pi_2(X) + \psi^{-1}\xi_2(X),$$

where $\xi_2(X) = (\psi - 1)\left[\mu_{p,2}(X) + (1-\gamma)\sigma_{p,2}\sigma^\top\right]$

We can write the expression for the portfolio of the unconstrained investor as follows:

$$r_2(X) = \mu_{p,2}(X) + \sigma\sigma_{p,2}(X)^\top + (1-\psi^{-1})\gamma\|\sigma\|\|^2 \sum_{j=0}^{J} x_j \left[\hat{\gamma}_j \alpha_{j,1}(X) + \frac{\alpha_{j,1}^2(X)}{2}\right] - \pi_2(X),$$

Price-dividend ratio. The price-dividend ratio is given by

$$\frac{p_1^2(X)}{p_0^3(X)} - \frac{p_2(X)}{p_0^2(X)} = r_2(X) + \pi_2(X) - \mu_{p,2}(X) - \sigma\sigma_{p,2}^\top.$$

Rearranging the expression above, and using the expression for $r_2(X)$, we obtain

$$\frac{p_2(X)}{p_0(X)} = -p_0(X)(1-\psi^{-1})\gamma\|\sigma\|\|^2 \sum_{j=0}^{J} x_j \left[\hat{\gamma}_j \alpha_{j,1}(X) + \frac{\alpha_{j,1}^2(X)}{2}\right] + \left(\frac{p_1(X)}{p_0(X)}\right)^2.$$

Consumption-wealth ratio. The second-order correction for the consumption-wealth ratio is given by

$$c_{j,2}(X) = (1-\psi)\left[r_2(X) + \pi_2(X) + \pi_1(X)\alpha_{j,1}(X) + \pi_0(X)\alpha_{j,2}(X)\right] + \xi_{j,2}(X)$$
$$- (1-\psi)\gamma\|\sigma\|\|^2 \left[\alpha_{j,1}(X)\hat{\gamma}_j + \alpha_{j,2}(X) + \frac{\alpha_{j,1}^2(X)}{2} + \sum_{k=1}^{d} \frac{\sigma_k \sigma_{p,k,2}(X)}{\|\sigma\|\|^2}\right].$$

Wealth dynamics. The diffusion term of x_j is given by

$$\sigma_{x_j,2}(X) = x_j \alpha_{j,2} \sigma.$$

The drift of x_j is given by

$$\mu_{x_j,2} = x_j \left[r_2(X) + \pi_2(X) + \pi_1(X)\alpha_{j,1}(X) + \pi_0(X)\alpha_{j,2}(X) - c_{j,2}(X) - \mu_{p,2}(X) - \sigma\sigma_{p,2}(X)^\top - \alpha_{j,2}(X)\|\sigma\|^2 \right].$$

Aggregate market elasticity. The derivative of p with respect to $\overline{\alpha}_p$ is given by

$$\frac{1}{p(X,\epsilon)} \frac{\partial p(X,\epsilon)}{\partial \overline{\alpha}_p} = \frac{1}{p_0(X)} \frac{\partial p_2(X)}{\partial \overline{\alpha}_p} \epsilon^2 + O(\epsilon^3).$$

The market elasticity satisfies the condition

$$\frac{1}{p_0(X)} \frac{\partial p_2(X)}{\partial \overline{\alpha}_p} = -p_0(X)(1-\psi^{-1})\gamma\|\sigma\|\|^2 \left[x_0(\hat{\gamma}_0 + \hat{\alpha}_p) + \sum_{j \in \mathcal{J}^u} x_j \left(\hat{\gamma}_j + \alpha_{j,1}(X) \right) \left(-\frac{x_0}{x_u} \right) \right] \epsilon^2.$$

From the market clearing for the risky asset, we have $x_0 \hat{\alpha}_p + \sum_{j \in \mathcal{J}^u} x_j \alpha_{j,1}(X) + x_c \frac{\hat{\sigma}}{\|\sigma\|} = 0$, so we can write the expression above as follows:

$$\frac{1}{p_0(X)} \frac{\partial p_2(X)}{\partial \overline{\alpha}_p} = p_0(X)(1-\psi^{-1})\gamma\|\sigma\|\|^2 \left[\hat{\gamma}_u(X) - \hat{\gamma}_0 + \frac{1-x_c}{1-x_0-x_c}(1-\overline{\alpha}_p) - \frac{x_c \frac{\hat{\sigma}}{\|\sigma\|\|}}{1-x_0-x_c} \right] x_0 \epsilon^2.$$

F.3 Third-order approximation

F.3.1 Passive demand

Suppose there is no preference heterogeneity and no leverage constraint. Without loss of generality, set $J = 1$. In this case, the price-dividend ratio is given by

$$
\begin{aligned}
p(X,\epsilon) &= p^\star - (p^\star)^2(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\frac{x_0\hat{\alpha}_p^2}{x_1}\epsilon^2 + O(\epsilon^3)\\
\pi(X,\epsilon) &= \pi_0(X) - \gamma\|\sigma\|^2\frac{x_0\hat{\alpha}_p}{x_1}\epsilon + O(\epsilon^3)\\
r(X,\epsilon) &= r_0(X) + \gamma\|\sigma\|^2\frac{x_0\hat{\alpha}_p}{x_1}\epsilon + (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\frac{x_0\hat{\alpha}_p^2}{x_1}\epsilon^2 + O(\epsilon^3)\\
\alpha_0(X,\epsilon) &= 1 + \hat{\alpha}_p\epsilon + O(\epsilon^3)\\
\alpha_1(X,\epsilon) &= 1 - \frac{x_0}{x_1}\hat{\alpha}_p\epsilon + O(\epsilon^3)\\
c_j(X,\epsilon) &= c_{j,0}(X) + (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2}\frac{x_0\hat{\alpha}_p^2}{x_1}\epsilon^2 + O(\epsilon^3)\\
\sigma_{x_1}(X,\epsilon) &= -\frac{x_0}{x_1}\hat{\alpha}_p\sigma\epsilon + O(\epsilon^3)\\
\mu_{x_1}(X,\epsilon) &= \left[(1-\gamma)\|\sigma\|^2(1-x_1)\hat{\alpha}_p + \hat{\kappa}(\omega_j - x_1)\right]\epsilon + O(\epsilon^3),
\end{aligned}
$$

where $\mu_p, \sigma_p, \mu_{c_j}$, and σ_{c_j} are all equal to zero up to second order, and $x_0 = 1 - x_1$.

The law of motion of $x_{1,t}$ can be written as

$$
\frac{dx_{1,t}}{x_{1,t}} = \left[x_{0,t}(c_{0,t} - c_{1,t}) + x_{0,t}(\alpha_{0,t} - \alpha_{1,t})\left(\|\sigma_{R,t}\|^2 - \pi_t\right) + \kappa\frac{\omega_1 - x_{1,t}}{x_{1,t}}\right]dt + (\alpha_{1,t} - 1)\sigma_{R,t}dZ_t.
$$

Diffusion and drift terms. The derivatives of $p(X,\epsilon)$ with respect to x_1 and $\overline{\alpha}_p$ are given by

$$
\frac{p_{x_1}(X,\epsilon)}{p^\star} = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2y^\star}\frac{1}{x_1^2}\hat{\alpha}_p^2\epsilon^2 + O(\epsilon^3), \qquad \frac{p_{\overline{\alpha}_p}(X,\epsilon)}{p^\star} = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y^\star}\frac{x_0}{x_1}\hat{\alpha}_p\epsilon^2 + O(\epsilon^3)
$$

The diffusion term for $p(X,\epsilon)$ is then given by

$$
\sigma_{p,3}(X) = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y^\star}\frac{x_0}{x_1}\left[\frac{\hat{\alpha}_p^3}{2x_1^2}\sigma + \hat{\alpha}_p\sigma_p\sqrt{\overline{\alpha}_p}\right].
$$

and $\sigma_{c_j,3}(X) = \sigma_{p,3}(X)$. Notice that excess volatility depends on the market elasticity times the volatility of portfolio flows.

The drift of p is given by

$$\mu_{p,3}(X) = (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2y^\star}\frac{1}{x_1^2}\hat{\alpha}_p^2\mu_{x_1,1} - (1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{y^\star}\frac{x_0}{x_1}\hat{\alpha}_p\theta_p(\overline{\alpha}-\overline{\alpha}_{p,t}),$$

and $\mu_{c_j,3}(X) = \mu_{p,3}(X)$.

Risk premium. The risk premium is given by

$$\pi(X) = \gamma\|\sigma\|^2\left[1 - \frac{x_0(\overline{\alpha}_p - 1)}{x_1} - \frac{1-\gamma^{-1}}{\psi-1}\frac{\gamma\|\sigma\|^2}{y^\star}\frac{x_0}{x_1}\left[\frac{(1-\overline{\alpha}_p)^3}{2x_1^2}\sigma + \hat{\alpha}_p\sigma_p\sqrt{\overline{\alpha}_p}\right]\right]$$

G Higher-order perturbations

Suppose we have the $(n-1)$-th order perturbation of $c_j(X,\epsilon) = \sum_{k=0}^{n-1} c_{j,k}(X)\epsilon^k$ and the law of motion of X. Let $lc_j(X,\epsilon) = \sum_{k=0}^{n-1} lc_{j,k}(X)\epsilon^k$ denote the expansion of $\log c_j(X,\epsilon)$. Then we can compute $\sigma_j(X)$ up to order n:

$$\sigma_{c_j,n}(X) = \sum_{k=1}^{n-1} lc_{j,k,X}(X)\sigma_{X,n-k}(X),$$

which is independent of the n-th order term in $c_j(X,\epsilon)$ and $\sigma_X(X,\epsilon)$, as $lc_{j,0,X}(X) = 0$. Similarly, we can compute $\sigma_{y,n}(X)$. A similar argument gives $\mu_{p,n}(X)$ and $\mu_{c_j,n}(X)$. We can then compute $\pi_n(X)$ and $\alpha_{j,n}(X)$. The n-th term of the consumption-wealth ratio satisfies the condition

$$c_{j,t} = \psi\rho + (1-\psi)\left[\pi_t(\alpha_{j,t}-1) + \mu + \mu_{p,t} + \sigma\sigma_{p,t}^\top - \frac{\gamma_j}{2}\|\sigma_{R,t}\|^2\alpha_{j,t}^2 + \sum_{j=0}^{J} x_j c_{j,t}\right] + \xi_{j,t}$$

We can rewrite the system above in matrix form as follows:

$$[I - (1-\psi)\mathbf{1}_{J+1}x_t^\top]c_t = \zeta_t,$$

where $c_t = [c_{0,t},\ldots,c_{J,t}]^\top$, $x_t = [x_{0,t},\ldots,x_{J,t}]^\top$, $\mathbf{1}_{J+1}$ is a $(J+1)$-th dimensional vector of ones, and $\zeta_{j,t} \equiv \psi\rho + (1-\psi)\left[\pi_t(\alpha_{j,t}-1) + \mu + \mu_{p,t} + \sigma\sigma_{p,t}^\top - \frac{\gamma_j}{2}\|\sigma_{R,t}\|^2\alpha_{j,t}^2\right] + \xi_{j,t}$. Applying the Sherman-Morrison formula, we obtain

$$c_t = [I - (1-\psi^{-1})\mathbf{1}_{J+1}x_t^\top]\zeta_t,$$

or $c_{j,t} = \zeta_{j,t} - (1-\psi^{-1})x_t^\top\zeta_t$. Notice that ζ_t can be computed at order n based on the coefficients

of order $n-1$ and their derivatives.

Computing the derivatives. The derivation above shows that, given the order $n-1$ expansion of $\zeta_j(X,\epsilon)$ and its derivatives, we can compute the expansion of order n. Suppose the expansion of $\zeta_j(X,\epsilon)$ is given by

$$\zeta_j(X,\epsilon) = \sum_{k=0}^{n-1} \zeta_{j,k}(X)\epsilon^k,$$

where $\zeta_{j,k}(X)$ takes the form:

$$\zeta_{j,k}(X) = A_{j,k} + B_{j,k}^\top(X-\overline{X}) + \frac{1}{2}(X-\overline{X})^\top C_{j,k}(X-\overline{X}),$$

where $\overline{X}$ is a reference point, $A_{j,k}$ is a scalar, $B_{j,k}$ is a vector, and $C_{j,k}$ is a matrix. Notice that . $\zeta_{j,k}(\overline{X}) = A_{j,k}$, $\zeta_{j,k,X}(\overline{X}) = B_{j,k}$ and $\zeta_{j,k,XX} = C_{j,k}$. Given this expansion, we can compute $c_j(X,\epsilon) = \sum_{k=0}^{n-1} c_{j,k}(X)\epsilon^k$.

G.1 Inner region

Consider the case of no preference heterogeneity and no leverage constraints. Consider the following change of variables: $x_1 = \epsilon\tilde{x}_1$. Define $\tilde{c}_j(\tilde{x}_1,\overline{\alpha}_p) = c_j\left(x_1,\overline{\alpha}_p\right)$, so $c_{j,x_1} = \frac{1}{\epsilon}\tilde{c}_{j,\tilde{x}_1}$ and $c_{j,x_1x_1} = \frac{1}{\epsilon^2}\tilde{c}_{j,\tilde{x}_1\tilde{x}_1}$. This implies the following is true:

$$\sigma_{c_j} = \frac{1}{\epsilon}\frac{\tilde{c}_{j,\tilde{x}_1}}{\tilde{c}_j}\sigma_{x_1},$$

where $\sigma_{x_1} = -\frac{1-\epsilon\tilde{x}_1}{\tilde{x}_1}\hat{\alpha}_p\sigma_R$. Similarly, we can write σ_y

$$\sigma_y = \frac{1}{\epsilon}\frac{\tilde{y}_{x_1}}{y}\sigma_{x_1}.$$

The drift of y is given by

$$\mu_y = \frac{1}{\epsilon}\frac{\tilde{y}_{\tilde{x}_1}}{y}\mu_{\tilde{x}_1} + \frac{1}{2\epsilon^2}\frac{\tilde{y}_{\tilde{x}_1\tilde{x}_1}}{y}\sigma_{\tilde{x}_1}^2.$$

The term of order 0 is the same as before.

H Volatility

Consider the case without preference heterogeneity or leverage constraints. The price-dividend ratio is given by:

$$\hat{p}_t = -(1-\psi^{-1})\frac{\gamma\|\sigma\|^2}{2y^\star}\,\frac{(1-x_{a,t})(1-\overline{\alpha}_{p,t})^2}{x_{a,t}}.$$

where, given $\overline{\alpha}_{a,t} \equiv \frac{1-(1-x_{a,t})\overline{\alpha}_{p,t}}{x_{a,t}}$, we have

$$\sigma_{x_a,t} = x_{a,t}(\overline{\alpha}_{a,t}-1)\sigma, \qquad \sigma_{\overline{\alpha}_p,t} = \sigma_{\overline{\alpha}_p}$$

The diffusion of $\hat{p}_t$ is given

$$\sigma_{p,t} = \hat{p}_{x_a,t}\sigma_{x_a,t} + \hat{p}_{\overline{\alpha}_p,t}\sigma_{\overline{\alpha}_p}.$$

$$\hat{c}_t \;\equiv\; c_t - c^\star \;=\; \chi_{0,t} \;+\; \chi_p\,\hat{p}_t \;+\; O(\epsilon^3),$$

with coefficients

$$\chi_p = (\psi-1)\,\frac{1}{p^\star}, \qquad \chi_{0,t} = (\psi-1)\,\frac{\gamma\|\sigma\|^2}{2}\,\frac{f_t^2}{x_{0,t}(1-x_{0,t})}.$$

I Flow of Funds aggregation

This appendix describes the procedure used in Section 5.1 to aggregate Flow of Funds (FoF) sectors into the three ultimate sectors of the model: passive, constrained active, and unconstrained active. We follow the aggregation in Koijen et al. (2024), drop sectors with no or minimal equity holdings, and route pass-through vehicles to their ultimate end investors. ETFs (L.124) and closed-end funds (L.123) are aggregated onto the household balance sheet (L.101); mutual-fund holdings (L.122) are split across end investors using the FoF's breakdown, with the household portion kept separate. Defined-contribution pensions (L.118c, L.119c, L.120c) are routed to households. Hedge funds are separated out using Table B.101.f for domestic funds and the Enhanced Financial Accounts for foreign funds, and the corresponding holdings are subtracted from the household (L.101) and rest-of-the-world (L.133) balance sheets. The remaining insurance and pension sectors (L.115, L.116, L.118b, L.119b, L.120b) are aggregated into a single long-horizon sector. The three ultimate sectors correspond, respectively, to *passive* (households and non-profits net of hedge funds, government, monetary authority, and money-market funds), *constrained* (hedge funds, broker-dealers, and other short-horizon intermediaries), and *unconstrained* (insurance, pensions,

and rest-of-the-world net of hedge funds).

J Numerical methodology

This appendix describes the numerical procedure used to solve the dynamic model in Section 3. We follow a two-step strategy where we first solve a 'simplified' model with an aggregate claim in positive net supply and a derivative security in zero net supply. In the second step (valuation step), we solve the valuation equation for the dividend and risky debt claims. This ordering is justified since, once the equilibrium is computed in the first step, the market prices of risk and interest rates are known functions of the simplified model, and the individual valuation ratios can be recovered from the respective pricing equations.

J.1 Simplified economy

The state variables in the simplified economy are

$$X = (x_c, x_u, \alpha_p) \in \mathcal{S}, \qquad x_p = 1 - x_c - x_u,$$

where x_c and x_u are the wealth shares of the constrained and unconstrained active investors, and α_p is the exogenous risky-share demand of the passive sector. The aggregate claim is in positive net supply and the derivative is in zero net supply. Denote the active portfolio exposures by $(\alpha_{j,Y}, \theta_j)$, where $\alpha_{j,Y}$ is the exposure to the aggregate claim and $\alpha_{j,F}$ is the exposure to the derivative. The passive investor has exposure $(\alpha_{p,t}, 0)$. We approximate

$$\xi_c(X), \quad \xi_u(X), \quad \xi_p(X), \quad \alpha_{c,Y}(X), \quad \alpha_{c,F}(X), \quad \alpha_{u,Y}(X), \quad \alpha_{u,F}(X)$$

with neural networks parameterized by Θ:

$$\begin{aligned} \hat{\xi}_j &: \mathcal{S} \to \mathbb{R}_+, \qquad j \in \{c, u, p\}, \\ \hat{\alpha}_{j,Y} &: \mathcal{S} \to \mathbb{R}, \qquad j \in \{c, u\}, \\ \hat{\alpha}_{j,F} &: \mathcal{S} \to \mathbb{R}, \qquad j \in \{c, u\}. \end{aligned}$$

and derive for all other equilibrium quantities as functions of the state variables and the approximated objects. Note that the passive investor does not trade the derivative, so $\alpha_{p,F}(X) \equiv 0$, while the passive exposure to the aggregate claim is the state variable itself, $\alpha_{p,Y}(X) = \alpha_p$. Thus the passive portfolio is imposed directly rather than approximated. We use fully connected feed-forward neural networks with 8 hidden layers and 10 neurons in each layer. Each neuron in the hidden and final

layer is represented by a softplus activation function for the consumption-wealth ratios $\hat{\xi}_j$'s, and a linear function for the others. The state space is sampled globally using Latin-hypercube points on the simplex

$$\{(x_c, x_u, \alpha_p) : x_c > 0,\ x_u > 0, \alpha_p > 0,\ x_c + x_u < 1\}$$

Given a candidate collection of network outputs at X, the remaining equilibrium objects are recovered analytically.

First, goods-market clearing implies the aggregate claim dividend yield

$$\hat{y}(X) = x_c\hat{\xi}_c(X) + x_u\hat{\xi}_u(X) + x_p\hat{\xi}_p(X).$$

Second, the volatility of the dividend yield is recovered from the Itô consistency condition. Third, once $\sigma_y(X)$ is known, the excess-return volatilities of the aggregate claim and the derivative, denoted $(\sigma_{R,Y}(X), \sigma_{R,F}(X))$, are recovered. The unconstrained active investor's first-order conditions then pin down the two risk premia $\pi_Y(X), \pi_F(X)$. Fourth, conditional on $(\hat{y}, \pi_Y, \pi_F, \sigma_y)$, the risk-free rate $r(X)$, the state drift $\mu_X(X)$, the state diffusion matrix $\Sigma_X(X)$, the consumption-wealth ratio drifts $\mu_{c_j}(X)$, and the market price of risk vector $\eta(X)$ are all recovered from the equilibrium identities and Itô's lemma. These calculations are explicit once the neural-network outputs and their first and second derivatives are known. The problem reduces to minimizing the following loss functions

1. **HJB residuals.** For each investor type $j \in \{c, u, p\}$, we construct the residual from the investor's Hamilton–Jacobi–Bellman equation after replacing all equilibrium objects by either neural-network outputs or the derived quantities defined above. We denote these residuals by

$$\mathcal{L}_j^{\text{HJB}}(X; \Theta).$$

2. **Portfolio FOCs.** The constrained investor's first-order conditions for the aggregate claim and derivative lead to two residuals denoted as:

$$\mathcal{L}_{c,Y}^{\text{FOC}}(X; \Theta), \qquad \mathcal{L}_{c,F}^{\text{FOC}}(X; \Theta).$$

   Note that the unconstrained investor's first-order conditions are used to recover (π_Y, π_F) exactly and therefore do not appear separately in the loss.

3. **Market-clearing residuals.** We impose clearing in the aggregate-claim and derivative markets as residuals:

$$\mathcal{L}_Y^{\text{MC}}(X; \Theta), \qquad \mathcal{L}_F^{\text{MC}}(X; \Theta).$$

The resulting loss is

$$\mathcal{L}(X;\Theta) = \sum_{j\in\{c,u,p\}} \left(\mathcal{L}_j^{\text{HJB}}(X;\Theta)\right)^2 + \left(\mathcal{L}_{c,Y}^{\text{FOC}}(X;\Theta)\right)^2 + \left(\mathcal{L}_{c,F}^{\text{FOC}}(X;\Theta)\right)^2 + \left(\mathcal{L}_Y^{\text{MC}}(X;\Theta)\right)^2 + \left(\mathcal{L}_F^{\text{MC}}(X;\Theta)\right)^2.$$

For a training batch $\{X^n\}_{n=1}^N$, the training objective is the empirical mean

$$\widehat{\mathcal{L}}(\Theta) = \frac{1}{N}\sum_{n=1}^{N} \mathcal{L}(X^n;\Theta). \tag{J.1}$$

The full algorithm is given in Algorithm (1).

Algorithm 1 First-stage training.

1: Initialize NNs $\{\hat{\xi}_c, \hat{\xi}_u, \hat{\xi}_p, \hat{\alpha}_{c,Y}, \hat{\alpha}_{c,F}, \hat{\alpha}_{u,Y}, \hat{\alpha}_{u,F}\}$ with parameters Θ.
2: **for** $t = 1, 2, \ldots, T_{\max}$ **do**
3: Sample a batch of aggregate states $\{X^n = (x_c^n, x_u^n, \alpha_p^n)\}_{n=1}^N$.
4: Evaluate the seven network outputs at each sampled state.
5: Recover the derived equilibrium objects state-by-state:

(a) compute $\hat{y}(X^n)$ from goods-market clearing;

(b) compute $\sigma_y(X^n)$ using Itô's lemma;

(c) recover $(\sigma_{R,Y}(X^n), \sigma_{R,F}(X^n))$;

(d) compute the hedging terms from the gradients of $(\hat{\xi}_c, \hat{\xi}_u, \hat{\xi}_p)$;

(e) recover $(\pi_Y(X^n), \pi_F(X^n))$ from the unconstrained investor's first-order conditions;

(f) get $(r(X^n), \mu_X(X^n), \Sigma_X(X^n), \mu_{\xi_j}(X^n))$ from equilibrium identities and Itô's lemma.

6: Construct the residual blocks $\mathcal{L}_j^{\text{HJB}}$, $\mathcal{L}_{c,Y}^{\text{FOC}}$, $\mathcal{L}_{c,F}^{\text{FOC}}$, $\mathcal{L}_Y^{\text{MC}}$, and $\mathcal{L}_F^{\text{MC}}$.
7: Form the batch loss $\widehat{\mathcal{L}}(\Theta)$ in (J.1).
8: Update Θ using ADAM, and update learning-rate scheduler using the current training loss.
9: **end for**

J.2 Valuation ratios

After solving the simplified economy, we price the dividend and risky-debt claims as a function of the state variables (X, s), where $s \in [0, 1]$ denote the dividend share of the aggregate claim. We solve for the valuation ratio of the dividend claim, denoted by $p_D(X, s)$, and the valuation ratio of risky-debt claim, denoted by $p_B(X, s)$. We approximate these functions by two additional neural

networks parameterized by Θ_D and Θ_B, respectively:

$$\hat{p}_D : \mathcal{S} \times [0,1] \to \mathbb{R}_+,$$
$$\hat{p}_B : \mathcal{S} \times [0,1] \to \mathbb{R}_+,$$

At this stage the first-stage simplified equilibrium is held fixed. For each state (X, s), the first-stage solution supplies the coefficients entering the claim-pricing PDEs: the risk-free rate $r(X)$, the market price of risk $\eta(X)$, the state drift $\mu_X(X)$, the state diffusion matrix $\Sigma_X(X)$, and the aggregate-claim return volatility. The remaining coefficients associated with the s-state are known analytically from the decomposition of the aggregate claim into dividend and risky-debt components. Given a candidate valuation ratio $\hat{p}_k(X, s)$ for claim $k \in \{D, B\}$, we compute its first and second derivatives with respect to (X, s) using automatic differentiation, leading to the residual

$$\mathcal{L}_k^{\text{val}}(X, s; \Theta_k), \qquad k \in \{D, B\}.$$

The training objective is simply the empirical mean squared PDE residual,

$$\widehat{\mathcal{L}}_k^{\text{val}}(\Theta_k) = \frac{1}{N} \sum_{n=1}^{N} \big(\mathcal{L}_k^{\text{val}}(X^n, s^n; \Theta_k)\big)^2, \qquad k \in \{D, B\}.$$

The full algorithm is given in Algorithm (2).

Algorithm 2 Valuation-ratio step.

Load the trained first-stage networks.
Initialize the valuation-ratio network $\hat{p}_k$ for claim $k \in \{D, B\}$ with parameters Θ_k.
for $t = 1, 2, \ldots, T_{\max}^{\text{val}}$ **do**
    Sample a batch of states $\{(X^n, s^n)\}_{n=1}^N$.
    Evaluate $\hat{p}_k(X^n, s^n)$ and compute its first and second derivatives with respect to (X, s).
    Recover the drift and volatility of the valuation ratio from Itô's lemma.
    Use the fixed baseline solution to evaluate the coefficients entering the pricing PDE.
    Construct the claim-pricing residual $\mathcal{L}_k^{\text{val}}(X^n, s^n; \Theta_k)$.
    Compute batch loss $\widehat{\mathcal{L}}_k^{\text{val}}(\Theta_k)$.
    Update Θ_k using ADAM step, update the learning-rate scheduler using current training loss.
end for

We follow Gopalakrishna and Wu (2026) and use vectorized automatic differentiation to compute the gradients that speeds up the computation. The total time to run the baseline model on a single-threaded computer with 128GB RAM and NVIDIA Ada 4500 GPU is 8 hours. The HJB loss in the first-stage training problem converges to a mean-squared loss of 3.7×10^{-8}. The corre-

sponding valuation-ratio training problems converge to a normalized mean-squared training loss of 1.4×10^{-5}. These losses are computed from the batch objectives in (J.1) and in the valuation-ratio objective, respectively.